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ABSTRACT

This thesis is concerned with addressing two main questions: what is the expected lifetime of a dual spacecraft cooperative GNSS-R formation flying mission, and how accurate are its predictions adjusted for its underlying Basilisk framework prediction accuracy? To address these questions, two distinct simulators were developed, the Comparative Propagation Simulator, and the GNSS-R Formation Flight Simulator. In order to accurately estimate the formation lifetime, its operational constraints had to be modeled, including the satellite configuration, electronic power system, autonomous mode switching, attitude- and formation control. The estimated lifetime was predicted using modes derived from two simulation campaigns, studying the fuel consumption's sensitivity to the leader and follower deployment velocities, and the steady-state formation-maintenance fuel consumption, respectively. The final lifetime estimates derived from the combination of these two models were 413.6 days and 366.5 days for the most optimistic and pessimistic deployment cases, respectively. However, due to modelling uncertainty and observed formation drift, the estimate was evaluated to be conservative.

The Basilisk propagation configuration used in the lifetime analysis was developed as a continuation of the specialization project, where the previous exponential atmosphere model was replaced by the higher-fidelity NRLMSISE-00 atmosphere model. This thesis configuration was benchmarked against a continuously updating Skyfield-Simplified General Perturbations 4 (SGP4) baseline by evaluating both same-spacecraft state differences and same-formation drift differences. The same-spacecraft comparison showed that the thesis configuration diverged substantially less from the Skyfield-SGP4 reference than the specialization-project configuration. However, the same-formation comparison showed that the thesis configuration predicted larger passive leader-follower drift than the Skyfield-SGP4 baseline. Since the lifetime campaigns were therefore likely subject to higher relative drift than predicted by the comparison baseline, it supports the claim of conservative mission lifetime estimate.

SAMMENDRAG

Denne masteroppgaven tar for seg to hovedspørsmål: hva er den forventede levetiden til et sammarbeidende GNSS-R-formasjonsflygingsoppdrag med to satellitter, og hvordan bør denne estimerte levetiden tolkes sett i lys av prediksjonsnøyaktigheten til det underliggende Basilisk-rammeverket? For å besvare disse spørsmålene ble det utviklet to separate simulatorer: en simulator for å sammenlikne propagasjonssimuleringer, og en GNSS-R-formasjonsflygingssimulator. For å estimere levetiden til formasjonsoppdraget måtte begrensningene tilknyttet GNSS-R oppdraget modelleres, inkludert satellittkonfigurasjon, det elektriske systemet, autonom modusbytting, attityde- og formasjonskontroll. Den estimerte levetiden ble predikert ved hjelp av modeller utledet fra to simuleringsrunder, som henholdsvis undersøkte drivstofforbrukets sensitivitet for leder- og følgersatellittenes utskytningshastighet fra et delt romfartøy, og drivstofforbruket knyttet til formasjonsvedlikehold i stasjonær tilstand. De endelige levetidsestimatene, utledet ved bruk av disse modellene, var 413.6 dager og 366.5 dager for henholdsvis det mest optimistiske og det mest pessimistiske utskytningsscenarioet. På grunn av modelleringsusikkerhet og observert formasjonsdrift ble estimatet vurdert som konservativt.

Basilisk-propagasjonskonfigurasjonen som ble brukt i levetidsanalysen, ble utviklet som en videreføring av spesialiseringsprosjektet, hvor den tidligere eksponentielle atmosfæremodellen ble erstattet av NRLMSISE-00 atmosfæremodellen. Denne konfigurasjonen ble sammenlignet med en kontinuerlig oppdatert Skyfield-SGP4-referanse ved å evaluere både tilstandsforskjeller for samme satellitt og formasjonsdriftsforskjeller. Sammenligningen av samme satellitt viste at konfigurasjonen brukt i denne masteroppgaven divergerte betydelig mindre fra Skyfield-SGP4-referansen enn konfigurasjonen fra spesialiseringsprosjektet. Sammenligningen av formasjonsdrift viste imidlertid at konfigurasjonen i denne masteroppgaven predikerte større passiv leder-følger-drift enn Skyfield-SGP4-referansen. Siden levetidskampanjene dermed sannsynligvis var utsatt for større relativ drift enn det som ble predikert av sammenligningsreferansen, støtter dette vurderingen av levetidsestimatet som konservativt.

PREFACE

This master's thesis marks the completion of my studies in the two-year Master's Degree Program in Cybernetics and Robotics at the Norwegian University of Science and Technology (NTNU), Department of Engineering Cybernetics.

The work has been both challenging and rewarding, and has given me the opportunity to apply theory from cybernetics, orbital mechanics, simulation, and control to a complex space mission analysis problem. Developing and analyzing the Basilisk-based simulation framework has provided valuable insight into both the technical modelling process and the practical limitations that arise when building long-horizon spacecraft simulations.

I would like to express my sincere gratitude to my supervisor, Jan Tommy Gravdahl, for his guidance, feedback, and support throughout the thesis work. I would also like to thank my co-supervisor, Corrado Chiatante, for valuable discussions and academic input during the project.

A special thanks goes to Oliver Keving Hasler for valuable input related to the implementation of the simulations in Basilisk. His insights were helpful in addressing several practical modelling and implementation challenges.

Finally, I would like to thank my family, friends, and fellow students for their support, encouragement, and discussions throughout the semester.

Contents

List of Figures

List of Tables

ABBREVIATIONS

CM Center of Mass. 6, 7

CPO Circular Projected Orbit. 3, 18, 79

ECEF Earth-Centered Earth-Fixed. 6, 11

ECI Earth-Centered Inertial. vii, 6, 9, 12–14, 24, 37, 48, 71

EPS Electronic Power System. vii, 27, 34, 51, 52, 77

GEO Geostationary Earth Orbit. 14

GNC Guidance Navigation and Control. 20, 27, 29, 36, 51, 52

GNSS Global Navigation Satellite Systems. 1, 2, 17, 18, 27, 30, 31, 37, 41, 77

GNSS-R GNSS Reflectometry. 2–4, 17–19, 27, 30, 31, 37, 41, 48, 53, 77–83

LEO Low Earth Orbit. 10, 12, 14–16, 45, 48, 50, 80

MC Monte Carlo. 27–29

MEO Medium Earth Orbit. 14

MRP Modified Rodrigues Parameters. 7, 8, 27, 37, 38

OBC On-Board Computer. 27, 35, 77

OED Orbital Element Difference. viii, x, 59–61, 73, 74, 77, 80

RAAN Right Ascension of the Ascending Node. 8, 9, 15, 16, 74

RTN Radial-Transverse-Normal. 6, 16, 18, 46, 83

RW Reaction Wheel. 27

SAR Sparse Aperture Radar. 18, 78, 80

SGP4 Simplified General Perturbations 4. i, ii, 3, 81, 82

SRP Solar Radiation Pressure. 10, 13, 14, 48

TLE Two-Line Element. 3, 22–24, 44–47, 54, 71, 72, 81, 82

TT Terrestrial Time. 6

CHAPTER

ONE

INTRODUCTION

1.1 Background & Motivation

Global Navigation Satellite Systems (GNSS), continuously broadcast signals from space. According to the European Union Agency for the Space Programme, GNSS constellations broadcast ranging and timing data that allow receivers to determine their position, and several global systems are currently available, including Galileo, GPS, GLONASS, and BeiDou [1]. These signals are therefore not only useful for navigation, but also represent a globally available source of radio signals that can be exploited for other applications.

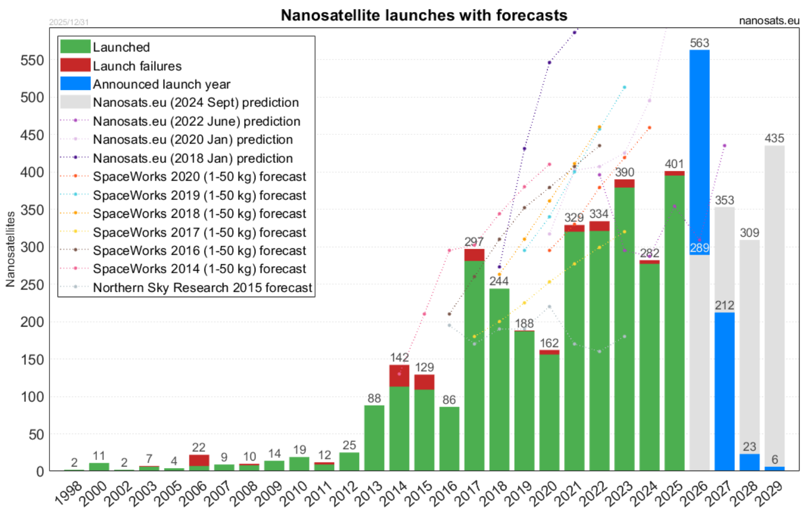

Figure 1.1.1: Nanosatellite launces and forcasts over time. Courtesy of [2]

.

At the same time, small satellites have become increasingly important in space mission design. BryceTech reports that nearly 2800 small satellites were launched in 2024 alone, representing 97% of all spacecraft launched that year [3]. This development is relevant for GNSS-based remote sensing because small satellites make it more practical to deploy a larger quantity of smaller receivers, constellations, and formation-flying missions. Figure 1.1.1 illustrates the corresponding increase in nanosatellite launches and expected future growth.

GNSS Reflectometry (GNSS-R) is an emerging passive remote-sensing technique that has become increasingly relevant due to these two developments. Zavorotny et al. describe GNSS-R as a bistatic radar configuration, where existing GNSS transmitters are used as signals of opportunity and a receiver processes the signals scattered from the Earth's surface [4]. The same source describes how the reflected signals can be used to estimate surface properties such as height, roughness, and dielectric properties. This has enabled applications such as ocean wind sensing, sea-state monitoring, soil moisture estimation, vegetation monitoring, and sea-ice observation. Additional proposed applications include maritime surveillance and ship detection using reflected GNSS signals, as discussed by Di Simone et al. for spaceborne GNSS-R ship-detection applications [5], and detection or localization of GNSS interference, where NTNU's small-satellite GNSS-R project description notes that jamming and spoofing signals can influence both reflected and direct signals measured by a GNSS-R satellite [6].

Despite this potential, GNSS-R measurements are constrained by the weak power of the reflected signals and by the available transmitter-surface-spacecraft geometry. Zavorotny et al. emphasize that GNSS-R relies on scattered signals of opportunity, which makes the measurement dependent on the reflection geometry and the properties of the reflecting surface [4]. Hill et al. further motivates the need for enhanced GNSS-R concepts by pointing out that single-satellite GNSS-R measurements are limited to available reflection tracks, which restricts coverage even when high-resolution coherent reflections are available [7].

Cooperative GNSS-R formation flying has been proposed as one way to improve these limitations. The *HydroSwarm* mission concept uses a swarm of CubeSats to improve GNSS-R measurement capability through three cooperative modes: an along-track coarse mode for increased spatial coverage, a beamforming mode for improved sensitivity at specular points, and a close-formation imaging mode for reducing ambiguities and improving imaging capability [7]. In all cases, the benefit comes from using several spacecraft in a maintained formation, which motivate the need for developing an end-to-end GNSS-R formation flight mission simulator.

1.2 Thesis Formulation

The ultimate goal of this thesis is to estimate the fuel-limited lifetime of a realistic GNSS-R formation flying mission. This is done by developing an end-to-end simulation framework using the Basilisk astrodynamics framework, where orbital perturbations, spacecraft subsystems, attitude control, formation control, propulsion, power consumption, and operational mode switching are combined in one simulation environment. The lifetime estimate is based on the fuel required to acquire and maintain a selected leader-follower formation geometry.

The project is divided into two main parts. The first part continues the work performed in the specialization project in [8]. That project developed an initial framework for comparing a Basilisk numerical propagator against a Skyfield-SGP4 propagation baseline. However, the comparison was limited by the use of a static Two-Line Element (TLE) propagated over the full comparison interval, and by an exponential atmosphere model that was not sufficiently representative for the orbital altitude considered. The first part of this thesis therefore improves both the comparison baseline and the Basilisk atmospheric modelling. A continuously updating TLE Skyfield-SGP4 reference is used, and the final Basilisk configuration includes a higher-fidelity atmospheric density model. This part of the thesis is dedicated to evaluating how the selected high-fidelity Basilisk configuration diverges from the updated SGP4 reference over a long propagation horizon.

The second part builds on the high-fidelity propagator by expanding the simulator into a GNSS-R formation flight mission simulator. This includes a realistic spacecraft configuration, autonomous operational mode switching, electrical power modelling, attitude guidance and control, propulsion-based formation control, and fuel consumption. These modules are included enable the functionalities required by a cooperative GNSS-R formation flight mission, while also imposing the realistic constraints that would restrict GNSS-R operations. Based on these capabilities, the simulator is then used to estimate the fuel required to acquire and maintain a cooperative concentric Circular Projected Orbit (CPO) formation. These fuel contributions are combined to estimate the fuel-limited mission lifetime.

Based on this project formulation, the thesis is guided by two research questions:

1. How does the final high-fidelity Basilisk propagation configuration compare against a continuously updating-TLE Skyfield-SGP4 baseline over a long-horizon propagation interval?

2. What fuel-limited mission lifetime is predicted for the selected GNSS-R 6U CubeSat leader-follower concentric formation when subject to realistic perturbations, operational constraints?

1.3 UN Sustainability Goals

The remote sensing capabilities of GNSS-R are directly relevant to several of the United Nations Sustainable Development Goals. By enabling observations of soil moisture, biomass, wetland inundation, and other climate-sensitive environmental variables, GNSS-R can support SDG 13, *Climate Action*. Since the measurement technique is spaceborne, it can provide repeated observations over large and remote areas. In addition, GNSS-R uses long-wavelength radio signals, which makes it less affected by cloud cover than optical remote sensing methods.

GNSS-R can also contribute to SDG 11, *Sustainable Cities and Communities*, through improved monitoring of flooding and inundation. Spaceborne flood monitoring can support disaster preparedness, emergency response, and long-term resilience planning for communities exposed to extreme weather events. In this context, improved GNSS-R capability through formation flying may contribute to more frequent or higher-resolution observations of flood-prone regions.

CHAPTER

TWO

BACKGROUND THEORY

2.1 Reference Frames

A reference frame $\mathcal{F}^{rf} : \left\{\mathcal{O}_{rf};\ \hat{\mathbf{x}}_{rf}, \hat{\mathbf{y}}_{rf}, \hat{\mathbf{z}}_{rf}\right\}$ is uniquely defined by three dextral orthogonal unit vectors $\{\hat{\mathbf{x}}_{rf}, \hat{\mathbf{y}}_{rf}, \hat{\mathbf{z}}_{rf}\}$ and the origin placement $\mathcal{O}_{rf}$ [9]. This section will present the definitions for the most important reference frames used in the thesis. The reader should note that the individual reference frame subscripts are assigned to be consistent with the conventional Basilisk notation.

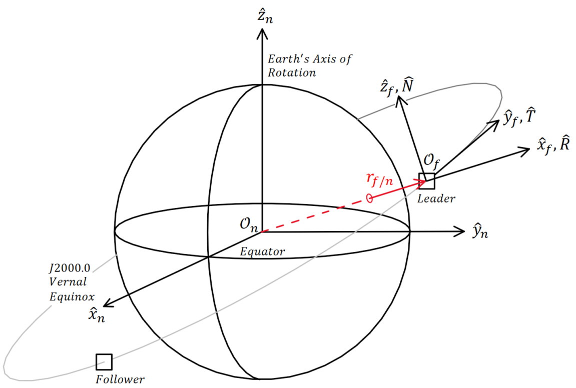

Figure 2.1.1: Illustration of the ECI ($\mathcal{F}^n$)and RTN ($\mathcal{F}^f$) reference frame definitions. The basis unit vectors are displayed with different lengths for illustrative purposes. Adopted from [10, 11] and modified from [8]

.

2.1.1 J2000 Earth-Centered Inertial (ECI) Frame

The ECI reference frame, $\mathcal{F}^n : \{\mathcal{O}_n;\ \hat{\mathbf{x}}_n, \hat{\mathbf{y}}_n, \hat{\mathbf{z}}_n\}$, is fixed relative to the distant stars, and is therefore commonly approximated as an inertial frame in which Newtonian mechanics are valid [12]. The frame is defined with its origin in the Earth's Center of Mass (CM), with $\hat{\mathbf{z}}_n$ pointing along its rotational axis towards the geographical North Pole, $\hat{\mathbf{x}}_n$ pointing towards the vernal equinox through the equator, and $\hat{\mathbf{y}}_n$ defined according to the right-hand rule. Multiple variants of the ECI definition exists, primarily varying in the epoch used to define the Earth's equatorial plane and equinox direction. In this thesis, the J2000 ECI frame is adopted. It is defined using the J2000.0 reference epoch, equivalent to 1st January 2000, at 12:00 Terrestrial Time (TT) [13]. The frame definition is illustrated in Figure 2.1.1.

2.1.2 Earth-Centered Earth-Fixed (ECEF) Frame

The Earth-Centered Earth-Fixed (ECEF) reference frame $\mathcal{F}^e : \{\mathcal{O}_e;\ \hat{\mathbf{x}}_e, \hat{\mathbf{y}}_e, \hat{\mathbf{z}}_e\}$ is fixed relative to the Earth's surface, with the frame origin positioned in its CM. This thesis adopts the WGS84 ECEF formulation as presented by Stryjewski in [14], where the $\hat{\mathbf{x}}_e$ axis points through the Prime Meridian in the equatorial plane, $\hat{\mathbf{z}}_e = \hat{\mathbf{z}}_n$ points along the Earth's rotational axis, and $\hat{\mathbf{y}}_e$ is defined according to the right-hand rule. Because of the Earth's angular velocity, the transformation $\mathbf{R}_n^e$ between ECI to ECEF can be expressed using a time-dependent rotation angle $\theta(t)$ about the shared $\hat{\mathbf{z}}_e$ axis [8]

$$\mathbf{R}_n^e = \begin{bmatrix} \cos\theta(t) & -\sin\theta(t) & 0 \\ \sin\theta(t) & \cos\theta(t) & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{R}_e^n = (\mathbf{R}_i^e)^T \tag{2.1}$$

2.1.3 Radial-Transverse-Normal (RTN) Frame

The Radial-Transverse-Normal (RTN) reference frame, $\mathcal{F}^f : \left\{\mathcal{O}_f;\ \hat{\mathbf{x}}_f, \hat{\mathbf{y}}_f, \hat{\mathbf{z}}_f\right\}$, is a local orbital frame that is useful for expressing relative motion between spaceborne objects. Vallado presents the frame as the *Satellite coordinate system, RSW* in [15], but the *RTN* name is chosen for this thesis to better convey each component's physical significance. For use in formation flight, the coordinate frame origin is assigned to the leader satellite, l's CM. The dextral orthogonal unit vectors are defined with the $\hat{\mathbf{x}}_f$ axis pointing in the radial direction along $\mathbf{r}_{l/n}^n$, the $\hat{\mathbf{z}}_f$ axis points normal to the orbital plane, and $\hat{\mathbf{y}}_f$ fulfills the right-hand rule. Note that the latter will always point in direction of positive velocity, but will only be perpendicular to the velocity for circular orbits [15, 16]. Using this definition, the frame can be expressed mathematically as:

$$\hat{\mathbf{R}}^n = \hat{\mathbf{x}}_f^n = \frac{\mathbf{r}_{l/n}^n}{||\mathbf{r}_{l/n}^n||}, \qquad \hat{\mathbf{N}}^n = \hat{\mathbf{z}}_f^n = \frac{\mathbf{r}_{l/n}^n \times \mathbf{v}_{l/n}^n}{||\mathbf{r}_{l/n}^n \times \mathbf{v}_{l/n}^n||}, \qquad \hat{\mathbf{T}}^n = \hat{\mathbf{y}}_f^n = \hat{\mathbf{z}}_f^n \times \hat{\mathbf{x}}_f^n \tag{2.2}$$

Here, $\mathbf{r}_{l/n}^n$ and $\mathbf{v}_{l/n}^n$ are used to denote the Earth-relative leader position and velocity expressed in ECI, while $\hat{\mathbf{y}}_f$ and $\hat{\mathbf{z}}_f$ correspond to the along- and cross-track directions, respectively. An illustration of the leader satellite's position relative to $\mathcal{O}_n$ and its affixed RTN frame is shown in Figure 2.1.1.

2.1.4 Body-Fixed Frame

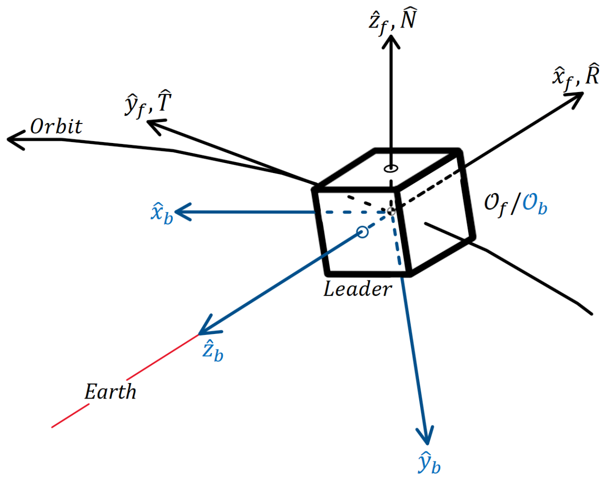

Figure 2.1.2: Illustration of the body-fixed ($\mathcal{F}^b$)and RTN ($\mathcal{F}^f$) reference frame definitions for a 1U CubeSat. Adopted from [17].

The body-fixed reference frame, $\mathcal{F}^b : \{\mathcal{O}_b;\ \hat{\mathbf{x}}_b, \hat{\mathbf{y}}_b, \hat{\mathbf{z}}_b\}$, is a moving reference frame that is rigidly attached to the spacecraft body. Because of this property, the body-fixed frame is used for expressing satellite attitude relative to an inertial frame, actuator torques and sensor measurements, since these quantities are naturally expressed relative to the body itself. The orientation of its axes are typically chosen to best represent the spacecraft geometry, and to coincide with the principal axes of inertia [18]. Consequentially, CubeSat body axes are commonly selected to be perpendicular to its side panels. The frame origin is commonly placed at the CM in order to simplify the rotational equations of motion, and to reduce translational-rotational coupling [19]. An illustration of the body frame definition for a 1U CubeSat is shown in Figure 2.1.2.

2.2 Modified Rodrigues parameters

Modified Rodrigues Parameters (MRP)s is a 3-element attitude parameterization commonly used for spacecraft attitude control. The representation can be derived from the Euler-Rodrigues symmetric parameters, better known as the unit quaternion:

$$\mathbf{q} = \begin{bmatrix} \boldsymbol{\epsilon} \\ \eta \end{bmatrix} = \begin{bmatrix} \epsilon_1 & \epsilon_2 & \epsilon_3 & \eta \end{bmatrix}^T, \qquad \mathbf{q}^T\mathbf{q} = |\boldsymbol{\epsilon}|^2 + \eta^2 = 1 \tag{2.3}$$

where η is the scalar component and $\boldsymbol{\epsilon}$ is the imaginary vector. Using this definition, Shuster writes in [20] that the MRPs are obtained through a stereographic projection of the unit quaternion onto $\mathbb{R}^3$

$$\mathbf{p} = \frac{\boldsymbol{\epsilon}}{1+\eta}, \qquad \mathbf{m} = \frac{\boldsymbol{\epsilon}}{1-\eta} \tag{2.4}$$

Here, $\mathbf{p}$ and $\mathbf{m}$ are the positive and negative forms, differing only in the singularity placement. This thesis utilizes the positive form, and will hereby be referred to as $\boldsymbol{\sigma}$. Using the axis-angle representation of a rotation, where $\hat{\mathbf{n}}$ represents the unit rotation axis and θ is the angle of rotation about the axis, the positive-form MRPs can be expressed as:

$$\mathbf{q} = \begin{bmatrix} \hat{\mathbf{n}} \sin(\theta/2) \\ \cos(\theta/2) \end{bmatrix} \quad \Rightarrow \quad \boldsymbol{\sigma}(\theta, \hat{\mathbf{n}}) = \hat{\mathbf{n}} \tan\left(\frac{\theta}{4}\right), \tag{2.5}$$

From this formulation, it follows that the MRPs become singular as $\theta \to 2\pi$, since

$$\tan\left(\frac{\theta}{4}\right) \to \infty \quad \text{for} \quad \theta \to 2\pi \tag{2.6}$$

To avoid numerical issues near the singularity, an equivalent MRP shadow set, $\boldsymbol{\sigma}^s$, may be used [21]. The shadow set transformation is defined as:

$$\boldsymbol{\sigma}^s = -\frac{\boldsymbol{\sigma}}{\boldsymbol{\sigma}^T \boldsymbol{\sigma}} \tag{2.7}$$

Switching to the shadow set when $||\boldsymbol{\sigma}|| > 1$ effectively bounds the MRP set by the unit sphere, and prevents the parameterization from approaching the singularity numerically [22, 21, 20].

2.3 Classic Orbital elements

The classic orbital elements, also known as Keplerian elements, are a set of six parameters used to uniquely define the shape, size and orientation of an orbital plane, as well as the position of an orbiting body [23, 15]. In astrodynamics, these parameters provide a compact and intuitive representation of orbital motion, commonly used for orbital analyses and spacecraft guidance. The classic orbital elements used in this thesis are:

$$\left.\begin{array}{ll} a: & \text{semimajor axis} \\ e: & \text{eccentricity} \end{array}\right\} \text{orbital plane size and shape}$$

$$\left.\begin{array}{ll} i: & \text{inclination} \\ \Omega: & \text{right ascension of the ascending node (RAAN)} \\ \omega: & \text{argument of perigee} \end{array}\right\} \text{orbital plane orientation}$$

$$\left.\begin{array}{ll} M: & \text{mean anomaly} \end{array}\right\} \text{orbital body position}$$

It is important to note that the semimajor axis, and mean anomaly are in some orbital element configurations replaced with the angular momentum h, and the true anomaly θ, respectively. In any case, these six parameters define a Keplerian orbit, and they can be grouped according to the orbital property they describe: The semimajor axis and eccentricity define the shape and size of the orbital plane, while the inclination, Right Ascension of the Ascending Node (RAAN) and argument of perigee describe the plane's orientation. Finally, the mean anomaly is used to describe the spaceborne object's position along the orbital trajectory. An illustration of the classic orbital elements is shown in Figure 2.3.1.

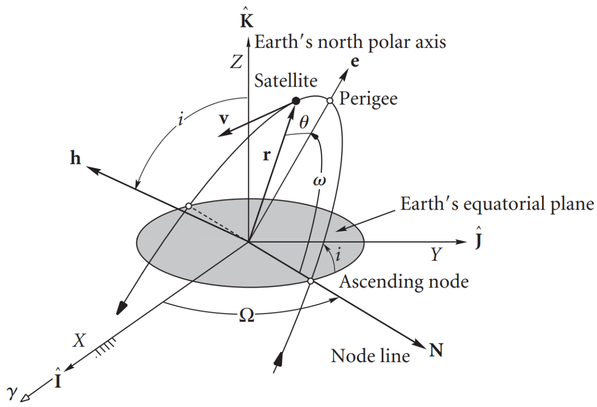

Figure 2.3.1: Orbital elements relative to the ECI frame. Courtesy of [23]

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The semimajor axis is defined as half of the longest ellipse diameter, and determines the overall size of the orbit. For elliptical orbits, this parameter is directly coupled with the orbital period through Kepler's third law. The eccentricity is a measure of how much the orbital trajectory deviates from a circle. It takes values in the span between zero and one, where an eccentricity of zero corresponds to a perfectly circular orbit, and an increasing eccentricity correspond to an elliptical orbit of increasing elongation.

The inclination is the intersection angle between the orbital plane and the reference plane, commonly the equatorial plane spanned by the $\hat{x}_n$ and $\hat{y}_n$ axes in ECI. This angle determines the tilt of the orbital plane, while its orientation about the $\hat{z}_n$ axis is determined by the RAAN parameter. For nonzero inclinations, the orbital trajectory intersects the reference plane at two points, the ascending and descending nodes. The ascending node is where the object travels from the Southern to the Northern Hemisphere, and RAAN is defined as the angle between this node and the $\hat{x}_n$ axis in the equatorial plane.

The argument of perigee specifies the orientation of the ellipse within the orbital plane. It is defined as the angle from the ascending node to the perigee along the orbital trajectory, where perigee is a term specific to Earth satellites, describing the point along the trajectory closest to Earth. Finally, the mean anomaly defines the orbiting body's position along the orbit as a function of time. It is defined as the elapsed fraction of the orbital period since the orbiting body passed the perigee [23, 24].

2.4 Orbital Perturbations

D.A. Vallado defines orbital perturbations as deviations from the ideal Keplerian two-body motion caused by additional gravitational or non-gravitational accelerations acting on a spacecraft [15]. For an Earth-orbiting satellite, the Earth's oblateness, atmospheric drag, Solar Radiation Pressure (SRP) and third-body attraction from the Sun and Moon all cause its trajectory to deviate from the ideal point-mass solution. The following subsections will therefore address each major perturbation effect in approximate decreasing order of significance for Low Earth Orbit (LEO) satellites in accordance with Montenbruck & Gill in [25].

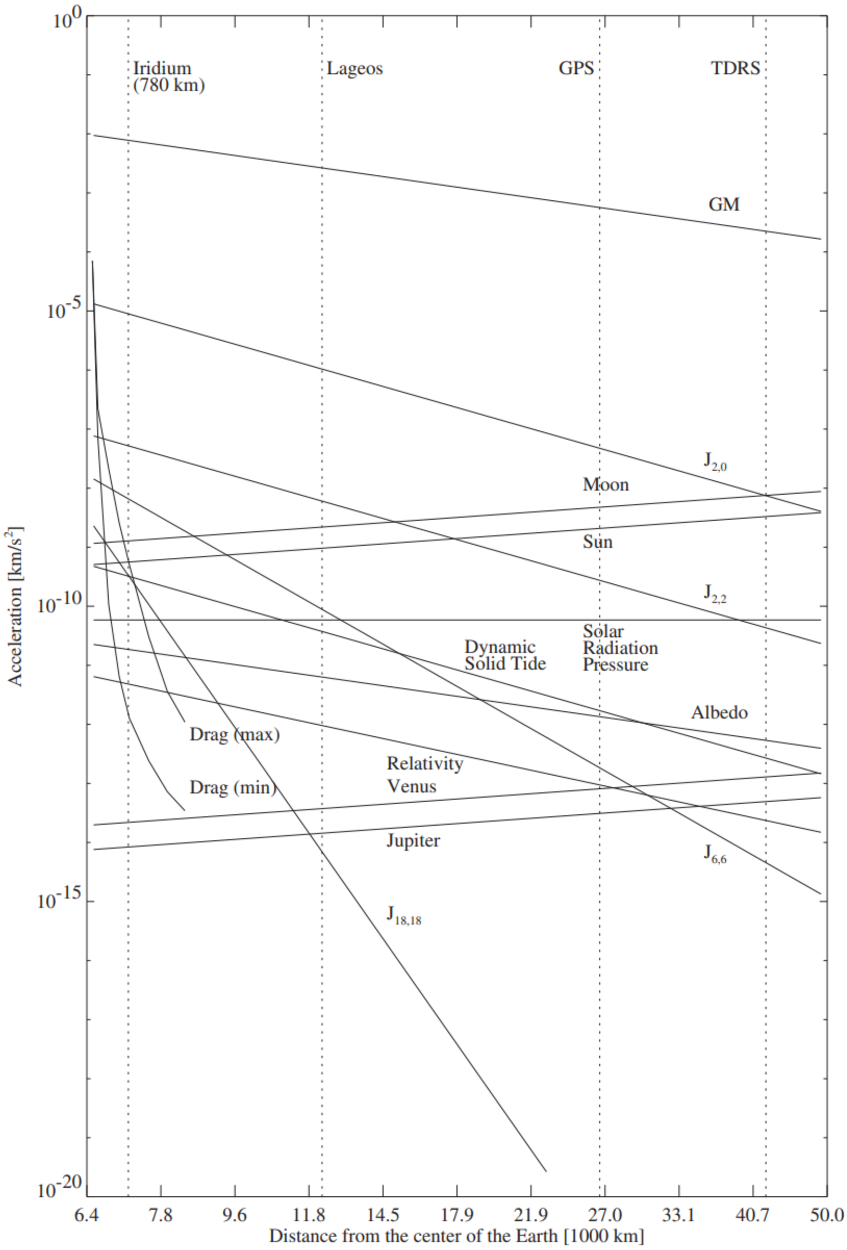

Figure 2.4.1: Acceleration magnitudes caused by various orbital perturbations. Courtesy of [25]

2.4.1 Earth's Oblateness

Keplerian motion assumes that the Earth can be represented as a perfectly spherical body with all its mass concentrated in the earth-centered inertial frame origin. Under this assumption, the gravitational force acting on a spacecraft, s, can be modeled using Newton's law of gravity

$$\mathbf{a}_{grav} = \left(\ddot{\mathbf{r}}_{s/n}\right)_{grav} = -\frac{\mu}{||\mathbf{r}_{s/n}||_2^3}\mathbf{r}_{s/n} \iff \left(\ddot{\mathbf{r}}_{s/n}\right)_{grav} = \nabla U, \quad \text{where } U = \mu\frac{1}{||\mathbf{r}_{s/n}||_2} \tag{2.8}$$

Using the notation from [8], $\mu = GM$ is the Earth's gravitational constant, and ∇U is the gradient of the gravity potential. In accordance with the spherical assumption, the gravity potential is only dependent on the satellite's geocentric distance, implying uniform density. In actuality, the Earth's mass is unhomogeneously distributed through its volume, and centrifugal forces causes the planet to bulge outwards around the equator due to its angular velocity. These irregularities produce perturbing accelerations that gradually alter the orbital trajectories over time.

Montenbruck & Gill explain that these deviations from spherical symmetry can be modeled by expanding the Earth's gravity potential into a spherical harmonic series representation [25]. The resulting series expresses the geopotential as a weighted sum of spherical harmonic terms, describing gravitational irregularities caused by the Earth's unhomogeneous mass distribution. Here, the spacecraft position is represented in ECEF spherical coordinates consisting of the geocentric distance $r = ||\mathbf{r}^e_{s/n}||_2$, the geocentric latitude ϕ^e, and the longitude λ^e. Using these coordinates together with the associated Legendre polynomials, the geopotential may be written in the general spherical harmonic form as

$$U(r, \phi^e, \lambda^e) = \frac{\mu}{R}\sum_{n=0}^{\infty}\sum_{m=0}^{n}\left(C_{nm}V_{nm} + S_{nm}W_{nm}\right) \tag{2.9}$$

where,

$$V_{nm} = \left(\frac{R}{r}\right)^{n+1} P_{nm}(\sin\phi^e)\cos(m\lambda^e), \qquad W_{nm} = \left(\frac{R}{r}\right)^{n+1} P_{nm}(\sin\phi^e)\sin(m\lambda^e). \tag{2.10}$$

Here, R denotes the Earth's radius, P_{nm} are the associated Legendre polynomials, while C_{nm} and S_{nm} are the normalized geopotential coefficients that encode the Earth's non-uniform mass distribution. The $_{nm}$ subscript denotes the degree n and order m, where increasing value correspond to decreasingly significant perturbation terms. The harmonic degree relates to longitudinal mass variation, while the order determines the longitudinal dependency on the harmonic term.

The reader should note that a common simplification is to assume order $m = 0$ due to comparatively small variations in the Earth's gravity field along the longitude. This yields the zonal harmonic expression, where the S_{n0} terms vanish, and the geopotential coefficients are commonly expressed as $J_n = -C_{n0}$. This gives the following expression derived from [25, 15]:

$$U(r, \phi^e) = \frac{\mu}{r}\left[1 - \sum_{n=2}^{4} J_n \left(\frac{R}{r}\right)^n P_n(\sin\phi^e)\right] \tag{2.11}$$

The resulting zonal model captures the dominant long-term perturbation effects while significantly reducing the complexity of the gravity model. Amongst the zonal coefficients, the second degree term J_2 dominates the non-spherical effects, as is clearly evident by the substantially larger acceleration magnitude of $J_{2,0}$ in Figure 2.4.1 compared to the other terms. The Figure also illustrates that the higher degree and order terms rapidly decrease in magnitude. However, to include the dominant lateral and longitudinal non-spherical effect, this thesis employs the general spherical harmonic formulation in Equation 2.9, with truncated degree and order.

2.4.2 Atmospheric Drag

Orbital atmospheric drag is caused by a momentum exchange between the orbiting body's surface and colliding atmospheric molecules along its trajectory. According to Montenbruck & Gill, this effect is the dominant non-gravitational perturbation acting on a spacecraft in LEO, as is also illustrated in Figure 2.4.1. Since the drag force acts opposite to the spacecraft's velocity, it continuously removes orbital energy, and causes a gradual reduction in orbital altitude and velocity over time [15, 25, 8].

For spacecraft operating in LEO, the interaction between the spacecraft surface and surrounding atmosphere is commonly modeled using the free-molecular flow assumption. In this regime, the atmospheric density is sufficiently low such that the inter-molecular collisions are negligible compared to the molecule-surface interaction. The drag acceleration experienced by a spacecraft s can then be expressed using the classical aerodynamic formulation, as presented in [25]

$$\mathbf{a}^n_{\mathrm{D}} = \left(\ddot{\mathbf{r}}^{\,n}_{s/n}\right)_D = -\tfrac{1}{2} C_D \frac{A_D}{m_s}\, \rho\, v^2_{\mathrm{rel}}\, \hat{\mathbf{v}}^n_{rel} \tag{2.12}$$

where C_D is the dimensionless drag coefficient, A_D is the effective cross-sectional area normal to the incoming flow, m_s is the spacecraft mass and ρ is the local atmospheric density. v_{rel} denotes the relative speed, while $\hat{\mathbf{v}}^n_{rel}$ is the unit vector in the direction of relative velocity. It is important to note that the relative velocity differs from the inertial spacecraft velocity because it represents its velocity relative to the local atmosphere [8]. Following Montenbruck & Gill in [25], the relative velocity can therefore be approximated by assuming the atmosphere co-rotates with the Earth

$$\mathbf{v}^n_{rel} = \mathbf{v}^n_{s/n} - \boldsymbol{\omega}^n_{e/n} \times \mathbf{r}^n_{s/n} \tag{2.13}$$

where $\{\mathbf{r}^n_{s/n}, \mathbf{v}^n_{s/n}\}$ are the spacecraft position and velocity expressed in the ECI frame, and $\boldsymbol{\omega}^n_{e/n}$ is the Earth's angular velocity vector. Amongst the parameters in the classic drag formulation, the atmospheric density ρ is generally the largest source of modeling uncertainty. In this thesis, two density models will be evaluated: the NRLMSISE-00 model and the exponential density model. The exponential density model is a heavily simplified alternative, and assumes exponentially decreasing density with increasing altitude.

$$\rho(h) = \rho_0 \exp\left(-\frac{h-h_0}{H_0}\right), \qquad H_0 = \frac{\mathcal{R}T}{\mu_m g} \tag{2.14}$$

Here, ρ_0 is the reference density at some geocentric reference height h_0, and h is the spacecraft altitude above the reference height. Finally, H_0 is the density scale height, whose equation in 2.14 can be derived from the hydrostatic equation and gas law from thermodynamics. This model captures the approximate atmospheric decay with altitude, but fails to account for major spatial and temporal effects, such as solar activity, geomagnetic disturbances, longitudinal/lateral variations and long-term solar cycles [25].

The NRLMSISE-00 empirical model accounts for these spatial and temporal atmospheric variations, and has been developed using several decades of atmospheric observations and satellite measurements. In addition to the density, the model also estimates other atmospheric parameters, such as temperature and individual molecule densities. Because of the spatial and temporal dependency of these parameters, the empirical model requires additional inputs describing both the spacecraft environment and current atmosphere conditions. These include the spacecraft altitude, like the exponential density model, but also the geographic location, current time and different time-averaged measurements of solar activity represented through the $F_{10.7}$ solar radio flux, and the geomagnetic index A_p [26]. Together, these parameters allow the model to account for short- and long-term variations in the atmospheric density.

2.4.3 Solar radiation pressure

Like atmospheric drag, SRP is a non-gravitational perturbation caused by momentum transfer between the spacecraft and incident particles. However, in the case of SRP, this transfer originates from photons rather than residual atmospheric molecules. When photons are absorbed or reflected by the spacecraft surface, they transfer their momentum to the spacecraft, generating a small force in the approximate direction away from the Sun. Because the force acts continuously over the illuminated surface area, it is referred to as a radiation pressure [25, 8].

As discussed by Montenbruck & Gill in [25], the radiation pressure does not vary with the geocentric altitude directly, but rather the distance from the Sun according to the inverse-square law. Because the variation in Sun-spacecraft distance is negligible compared to the Earth orbital altitude, the resulting SRP acceleration experienced by an Earth-orbiting spacecraft is approximately constant, as illustrated in Figure 2.4.1. Although the SRP acceleration remains approximately constant with altitude, its relative significance increases at higher altitudes where atmospheric drag becomes progressively weaker, especially for spacecraft with large area-to-mass ratios.

For a flat spacecraft surface, the resulting SRP acceleration depends on the relative geometry between the solar radiation direction $\mathbf{r}^n_{Sun/s}$, and the spacecraft surface normal $\hat{\mathbf{n}}$. Using the formulation presented by Montenbruck & Gill, the acceleration in the ECI frame can be written as

$$\mathbf{a}^n_R = \left(\ddot{\mathbf{r}}^n_{s/n}\right)_R = -P_{Sun}\frac{\mathrm{AU}^2}{||\mathbf{r}^n_{Sun/s}||_2^2}\frac{A_R}{m_s}\cos(\theta)\left[(1-\varepsilon)\hat{\mathbf{r}}^n_{Sun/s} + 2\varepsilon\cos(\theta)\hat{\mathbf{n}}_s\right] \tag{2.15}$$

where P_{Sun} denotes the solar radiation pressure at one astronomical unit (AU), the parameter A_R/m_s represents the spacecraft sun-normal cross-sectional area-to-mass ratio, and θ is the angle between the incident radiation direction and $\hat{\mathbf{n}}$. Finally, ε denotes to the fraction of reflected incoming radiation, where $\varepsilon = 0$ and $\varepsilon = 1$ correspond to perfect absorption and reflection, respectively.

For many orbit propagation applications, Montenbruck & Gill further argue that the simplified assumption $\theta = 0$ provides sufficient accuracy[25]. Under this assumption, the SRP model reduces to the commonly used cannonball model

$$\mathbf{a}_R^n = \left(\ddot{\mathbf{r}}_{s/n}^n\right)_R = -P_{Sun} C_R \frac{A_R}{m_s} \frac{\mathbf{r}_{Sun/s}^n}{||\mathbf{r}_{Sun/s}^n||_2^3} \, \mathrm{AU}^2 \tag{2.16}$$

where C_R is the radiation pressure coefficient. This approximation models the spacecraft as an idealized spherical body with uniform reflective properties, thus simplifying the SRP model while keeping the dominant perturbation behaviour [15, 8].

2.4.4 Third-Body Perturbations

In addition to the Earth's gravitational field, Earth-orbiting spacecraft are subjected to gravitational attraction from other massive bodies within the solar system. For satellites orbiting around the Earth, the Sun and Moon are the dominant sources of third-body perturbation. Although these bodies are located far from the spacecraft compared to the Earth, their gravitational influence still produce measurable orbital perturbations, particularly for satellites operating in Medium Earth Orbit (MEO), Geostationary Earth Orbit (GEO), or highly elliptical trajectories, as presented by Figure 2.4.1 [25, 8].

Third-body perturbations introduce additional gravitational accelerations that continuously vary with the relative geometry between the spacecraft, Earth and the perturbing body. Although the perturbing accelerations from the Sun and Moon are much smaller than the Earth's central gravitational attraction, Montenbruck & Gill note that the influence become increasingly significant with orbital altitude, becoming comparable to geopotential perturbations of low degree [25]. Because the perturbing acceleration changes direction throughout the orbit, the resulting effect manifests as gradual deviations from ideal Keplerian motion, and contributes both periodic and long-term variations in the spacecraft trajectory, which is consistent with the general three-body dynamics described by Curtis in [23].

Following the formulation presented by Montenbruck & Gill, the perturbing acceleration from a third massive body B is obtained by subtracting the gravitational acceleration excerted by the perturbing body on the Earth from the gravitational acceleration acting on the spacecraft. Using the same notation as in [8], the perturbing acceleration expressed in the ECI frame becomes

$$\mathbf{a}_B^n = \left(\ddot{\mathbf{r}}_{s/n}^n\right)_B = \mu_B \left(\frac{\mathbf{r}_{B/n}^n - \mathbf{r}_{s/n}^n}{||\mathbf{r}_{B/n}^n - \mathbf{r}_{s/n}^n||_2^3} - \frac{\mathbf{r}_{B/n}^n}{||\mathbf{r}_{B/n}^n||_2^3} \right) = \mu_B \left(\frac{\mathbf{r}_{B/s}^n}{||\mathbf{r}_{B/s}^n||_2^3} - \frac{\mathbf{r}_{B/n}^n}{||\mathbf{r}_{B/n}^n||_2^3} \right) \tag{2.17}$$

When multiple perturbing bodies are considered simultaneously, the total third-body acceleration is obtained by summing the individual contributions from each body. Using the major perturbing bodies for a LEO satellite, this becomes

$$\mathbf{a}_{B,\mathrm{tot}}^n = \mathbf{a}_{\mathrm{Sun}}^n + \mathbf{a}_{\mathrm{Moon}}^n \tag{2.18}$$

2.5 Differential Perturbations in Formation-Flying

Alfriend et al. refers to the generally agreed upon definition of spacecraft formation flying in [27] as: "The tracking or maintenance of a desired relative separation, orientation or position between or among spacecraft". It then follows that differential perturbations describe the relative effect of environmental accelerations acting on two satellites in formation, that cause the formation to naturally drift away from its desired geometry over time. The difference in experienced accelerations by the two spacecraft in formation can arise because the spacecraft occupy slightly different obits, positions, velocities and attitudes, or because they have different physical attributes. In formation-flying, even small prolonged differences in experienced acceleration can accumulate over many orbital periods to gradually change the radial, along- or cross-track separation. Differential perturbations therefore motivate the need for formation-control strategies to preserve the desired formation geometry within mission constraints. Since stronger natural drift generally requires more frequent or larger corrective measures, differential perturbations also become direct drivers for formation maintenance fuel consumption, and thus, also mission lifetime [25, 27]. The following subsections will therefore summarize the most important differential perturbation effects for LEO orbiting formations.

2.5.1 Natural Relative Drift Under Ideal Two-Body Motion

For a leader-follower spacecraft pair, the relative orbit can be described by the difference between their respective classic orbital elements. If the leader has orbital elements $\boldsymbol{\alpha}_L$, and the follower has $\boldsymbol{\alpha}_F$, the differential orbital elements can be expressed as

$$\Delta\boldsymbol{\alpha} = \boldsymbol{\alpha}_F - \boldsymbol{\alpha}_L = \begin{bmatrix}\Delta a & \Delta e & \Delta i & \Delta\Omega & \Delta\omega & \Delta M_0\end{bmatrix}^T \tag{2.19}$$

where the components represent the differences in semimajor axis, eccentricity, inclination, RAAN, argument of perigee and mean anomaly at a chosen epoch, respectively. Under the ideal two-body assumption, where the Earth is modeled as a spherical body and all perturbing forces from Section 2.4 are neglected, the only form of secular formation drift is caused by a difference in the semimajor axis. This is stated by Alfriend et al. in [27], who explain that the drift only manifest in the along-track direction, resulting in continuous differential mean anomaly growth. This interpretation is also consistent with Hammerl & Schaub, who show that the relative radial offset must be zero for bounded relative motion in [28].

Other differential orbital elements still contribute to natural relative motion, but they do not cause secular drift in the ideal two-body problem. Instead, they induce bounded periodic motion. For example, a differential inclination produces periodic cross-track motion as the two orbital planes are tilted with respect to each other. Similarly, differential eccentricity contributes to in-plane radial and along-track oscillations. These periodic relative motions may be important for the formation geometry, but they do not themselves cause the formation to drift apart. However, additional secular drift mechanisms arise from differential perturbations, especially differential J_2 effects and atmospheric drag [29].

2.5.2 Differential J_2 Perturbations

As presented in Section 2.4.1, J_2 is the second degree, zeroth order zonal coefficient in the spherical harmonic representation of the Earth's gravity field. The coefficient represents the dominant term associated with the Earth's oblateness, and the zonal model of the same degree and order is therefore commonly known as the J_2 perturbation model. In a formation-flying context, the differential J_2 perturbation refers to the difference between the J_2-induced accelerations acting on the two spacecraft, and is one of the most important causes of secular formation drift in LEO.

The J_2 perturbing acceleration causes a slow precession of the orbit, seen primarily as precession of the orbital plane and a rotation in the argument of perigee. Alfriend et al. gives the corresponding per-satellite secular rates for RAAN and the argument of perigee in [27], which can be expanded to the following expression [30]

$$\dot{\Omega} = -\frac{3\bar{n}J_2R^2}{2a^2\left(1-e^2\right)^{1/8}}\cos i \tag{2.20}$$

$$\dot{\omega} = -\frac{3\bar{n}J_2R^2}{2a^2\left(1-e^2\right)^{1/8}}\left(\frac{5}{2}\sin^2 i - 2\right) \tag{2.21}$$

where $\bar{n}$ is the mean motion, R is the Earth radius, and a, e and i are the satellite semimajor axis, eccentricity and inclination, respectively. Note that these per-satellite equations can be expressed as a function of each differential orbital elements by Taylor series expansion about the leader orbit, which has been derived for differential inclination in [27]. Examining Equations 2.20 and 2.21 shows that a difference in either semimajor axis, eccentricity or inclination between the satellites in a formation can cause J_2-induced secular rates, resulting in secular drift. This effect will manifest in the RTN frame as radial, cross- and along-track drift, depending on which secular rates differ. Because both secular rates are inversely proportional to $(1-e^2)$ a difference in eccentricity will cause much smaller secular rate than a corresponding difference in inclination for near-circular orbits where $e \approx 0$ [27].

2.5.3 Differential Atmospheric Drag

The atmospheric drag acceleration acting on a single spacecraft was introduced in Section 2.4.2. For formation-flying, however, the relative effect of drag is determined by the difference between the drag accelerations acting on each spacecraft. D'Amico & Montenbruck explain in [29], that atmospheric density variations can be neglected for close formations with sub-kilometer separation. Under this approximation, the main source of differential atmospheric drag is not a difference in local atmosphere, but a difference in the spacecraft ballistic coefficients. The ballistic coefficient is defined as

$$b = C_D\frac{A_D}{m_s} \tag{2.22}$$

where C_D, A_D and m_s are the drag coefficient, flow-normal cross-sectional area and spacecraft mass from equation 2.12. If the leader and follower spacecraft have ballistic coefficients b_L and b_F, the relative ballistic coefficient difference can be written as $\varepsilon = (b_F - b_L)/b_L$. The resulting relative along-track drift over a time duration Δt with a constant relative ballistic coefficient difference is given in [29]

$$\Delta r_{\hat{T}} = \frac{1}{2}\varepsilon ||\mathbf{a}_D||_2 \cdot \Delta t^2 \tag{2.23}$$

where $||\mathbf{a}_D||_2$ denotes the drag acceleration magnitude of the leader spacecraft. This expression shows that small ballistic coefficient differences can accumulate into substantial along-track drift if the difference is persistent throughout significant time. Finally, it is important to note that the ballistic coefficient is attitude dependent though the flow-normal cross-sectional area, which makes relative spacecraft attitude an important driver for relative along-track drift and subsequent fuel consumption [27, 29].

2.6 GNSS-R Mission Concepts

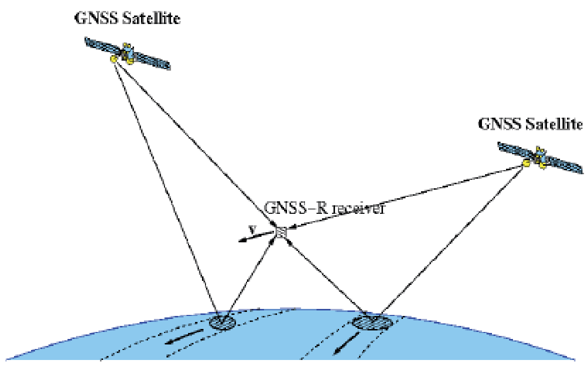

Figure 2.6.1: Illustration of the GNSS-R measurement principle. Courtesy of [31]

.

2.6.1 Remote Sensing Principles

GNSS-R is a passive remote sensing technique that uses reflected GNSS signals as signals of opportunity. As shown in Figure 2.6.1, GNSS-R forms a bistatic radar configuration where the GNSS satellite acts as a non-cooperative transmitter, while the observing spacecraft carries the receiver. Zavorotny et al. describes this concept in [4] as GNSS bistatic radar of opportunity, there the broadcasted GNSS signals illuminate the Earth's surface, and the receiver processes the scattered signal to infer surface properties such ass ocean wave height, wind speeds, soil moisture, etc.

2.6.2 Cooperative GNSS-R Measurements

Hill et al. propose three different formation flying modes in [7] that can enhance the GNSS-R capabilities beyond a single receiver spacecraft. First, the *coarse swarm mode*, or *along-track mode* uses several GNSS-R satellites separated by approximately 15 km in the along-track direction to enhance the spatial coverage by observing adjacent ground tracks simultaneously. Second, the *beamforming mode* places the GNSS-R satellites in a closer formation using a CPO with approximately 1000 m radius and 60° separation between its orbiting satellites. This formation type is defined as the relative orbit where the followers' motion around the leader or barycenter projected into two of the RTN basis-normal planes traces an approximate circle [32]. In this configuration, the receivers are treated as a sparse antenna array cooperatively measuring the same specular point reflection. The reflected signals measured by the separate spacecraft can then be phase-adjusted and combined in post-processing to increase the effective gain in the direction of a selected specular point, resulting in improved sensitivity [7].

The third and final operational mode is the *SAR imaging mode*, which uses an even tighter formation type. In the HydroSwarm concept, this is achieved using concentric CPOs, or simply a *concentric formation*, where the leader spacecraft is placed near the center and the remaining spacecraft occupy relative orbits separated by approximately 100 m in increasingly large orbital shells. The purpose of this configuration is to use the distributed receivers as a Sparse Aperture Radar (SAR). This enables post-processing to reduce data ambiguities, which increases both the cross-track coverage and resolution. While the processing step is more complex for this mode, it does not require cooperative measurements of the same specular point reflections [7].

2.6.3 Operational Constraints

The operational constraints of a formation-flying GNSS-R mission follow from both the bistatic observation geometry and the cooperative formation requirements proposed in [7]. The general GNSS-R measurement principle, described by Zavorotny et al., requires a receiver capable of processing reflected GNSS signals from the Earth surface while also maintaining access to direct GNSS signals for positioning, timing, and reference purposes [4]. From a spacecraft operations perspective, this imposes pointing and antenna-placement constraints: the payload side must be oriented such that reflected signals from the Earth-facing direction can be observed, while the navigation receiver must preserve sufficient visibility toward the GNSS constellation.

The cooperative modes described by Hill et al. introduce an additional formation-geometry constraint [7]. The spacecraft must not only observe individually, but must also maintain a known and controlled relative geometry. This requires relative navigation, autonomous formation-control decisions, and propulsive maneuvers to acquire and maintain the selected formation. The formation state therefore becomes an operational condition for the mission. The paper directly states that formation geometries are required to maintain a 10 m relative position accuracy. If the spacecraft cannot maintain the required separation within this tolerance, the cooperative observation mode can no longer be supported.

The same study also identifies CubeSat resource limitations as a central operational constraint. Payload operation, attitude control, communications, propulsion system operation, on-board computing, and battery thermal control all compete for the stored energy. A realistic formation-flying GNSS-R spacecraft can therefore not be assumed to observe, communicate, charge, and maneuver simultaneously without restriction. Instead, the spacecraft operation must be divided into modes that prioritize between payload observations, battery charging, ground-station communication, coasting, and formation-control burns.

Communication and data handling further constrain the mission. Hill et al. notes that raw GNSS-R data can be generated at high data rates and must be stored and relayed to the ground. Even when the observation geometry is favorable, the spacecraft may therefore be limited by on-board storage, downlink rate, and ground-station access. From an operations point of view, this motivates intermittent observation windows rather than continuous payload operation. Finally, Hill et al. highlights that formation transitions, station-keeping, missed maneuvers, and safe-mode recovery must be considered when designing cooperative CubeSat operations. Close formation geometries require sufficient propulsive capability to correct relative drift, but also require safe behavior if a maneuver is delayed or not executed.

2.6.4 GNSS-R Formation Flying Mission Lifetime Definition

A spacecraft mission lifetime can generally be defined as the time from deployment until the spacecraft can no longer provide its intended functionality. For a conventional remote-sensing spacecraft, this functionality is often limited by the ability to generate power, communicate with ground stations, maintain attitude pointing, and operate the payload. For a formation flying GNSS-R mission, however, the mission functionality also depends on the ability to maintain the required relative geometry between the spacecraft. If the formation geometry can no longer be maintained within the required tolerance, the cooperative GNSS-R measurement concept can no longer be supported.

As discussed in Section 2.5, differential perturbations cause spacecraft formations to drift away from their desired relative geometry over time. Formation-control maneuvers are therefore required to counteract this drift. For a propulsion-based formation maintenance strategy, this makes the available propellant a limiting resource. A spacecraft loses its ability to maintain the formation once the remaining usable fuel is depleted, or once it falls below a reserved fuel threshold. The fuel-limited lifetime of the formation can therefore be defined as the time from deployment until one of the required spacecraft can no longer perform the maneuvers needed to maintain the desired formation geometry.

Let $m_{\text{fuel}}(t)$ denote the remaining usable fuel mass of the limiting spacecraft at time t, and let m_{res} denote the reserved or unusable fuel mass. The fuel-limited mission lifetime is then defined as the time T_{life} that fulfills the following equation

$$m_{\text{fuel}}(T_{\text{life}}) = m_{\text{res}}. \tag{2.24}$$

For a formation consisting of multiple controlled spacecraft, the formation lifetime is determined by the first required spacecraft that reaches this condition. The mission lifetime can therefore be expressed as

$$T_{\text{life,form}} = \min_i T_{\text{life},i}, \tag{2.25}$$

where $T_{\text{life},i}$ is the fuel-limited lifetime of spacecraft i. This definition only describes the fuel-limited lifetime. Other end-of-life drivers, such as battery degradation, component failure, loss of communication capability, or orbital decay, may also limit an actual mission. End-of-life disposal or de-orbiting requirements may also reserve part of the propellant, effectively reducing the usable fuel available for formation maintenance.

2.7 Basilisk Process-Task-Model Hierarchy

Basilisk simulations are constructed from a hierarchy of processes, tasks and models. A model is the lowest-level simulation object, and represents an individual computational component, such as a spacecraft dynamics model, environmental effector, guidance module, controller, actuator model or data recorder. In this thesis, models are grouped according to their functional role. Environmental models describe the spatial conditions, such as gravity bodies, atmospheric density, eclipse conditions and ground-station access. Dynamics models describe the physical spacecraft and its interacting forces and subsystems, including translational and attitude dynamics, perturbation effectors and actuators. Flight software models describe onboard Guidance Navigation and Control (GNC) algorithms.

```
Generic Basilisk Simulator
├── [1] EnvironmentProcess
│   └── [-] EnvironmentTask (1Hz)
│       ├── [10] spiceModel
│       └── [20] atmosphereModel
└── [2] DynamicsProcess
    ├── [1] PerturbationTask (5Hz)
    │   ├── [20] spacecraftModel
    │   └── [20] dragEffector
    └── [2] SensorTask (20Hz)
        └── [20] imuModel
```

Figure 2.7.1: Hierarchical process-task-model structure of a generic Basilisk simulator. The square brackets denote the assigned object priority, and respective task update rates are displayed in the parentheses

Models are scheduled for execution by assigning them to tasks, where each task has a specified update rate. Tasks are in turn assigned to processes, which provide a higher-level grouping of related tasks. An illustration of a process-task-model hierarchical structure for a generic Basilisk simulator is shown in Figure 2.7.1. The execution order is determined by the assigned priority, where higher-order objects are executed first. Same-priority objects or object without a specified priority are updated according to their initialization order. Using these rules, the execution order of the generic Basilisk simulator would thus be:

`imuModel` → `spacecraftModel` → `dragEffector` → `atmosphereModel` → `spiceModel`

CHAPTER

THREE

COMPARATIVE PROPAGATION SIMULATOR

The Comparative Propagation Simulator developed in this thesis is a continuation of the comparative propagation framework from the specialization project [8]. In that work, a Basilisk numerical propagator was compared against a Skyfield SGP4 propagation baseline to assess the impact of individual perturbation effects and numerical integrator configuration to evaluate Basilisk long-horizon prediction accuracy. However, two important limitations motivated further development of the simulator. First, the Skyfield SGP4 baseline was itself subject to prediction inaccuracies, which made it a weak reference for assessing long-horizon prediction accuracy. TLE-based propagation, such as SGP4, is primarily intended for efficient near-epoch orbit prediction, and is known to diverge over longer propagation intervals. Second, the Basilisk exponential atmospheric model used in the specialization project was too simple and wrongly tuned, which resulted in significantly smaller drag accelerations than expected.

The Comparative Propagation Simulator developed for this thesis aims to address these limitations by extending both the Skyfield and Basilisk parts of the comparison. The following sections present the implementation of the simulator at a high level of abstraction to provide an understanding of how the separate propagators are initialized, and how the output data is generated and made comparable. First, the overall simulator architecture is introduced, including the configuration system, data flow, execution sequence, and output files. Then the simulator input parameters will be presented in further detail. Specific parameters or simulation configuration is outside the scope of this chapter, but are presented in 5. For future reference, the term *simulator* will be used to address the Comparative Propagation Simulator in its entirety, while *sub-simulation* will be used to denote the individual Skyfield or Basilisk propagators.

3.1 Simulation Architecture

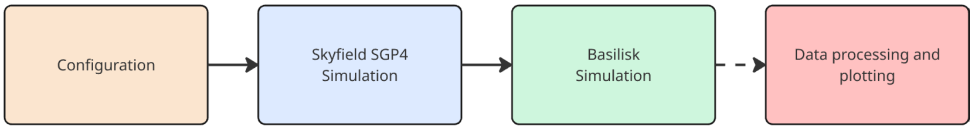

Figure 3.1.1: Simplified Comparative Propagation Simulator flow. Courtesy of [8]

The overall structure and execution order of the Comparative Propagation Simulator can be represented as a sequence of three main program blocks: initialization of the simulation environment, Skyfield SGP4 propagation, and Basilisk numerical propagation. This execution order is necessary because the Basilisk sub-simulation is initialized using the Skyfield-propagated spacecraft states at the simulation start time. A more detailed overview of the simulator architecture is shown in Figure 3.1.2. The figure illustrates the execution sequence from left to right, as well as the abstracted program flow within each program block. The vertically stacked layers separate file inputs, internal object instances, program blocks and output files. The top layer contains the user-defined input files, including configuration files, TLE data, and external model data. The middle layer shows the main program blocks, while the adjacent layers show how object instances and data are created and passed between blocks. Finally, the bottom layer shows the output files generated by the simulator.

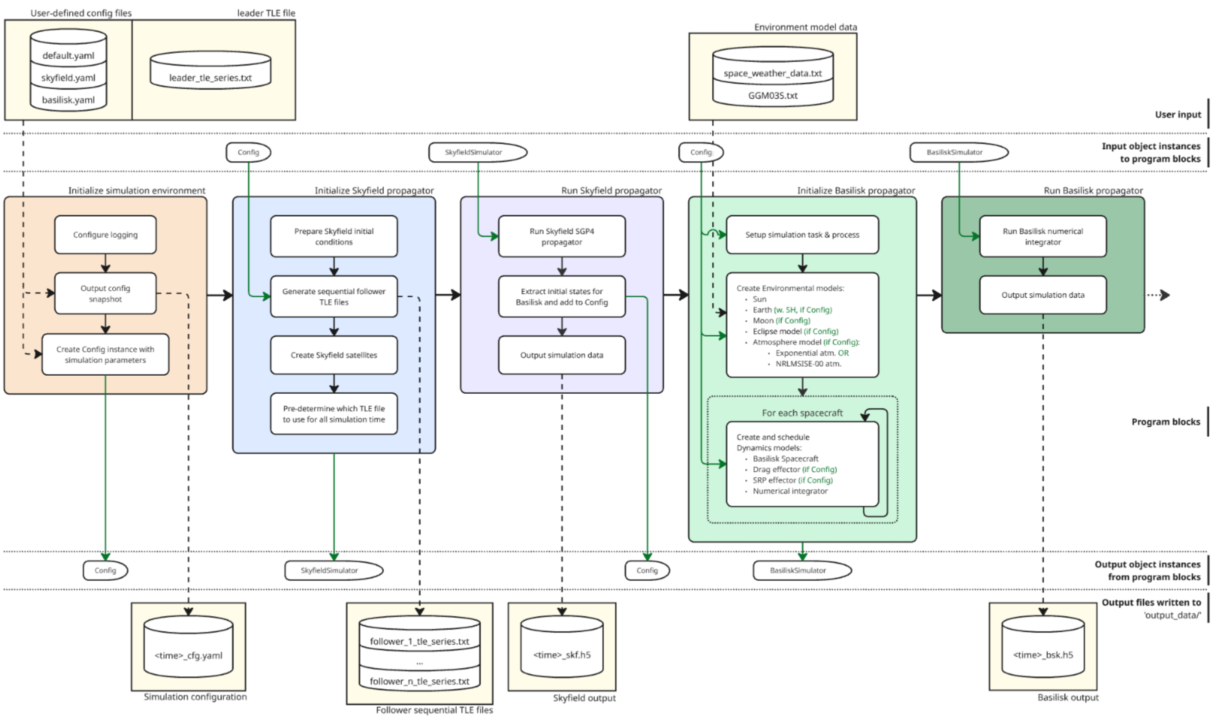

Figure 3.1.2: Detailed Comparative Propagation Simulator flow diagram. Adopted from [8]. A full-size version of this diagram is shown in the Appendix Figure B.0.2

3.1.1 Initialization of Propagator Simulation Environment

When the simulator is executed, the simulation environment is initialized by loading the configuration files and the leader TLE time series. The file `default.yaml` defines the parameters shared by both propagation methods, while `skyfield.yaml` and `basilisk.yaml` define settings specific to the Skyfield and Basilisk sub-simulations, respectively. The file `leader_tle_series.txt` contains a sequence of TLEs for the leader spacecraft, ordered by epoch, spanning the full simulation duration with a maximum epoch separation of 12 hours. The use of this TLE time series is described in more detail in Section ??. For traceability, a timestamped snapshot of the combined simulation configuration is written to file, allowing the exact configuration used to generate a given set of outputs to be inspected later. The validated configuration parameters are then collected into a shared configuration instance that is used by the subsequent Skyfield and Basilisk program blocks.

3.1.2 Skyfield SGP4 Propagator Flow

The Skyfield SGP4 sub-simulation is initialized after the shared configuration instance has been created. During this step, the leader TLE time series is loaded and used to construct the corresponding follower TLE time series with constant shifted mean anomaly relative to the leader. These are then used to create Skyfield satellite objects, which define the SGP4 propagation model used for each spacecraft. Finally, the simulator pre-determines which specific TLE from the time series should be used at any given simulation time. This produces a time-indexed mapping between simulation time and TLE set, allowing the subsequent Skyfield propagation block to evaluate each spacecraft state using the TLE closest to the requested propagation time.

After the Skyfield propagator has been initialized, the Skyfield SGP4 sub-simulation is executed for the full simulation duration. At each simulation time step, the simulator selects the pre-determined TLE for each spacecraft, and evaluates the corresponding SGP4 model to obtain the spacecraft position and velocity in the ECI frame. This produces a time series of translational states for both the leader and follower spacecraft, where the propagation remains tied to the continuously updated TLE series rather than a single fixed TLE epoch. When the Skyfield propagation is complete, the states at the initial simulation time are extracted, and the resulting state time series are written to a timestamped output file, `<time>_skf.h5`. This output represents the Skyfield comparison baseline used in the later comparison against the Basilisk numerical propagation.

3.1.3 Basilisk Propagator Flow

Once the Skyfield SGP4 sub-simulation has completed, the Basilisk propagator is initialized. The initialization first creates the Basilisk simulation process and task, which are required by the Basilisk framework to schedule and integrate all attached simulation models. The environmental and spacecraft dynamics models are then created according to the selected simulation configuration. In contrast to the Skyfield sub-simulation, where the propagation behavior is determined by the selected TLEs and the SGP4 model, the

Basilisk sub-simulation allows different combinations of explicitly modeled perturbation effects to be enabled or disabled. This configurability is indicated by the *(if config)* annotations in Figure 3.1.2, and makes it possible to run otherwise identical propagation cases with different environmental model combinations. Finally, the leader and follower spacecraft are initialized and the numerical integration method is configured.

After initialization, the Basilisk numerical propagation is run for the entire simulation duration. During this process, Basilisk integrates the spacecraft equations of motion using the configured numerical integration method and the environmental and dynamics models attached during initialization. When the propagation is complete, the translational states of each spacecraft is written to a timestamped output file, `<time>_bsk.h5`. This completes the Comparative Propagation Simulator execution and provides the Basilisk state time seris used for comparison against the Skyfield SGP4 baseline.

3.2 Simulator Inputs

As mentioned in Section 3.1.1, each simulation case is defined through three user-defined configuration files, which have been summarized in Table 3.2.1. This structure makes it easy to keep the simulated scenario fixed while independently changing the configuration of each propagation method. The file `default.yaml` contain all parameters required to define the simulation scenario. This includes the simulation start time and duration, the leader TLE time series, the number of satellites, the initial in-plane separation between the leader and follower, spacecraft physical parameters, and plotting options.

The file `skyfield.yaml` contains the Skyfield-specific propagation setting, which is only the time step between each SGP4 evaluation. Finally, the `basilisk.yaml` file defines the Basilisk propagation setup, including the numerical integrator, integration step size, spherical harmonics degree, and boolean flags for enabling or disabling individual perturbation models. In addition to the configuration files, the Basilisk sub-simulation uses external model data for selected environmental models. This includes the gravity coefficient file used by the spherical harmonics model and the space-weather data used by the NRLMSISE-00 atmosphere model.

Config file	**Input parameters**	**Unit**
default.yaml	Simulation start time	UTC
	Simulation duration	h
	Leader TLE	-
	In-plane separation angle	deg
	Number of satellites	-
	For each satellite:	-
	Mass	kg
	Drag coefficient	-
	Cross-section area perpendicular to velocity	m^2
	Radiation pressure coefficient	-
	Cross-section area perpendicular to the Sun-vector	m^2
	Data processing and plotting flags	-
skyfield.yaml	SGP4 step size	sec
basilisk.yaml	Basilisk step size	sec
	Numeric integrator	-
	Spherical harmonics degree	-
	Enable spherical harmonics flag	-
	Enable exponential density drag flag	-
	Enable MSIS drag flag	-
	Enable solar radiation pressure flag	-
	Enable Sun third-body pull flag	-
	Enable Moon third-body pull flag	-

Table 3.2.1: A complete list of all Comparative Propagation Simulator configuration input parameters. Adopted from [8]

CHAPTER

FOUR

GNSS-R FORMATION FLIGHT SIMULATOR

The GNSS-R Formation Flight Simulator is developed to estimate the operational lifetime of a realistic dual-satellite GNSS-R formation-flying mission. While the Comparative Propagation Simulator establishes confidence in the Basilisk translational propagation environment, the formation flight simulator extends this environment into a closed-loop mission simulation where orbital motion, attitude dynamics, power availability, actuator usage, and autonomous operational decisions are evaluated together. The following sections describe the simulator architecture and the implemented modules that enable mission lifetime analysis. These include an operational mode switching logic that selects spacecraft activities based on environmental, temporal, and battery constraints, an EPS model that tracks power generation, storage, and consumption, and the GNC functionality required for pointing and formation maintenance. The latter includes a pointing guidance system for generating attitude references from the active operational mode, an MRP feedback pointing controller for attitude tracking, and a formation controller for obtaining and maintaining the desired leader-follower geometry.

The implemented mission logic, EPS and GNC depend on the spacecraft components available to each satellite. This chapter therefore assumes that each spacecraft is equipped with a propulsion system consisting of a propellant tank and thruster, a three-axis controllable Reaction Wheel (RW) configuration, an On-Board Computer (OBC), solar panels, a battery pack with a built-in heating element, a power distribution module, a radio communication antenna, and a payload consisting of a high-gain GNSS-R receiver and a lower-gain GNSS receiver. These components are introduced here only to clarify the subsystem models referenced throughout the chapter, while their exact hardware values and mounting configuration are described later in Section 5.1. Finally, throughout this chapter, the term simulator refers to one complete Monte Carlo (MC) campaign, consisting of multiple GNSS-R formation-flight simulation runs. Each simulation run corresponds to one complete Basilisk execution with a resolved set of configuration parameters.

4.1 Simulation Architecture

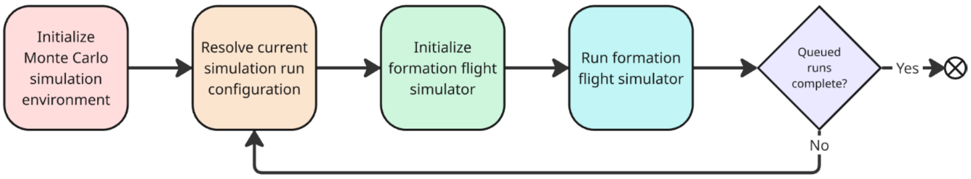

Figure 4.1.1: Simplified GNSS-R formation flight simulator flow diagram

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The high-level execution flow of the GNSS-R Formation Flight Simulator is shown in Figure 4.1.1. The simulator is organized into five main program blocks: initialization of the MC simulation environment, resolution of the current simulation run configuration, initialization of the formation flight simulator, execution of the formation flight simulator, and evaluation of whether additional MC runs remain. These blocks can be divided into two main parts, separated by the vertical dashed line in the detailed flowchart in Figure 4.1.2. The left part acts as a simulation run orchestrator, where the MC environment is configured, and all run-specific configuration files are prepared. The right part represents the iterative execution of individual simulation runs, where each run is initialized, executed, and written to file before the next queued run is started. As in the Comparative Propagation Simulator architecture presented in Section 3.1, the detailed flowchart is divided into vertically stacked layers representing input files, internal object instances, program blocks, transferred output objects, and generated output files. However, unlike the Comparative Propagation Simulator, the GNSS-R Formation Flight Simulator does not include a separate Skyfield propagation module, but is built entirely upon the Basilisk framework.

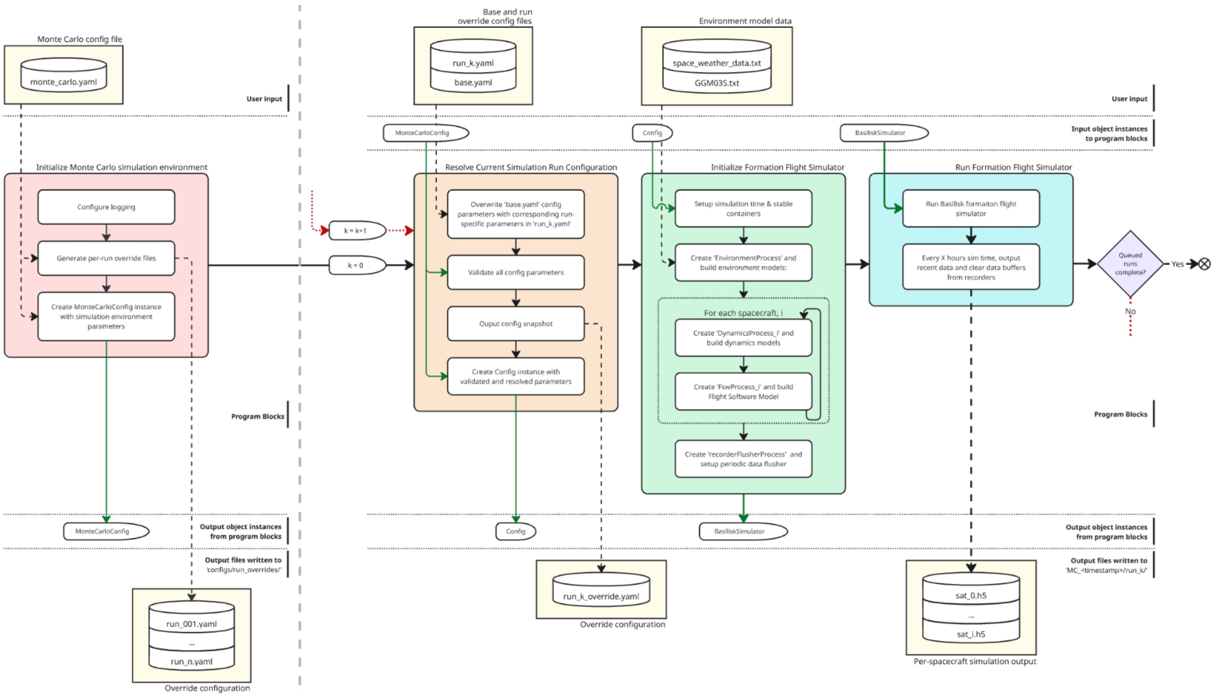

Figure 4.1.2: Detailed GNSS-R formation flight simulator flow diagram. A full-size version of this diagram is shown in the Appendix Figure B.0.2

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4.1.1 Monte Carlo Environment Initialization Flow

The first program block initializes the MC simulation environment from the user-defined configuration file `monte_carlo.yaml`, and is considered the simulation run orchestrator. In this thesis, MC execution is assumed to be enabled for all simulation cases, such that the simulator is structured around a queue of formation-flight simulation runs rather than a single isolated run. The MC configuration defines the number of queued runs and the run-specific parameter variations used to generate override configuration files. If n simulation runs are queued, $n-1$ override files are generated. Finally, the validated MC settings are collected into a global MC configuration instance, which is passed to the iterative part of the simulator and determines how the queued formation-flight simulations are executed.

4.1.2 Current Run Configuration Resolution Flow

The next program block resolves the configuration for the current simulation run. This step is similar to the configuration initialization used in the Comparative Propagation Simulator, described in Section 3.1.1, but is extended to support iterative MC execution. For the first simulation run, the configuration is defined directly by the default scenario in `base.yaml`. For all subsequent runs, the corresponding override file `run_k.yaml` is loaded and used to replace selected parameters in `base.yaml`, where k denotes the current simulation run index. This allows each queued run to modify a limited set of scenario parameters while preserving the remaining default configuration. Once the current run configuration has been resolved, all parameters are validated and written to a timestamped configuration snapshot. As in the Comparative Propagation Simulator, this snapshot provides traceability by allowing each generated output data set to be linked to the exact parameter values used to produce it. Finally, the validated parameters are collected into a run-specific `Config` instance, which is passed to the formation flight simulator initialization block.

4.1.3 Single Formation Flight Initialization & Execution Flow

The resolved run configuration is then used to initialize and execute one complete Basilisk GNSS-R formation flight simulation. At a high level, this program block follows almost the same structure as the Basilisk propagation flow described in Section 3.1.3, but extends it with two additional model clusters: flight software and recorder flushing. The simulation models are also distributed across separate Basilisk processes rather than being scheduled on a single shared process. Environmental models are scheduled on `EnvironmentProcess`, each spacecraft is assigned its own `DynamicsProcess_i` and `FswProcess_i` for its dynamical spacecraft components and GNC system, and the data flushing model is scheduled on `RecorderFlusherProcess`. A comprehensive overview of the process-task-model hierarchical structure for this simulator is shown in Figure D.1.1, which illustrates how the different models are distributed across all processes. Once initialization is complete, the Basilisk simulation is executed for the configured duration, and periodically write selected simulation data to file to clear data buffers. The resulting total per-spacecraft output contains various time series and mission-level data used in the later lifetime analysis.

4.2 Autonomous Mission Operational Modes

The GNSS-R Formation Flight Simulator implements the mission-level operational constraints through a rule-based operational mode switching logic. This logic determines which high-level activity each spacecraft should perform at a given time, and thereby replaces a predefined operations timeline with autonomous decision-making based on the simulated spacecraft state and environment. This is important because the feasibility of GNSS-R observations, communication sessions, charging periods, and formation-maintenance maneuvers depends on time-varying conditions such as battery state, illumination, ground-station visibility, and maneuver requests. By evaluating these conditions during simulation execution, each spacecraft can adapt its operational mode to the current mission state, providing a more realistic representation of autonomous small-satellite operations than a fixed sequence of scheduled activities. The following subsections will present how each operational mode is defined, how each mode effect the instantaneous mission objective, and finally, the logic governing the transitions between operational modes.

4.2.1 Mode Definitions

The simulator uses eight operational modes, each representing a distinct mission-related spacecraft activity. A summary of the modes, their enabled power-consuming modules, and their associated pointing objectives is provided in Figure 4.2.1. In *coast* mode, the spacecraft remains in a nominal idle state while keeping the reaction wheels enabled to maintain a Sun-oriented attitude, even when the Earth occludes the Sun. The *charge* mode similarly prioritizes Sun-pointing, but is used specifically to increase the stored battery energy. In *capture* mode, the payload is activated, and the spacecraft is oriented for GNSS-R observation. This entails pointing the GNSS-R receiver directly down towards the Earth's surface (nadir-direction), and the GNSS receiver in the opposite direction (zenith-direction) toward the GNSS constellation. In *communication* mode, the communication system is enabled and the spacecraft points toward a visible ground station. The *burn transit* and *burn* modes support formation-maintenance maneuvers by orienting the spacecraft thrust axis toward the desired maneuver direction. The former enables propulsion-system heating, while the latter represents active thrusting. Finally, *emergency* mode disables all nonessential loads except the onboard computer and battery heater, and does not impose an active pointing objective. A more detailed definition of each power-consuming module is described in Section 4.3, while the pointing modes are described in Section 4.4.1.

Table 4.2.1: Overview of operational modes and corresponding active variable power modules and pointing objectives.

Operational mode	Additional variable power modules	Primary objective	Secondary objective
`Coast`	–	Sun-pointing	Minimum drag
`Charge`	–	Sun-pointing	Minimum drag
`Capture`	Payload	Data capture	Sun-pointing
`Comms`	Communication system	Communication	Sun-pointing
`Burn transit`	Propulsion heating	Burn-pointing	Sun-pointing
`Burn`	Propulsion thrusting	Burn-pointing	Sun-pointing
`Emergency`	–	–	-

Reaction wheels are active in all nominal modes and are omitted from the module column for readability. The constant power modules: OBC and Battery pack heater are active for all modes

4.2.2 Mode Switching Logic

The mode switching logic is built around a set of logical parameters, here referred to as switching parameters, which determine whether transitions between operational modes are feasible. These parameters are divided into three groups depending on what part of the simulation state they depend on. The first group consists of environmental switching parameters, which are determined by spatial or illumination conditions. The parameter `canCap` indicates whether scientific data capture is feasible. In a full GNSS-R mission simulator, this would require a GNSS satellite to be within the GNSS receiver field of view and the corresponding specular reflection point to be within the GNSS-R receiver field of view. Since GNSS satellite constellations are not modeled in this simulator, data capture is assumed to be geometrically feasible whenever it is requested. The parameter `canCom` is true when at least one ground station has direct line of sight to the spacecraft, and its elevation angle is greater than 10° above the horizon. Finally, `canChar` indicates whether charging is possible, which requires the spacecraft to be fully or partially illuminated by the Sun.

The second group consists of battery switching parameters, which determine whether the remaining battery energy is sufficient to permit different operations. In this group, `capBat` and `comBat` indicate whether the battery level is high enough to allow science capture and communication, respectively, while `critBat` and `exitCriticalBat`, define entry into and recovery from critically low battery operation. The third group consists of operational state parameters, which depend on the internal mission state rather than directly on the external environment or battery level. These include `maxNoCom`, which becomes true if the time since the previous communication session exceeds a specified threshold, `burnRequested`, which indicates that a formation-maintenance maneuver is required and `burnAttitudeReached`, which indicates that the spacecraft is sufficiently aligned with the requested burn attitude. Finally, `capPossible` and `comPossible` are introduced to simplify switching logic visualization. The prior is a combination of `canCap` and `capBat`, and is true whenever both switching parameters are true. Similarly, the latter is true whenever its components, `canCom` and `comBat` are true.

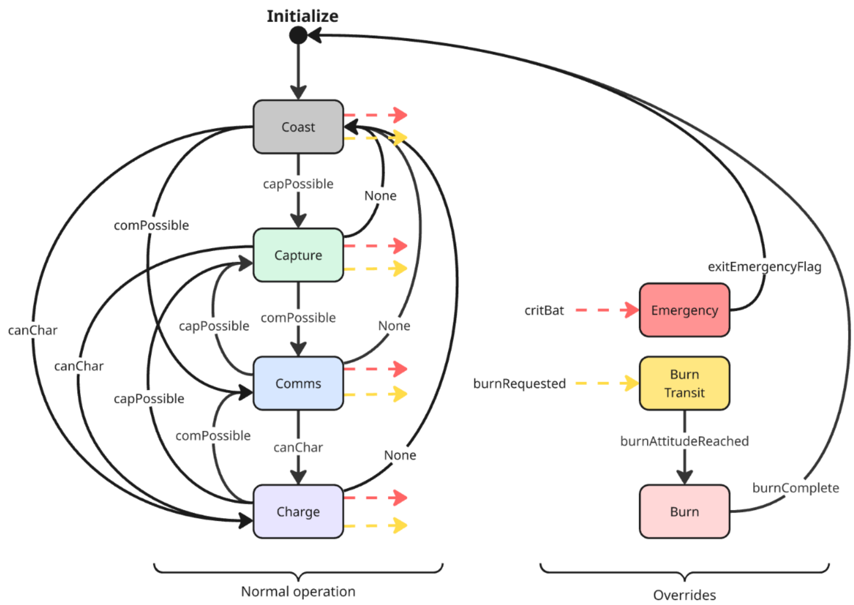

Figure 4.2.1: Mode switching logic flow diagram to visualize mode transitions. In cases where multiple feasible transitions from a mode exist, see the mode switching priority tables in Table 4.2.3 and 4.2.2 for normal operation and overrides, respectively

The resulting mode switching transition flow is illustrated in Figure 4.2.1. The left part of the diagram contains the nominal operational modes: coast, capture, communication and charge. These modes represent ordinary mission operation, where the spacecraft attempts to perform science observations, communicate with a ground station, charge the battery, or remain in a low-activity coast mode, depending on the currently feasible transitions. The labelled arrows indicate the switching parameters required for each transition. If none of the higher-priority nominal transitions are feasible, the spacecraft returns to or remains in coast mode. The right part of the diagram contains the override modes, which can interrupt the nominal operation logic. A critically low battery state triggers emergency mode, while a formation-maintenance request triggers burn transit, followed by burn once the required thrust-pointing attitude has been reached. After the emergency condition has cleared or the burn has completed, the logic returns to the nominal operation sequence.

Although Figure 4.2.1 shows the allowed transitions between operational modes and the switching parameters associated with each transition, it does not by itself define which mode is selected when several transitions are feasible at the same time. This ambiguity is resolved using the mode priority tables. If the battery has reached a critically low level or a formation-maintenance burn has been requested, the switching logic follows the override mode priority in Table 4.2.2. These override modes are evaluated before all nominal operations, ensuring that emergency battery protection and maneuver execution take precedence over capture, communication, charging and coasting. If neither emergency nor burn override conditions are active, the selected mode is instead determined by the normal operation mode priority in Table 4.2.3. In this case, the priority order depends on whether the maximum allowed time without communication has been exceeded, such that communication is prioritized over capture when `maxNoCom` is true.

Table 4.2.2: Override operation mode priority used when an override condition is active. The priority list is designed to resolve cases where multiple feasible mode transitions exist

Priority	Condition	Selected mode
1	`critBat` or (`Emergency` and not `exitEmergencyFlag`)	Emergency
2	`burnRequested` and not `burnAttitudeReached`	Burn transit
3	`burnRequested` and `burnAttitudeReached`	Burn
4	not `critBat` and not `burnRequested`	Use normal operation priority

Override conditions are evaluated before the normal operation mode priority logic. Emergency mode has the highest priority, followed by formation-maintenance burn modes.

Table 4.2.3: Normal operation mode priority used when neither emergency nor burn override is active. The priority list is designed to resolve cases where multiple feasible mode transitions exist

Priority	Condition when not `maxNoCom`	Selected mode	Condition when `maxNoCom`	Selected mode
1	`capPossible`	Capture	`comPossible`	Comms
2	`comPossible`	Comms	`capPossible`	Capture
3	`canChar`	Charge	`canChar`	Charge
Fallback	–	Coast	–	Coast

The selected mode is assigned by evaluating the listed conditions from highest to lowest priority. The priority between capture and communication depends on whether the maximum allowed time without communication has been exceeded.

4.3 Electronic Power System (EPS)

Power management is one of the key operational constraints for autonomous remote sensing satellite missions, since the available battery charge determines which operational modes can be safely executed. In the GNSS-R Formation Flight Simulator, the battery state therefore influences the spacecraft guidance logic by constraining transitions between internal operational modes, which is described in Section 4.2. This section, however, focuses on how electrical power generation, storage and consumption are modeled for each spacecraft in the formation. Since these quantities depend directly on the assumed spacecraft hardware, the EPS model assumes the spacecraft component configuration presented in Section 5.1.

4.3.1 EPS Model Overview

The EPS model is implemented to track the high-level electrical power flow for each spacecraft. Its purpose is to estimate the remaining battery energy during the mission, rather than to model detailed electrical bus dynamics. An overview of the per-spacecraft EPS structure is shown in Figure 4.3.1. Here, the green module represents power sources, red and yellow modules represent constant and variable power sinks, and the blue module represents battery energy storage. The power management module at the center represents the physical power distribution component included in the spacecraft configuration, but is not modeled with internal distribution logic, and does not contribute to the net power itself. The individual power generation, storage, and consumption models are described in the following subsections.

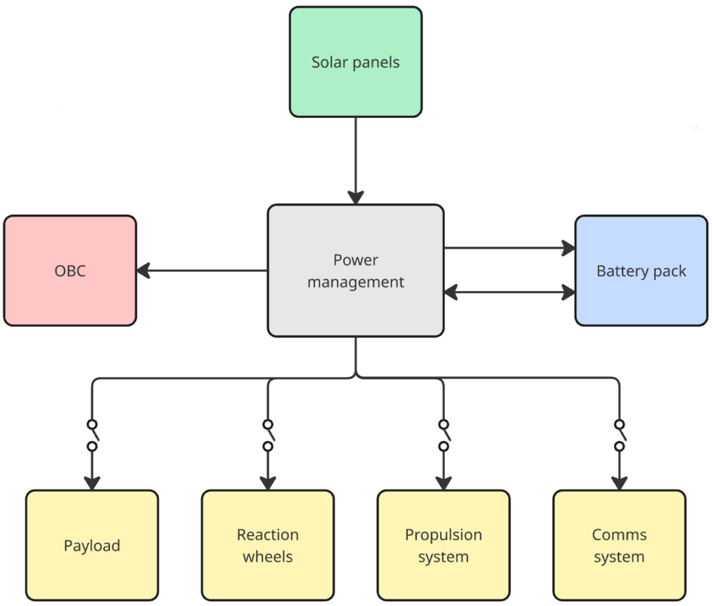

Figure 4.3.1: Overview of the per-spacecraft EPS

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4.3.2 Power Generation & Storage

Electrical power generation is modeled using the Basilisk solar panel module, where each panel computes its generated power according to the simple solar panel formulation from Wertz and Larson in [33]:

$$W_{\text{sp}} = W_{\text{base}} C_{\text{eclipse}} C_{\text{panel}} (\hat{\mathbf{n}} \cdot \hat{\mathbf{s}}) A_{\text{panel}}, \tag{4.1}$$

Here, W_{sp} is the per-panel generated electrical power, W_{base} is the incident solar power per unit area, C_{eclipse} is the eclipse or shadow factor, C_{panel} is the solar panel conversion efficiency, A_{panel} is the panel area, $\hat{\mathbf{n}}$ is the panel normal vector, and $\hat{\mathbf{s}}$ is the Sun direction vector. The generated power is proportional to the cosine of the angle between $\hat{\mathbf{n}}$ and $\hat{\mathbf{s}}$ through their dot product, which makes the panel output attitude dependant. In implementation, panel output is set to zero when the Sun is behind the panel or when the spacecraft is in eclipse.

The generated power from all solar panel modules is accumulated by the battery model. The battery energy rate of change is given by the instantaneous balance between total generated and consumed power,

$$\dot{E}_{\text{storage}} = W_{\text{generated}} - W_{\text{consumed}} \tag{4.2}$$

where $W_{\text{generated}}$ is the total electrical power produced by solar panels, W_{consumed} is the total power drawn by spacecraft subsystems, and E_{storage} is the stored battery energy. The net power $\dot{E}_{\text{storage}}$ is integrated over time by the Basilisk battery model to estimate the battery energy level throughout the simulation.

4.3.3 Power Consumption

The OBC is modeled as a constant power sink. This represents the baseline electrical load required to keep the spacecraft operational, independent of attitude, illumination conditions, or operational mode. The constant power draw C_{OBC} is therefore always included in the total consumed power.

$$W_{\text{OBC}} = C_{\text{OBC}} \tag{4.3}$$

The battery heater is modeled as an averaged constant power sink. Since no battery temperature sensor or thermal model is implemented, the heater duty cycle is roughly approximated by assuming that the battery heater is active for one half of each orbital period. Under this assumption, the heating contribution to the total consumed power is modeled using the nominal active heater power, $C_{\text{bat,heating}}$

$$W_{\text{bat,heating}} = \frac{1}{2} C_{\text{bat,heating}}, \tag{4.4}$$

The payload power consumption is modeled as a mode-dependent power sink. When the spacecraft is performing data capture, the payload is assumed to draw the constant power

C_{payload}. Outside this mode, the payload is treated as inactive and does not contribute to the power consumption.

$$W_{\text{payload}} = \begin{cases} C_{\text{payload}} & \text{during data capture} \\ 0 & \text{otherwise} \end{cases} \tag{4.5}$$

The reaction wheel power consumption is modeled using the Basilisk reaction wheel power module presented in [34]. The expression accounts for a constant baseline power draw W_{base} and an additional term associated with wheel actuation. The actuation-dependent part scales with the mechanical power required by the wheel W_{mech}, which depends on the commanded wheel torque and wheel speed, and is corrected by the assumed conversion efficiency η_{e2m}.

$$W_{\text{RW}} = \begin{cases} W_{\text{base}} + \dfrac{W_{\text{mech}}}{\eta_{e2m}}, & W_{\text{mech}} > 0 \\ W_{\text{base}}, & W_{\text{mech}} \leq 0 \end{cases} \tag{4.6}$$

The propulsion system power consumption is modeled using three discrete draw states. During propellant heating, the propulsion system draws a constant heater power to prepare the system for maneuver execution. During an active thrust maneuver, the power draw is set to the maneuver power level, representing the electrical load associated with firing the propulsion system. Outside heating and maneuver execution, the propulsion system remains in an idle state with a lower constant standby consumption.

$$W_{\text{prop}} = \begin{cases} C_{\text{prop,heating}} & \text{during propellant heating} \\ C_{\text{prop,thrusting}} & \text{during thrust} \\ C_{\text{prop,idle}} & \text{otherwise} \end{cases} \tag{4.7}$$

The communication system is modeled as a mode-dependent power sink. During communication mode, the transmitter and associated communication hardware are assumed to draw the constant power C_{comms}. When the spacecraft is not communicating with a ground station, this contribution is excluded from the total consumed power.

$$W_{\text{comms}} = \begin{cases} C_{\text{comms}} & \text{during communication} \\ 0 & \text{otherwise} \end{cases} \tag{4.8}$$

Finally, The total consumed power W_{consumed} used in Equation 4.2 is obtained by summing the active constant and mode-dependent power sinks at each simulation time. A comprehensive overview of when each power-consuming module is enabled is shown in Table 4.2.1.

4.4 Guidance Navigation & Control (GNC) System

The GNC system provides the flight software functionality required to translate operational mode decisions into spacecraft motion. In the simulator, this includes generating attitude references for the active operational mode, tracking these references using a reaction-wheel-based MRP feedback controller, and scheduling formation-maintenance maneuvers when the leader-follower geometry drifts from the desired configuration. The following subsections describe these three functions at a high level, focusing on their role in the closed-loop mission simulation rather than the detailed Basilisk message connections used to implement them.

4.4.1 Pointing Guidance System

The pointing guidance system maps the active operational mode to the desired attitude tracked by the attitude controller. This mapping is necessary because the operational modes in Table 4.2.1 impose different mission-critical pointing objectives. In Sun-pointing modes, the spacecraft is oriented to expose the largest possible solar panel area toward the Sun, increasing its power generation. In data-capture mode, the payload geometry is prioritized by pointing the GNSS-R receiver nadir, while the GNSS receiver is pointed zenith, consistent with the payload constraint introduced in Section 2.6.3. In communication mode, the radio antenna is pointed toward the selected ground station when a direct line of sight is available. During burn transit and burn modes, the attitude reference is instead supplied by the formation-keeping controller described in Section 4.4.3, which aligns the body-fixed thrust direction with the desired maneuver direction. Finally, for modes in which minimum drag is included as a secondary objective, the spacecraft is oriented to minimize the cross-sectional area normal to the velocity direction.

When a mode defines both a primary and secondary pointing objective, the primary objective is enforced first. In the simulator implementation, this corresponds to aligning a body-fixed vector with a desired inertial direction, which still leaves one rotational degree of freedom about the primary pointing direction. This remaining freedom is used to maximize the secondary objective. For example, in communication mode, the antenna vector is aligned with the selected ground station direction, while the remaining rotation about the antenna axis is chosen to maximize solar-panel illumination, provided that the relevant panel normal is not parallel to the antenna direction. However, the secondary objective can generally not be satisfied exactly, because the primary pointing constraint restricts the available attitude space. Let $\hat{\mathbf{v}}_p^n$ denote the desired primary pointing direction, and let $\hat{\mathbf{v}}_s^n$ denote the direction associated with the secondary objective, both expressed in ECI. The component of $\hat{\mathbf{v}}_s^n$ that can be used while preserving the primary pointing direction is obtained by projecting it into the $\hat{\mathbf{v}}_p^n$-normal plane,

$$\mathbf{v}_{s,\perp}^n = \hat{\mathbf{v}}_s^n - \frac{\left(\hat{\mathbf{v}}_s^n \cdot \hat{\mathbf{v}}_p^n\right)}{\left\|\hat{\mathbf{v}}_p^n\right\|^2}\hat{\mathbf{v}}_p^n \tag{4.9}$$

Using the normalized projected desired secondary pointing direction $\mathbf{v}_{s,\perp}^{\hat{n}}$, the transformation to the ECI frame from the desired frame $\mathcal{F}^R$, is then given by

$$\boldsymbol{R}_R^n = \begin{bmatrix}\hat{\mathbf{v}}_p^n & \hat{\mathbf{v}}_{s,\perp}^n & \left(\hat{\mathbf{v}}_p^n \times \hat{\mathbf{v}}_{s,\perp}^n\right)\end{bmatrix} \tag{4.10}$$

which can then be converted into the corresponding desired MRP parameterization used by the attitude controller described in Section 4.4.2. The resulting attitude reference therefore maximizes the secondary objective within the constraint imposed by the primary objective. Note that the guidance system only calculates the desired attitude, not the corresponding reference angular velocity or angular acceleration. Consequently, time-varying attitude references are tracked purely through feedback by the attitude controller, which can increase reaction-wheel actuation and therefore reaction-wheel power consumption.

4.4.2 MRP Feedback Pointing Controller

Each spacecraft must be able to track the attitude reference generated by the pointing guidance system for the active operational mode. This is achieved using Basilisk's `mrpFeedback` module, which implements a nonlinear attitude feedback controller based on the MRP attitude and body frame angular-rate tracking error. The commanded body torque is computed using the control law presented in [35]

$$\begin{aligned} \mathbf{L}^b = & -K\boldsymbol{\sigma_e} - [P]\boldsymbol{\omega_e} - [P][K_I]\mathbf{z_e} - [I_{RW}]\left(-\dot{\boldsymbol{\omega}}_{R/n} + [\tilde{\boldsymbol{\omega}}]\boldsymbol{\omega}_{R/n}\right) \\ & - \mathbf{L}_{\mathbf{ff}} + \left([\tilde{\boldsymbol{\omega}}_{R/n}] + [K_I\mathbf{z}]\right)\left([I_{RW}]\boldsymbol{\omega_{b/n}} + [G_s]\mathbf{h}_s\right) \end{aligned} \tag{4.11}$$

In this expression, $\mathbf{L}^b$ is the requested control torque expressed in the spacecraft body frame, $\boldsymbol{\sigma}_e$ is the MRP attitude tracking error, and $\boldsymbol{\omega}_e$ is the body-frame angular-rate tracking error. The controller gains are given by the proportional gain K, the rate-feedback gain matrix $[P]$, and the integral gain matrix $[K_I]$, while $\mathbf{z}_e$ denotes the integral error state. The matrix $[I_{RW}]$ represents the spacecraft inertia including the reaction-wheel contribution, $\boldsymbol{\omega}_{R/n}$ is the angular velocity of the reference frame relative to the inertial frame, and $\boldsymbol{\omega}_{b/n}$ is the spacecraft body angular velocity relative to the inertial frame. The reaction-wheel geometry matrix $[G_s]$ maps the wheel angular momentum vector $\mathbf{h}_s$ into the body frame. Finally, $\mathbf{L}_{\text{ff}}$ represents feedforward compensation for known external torques, such as modeled disturbance torques.

In the simulator implementation, the integral feedback is disabled by setting $[K_I] = 0$. In addition, the pointing guidance system only defines the reference attitude, while the reference-frame angular velocity and angular acceleration are set to zero. This gives $\boldsymbol{\omega}_{R/n} = \mathbf{0}$ and $\dot{\boldsymbol{\omega}}_{R/n} = \mathbf{0}$, which removes the reference-motion feedforward terms from Equation 4.11. The implemented attitude controller can therefore be written as the simplified PD-type feedback law

$$\mathbf{L}^b = -K\boldsymbol{\sigma_e} - [P]\boldsymbol{\omega_e} - \mathbf{L}_{\text{ff}}, \tag{4.12}$$

where the remaining feedforward term is only present when known compensation torques are supplied. Under the assumptions stated in the Basilisk module documentation, the tracking control law can be shown to be asymptotically stable using the provided Lyapunov candidate function.

4.4.3 Formation-Maintenance Controller

As described in Section 2.5, the absence of corrective maneuvers causes the relative geometry to drift away from the desired formation. Formation control is therefore required in the GNSS-R Formation Flight Simulator because the scientific mission concept depends on maintaining a prescribed leader-follower geometry over time. In this thesis, this is achieved using the Basilisk `spacecraftReconfig` module, which is an extended version of the spacecraft formation reconfiguration model described in [19] to account for the under-actuated control problem with only one static thrust vector. The control objective is formulated in terms of the orbital element difference between the two spacecraft, as introduced in Section 2.5.1. With $\boldsymbol{\alpha}_L$ and $\boldsymbol{\alpha}_{Fi}$ denoting the leader and follower i orbital element vectors, respectively, the formation-relative orbital element difference is written as

$$\Delta\boldsymbol{\alpha} = \boldsymbol{\alpha}_{Fi} - \boldsymbol{\alpha}_L = \begin{bmatrix}\Delta a & \Delta e & \Delta i & \Delta\Omega & \Delta\omega & \Delta M_0\end{bmatrix}^T. \tag{4.13}$$

Using this notation, the control objective can be written as

$$\Delta\boldsymbol{\alpha}(t) = \Delta\boldsymbol{\alpha}_R, \qquad t \to \infty. \tag{4.14}$$

The `spacecraftReconfig` module achieves this objective by computing a sequence of impulsive maneuvers that drive the current orbital element difference toward the desired target difference $\Delta\boldsymbol{\alpha}_R$ within approximately one orbital period [36]. The module should therefore be interpreted as an orbital element-based maneuver scheduler rather than a continuous Cartesian relative-motion controller. As will be described in further detail in Section 5.1, the simulated spacecraft is configured with a single body-fixed thruster. Therefore, the module also computes an attitude reference to align the body thrust direction with the desired impulse direction before the maneuver is executed. In the simulator, the resulting maneuver request is passed through the operational mode logic before the thruster is commanded, such that formation-control burns remain coupled to attitude pointing, fuel availability, propulsion-system state, and battery constraints.

The module also introduces several assumptions and limitations that constrain how the spacecraft orbits and components are configured, and influence how the results should be interpreted. First, the method is formulated using classical orbital elements, and is therefore not applicable to near-circular or near-equatorial orbits where these elements become poorly defined. Second, the reconfiguration strategy is based on impulsive maneuver planning, while the Basilisk implementation approximates each impulse using steady thrust over a finite time interval. This results in non-optimal orbital reconfigurations, especially for weaker thrust-to-weight ratio configurations. Thus, the maneuver fuel consumption should therefore be interpreted as a mission-level approximation of the formation-maintenance cost rather than an exact optimal maneuver solution.

CHAPTER

FIVE

SHARED SIMULATOR CONFIGURATION

To address the research objectives of this thesis, two distinct simulators were developed. The first simulator, referred to as the Comparative Propagation Simulator, is used to evaluate the long time translational prediction accuracy of the Basilisk numerical propagator by comparing its propagated spacecraft states against a Skyfield SGP4 baseline. The second simulator, referred to as the GNSS-R Formation Flight Simulator, extends the Basilisk-based simulation framework to model the autonomous operation of a dual-satellite GNSS-R formation mission subject to realistic operational constraints, with particular emphasis on formation-keeping fuel consumption and mission lifetime estimation.

Although the two simulators differ in purpose and implementation complexity, they share the same Basilisk spacecraft configuration, orbital environment models, and numerical integration setup. This chapter therefore presents the shared physical and numerical foundation in more detail. This common configuration is used to ensure that the propagation behavior assessed in the Comparative Propagation Simulator is directly transferable to the GNSS-R Formation Flight Simulator. However, it is important to note that the shared models presented in this chapter only apply to the Basilisk-based parts of the simulation framework. The Skyfield SGP4 propagation used in the comparative analysis is instead determined by the TLE data and the SGP4 model, and is therefore not configured using the same spacecraft or numerical integration method.

5.1 Shared Spacecraft Configuration

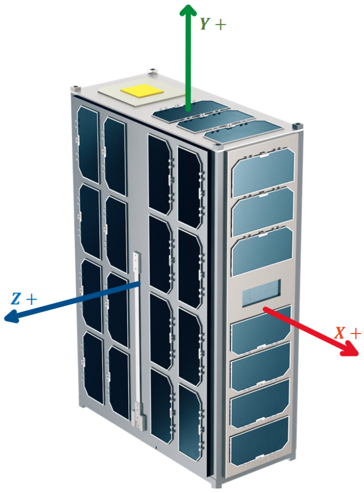

Figure 5.1.1: 6U spacecraft body frame definition used for both simulators. Adopted from [37]

The spacecraft component set was selected to represent the main subsystem functions required for a GNSS-R formation-flying mission. This includes components required for payload operation, direct GNSS reception, communication, power generation and storage, attitude control, propulsion, and on-board computation. The purpose of the component selection was not to produce a detailed flight design, but to construct a representative 6U spacecraft model whose mass, power consumption, and actuation capabilities are consistent with the operational requirements of the simulated mission.

Table 5.1.1: Mounting configuration for spacecraft components.

Subsystem	Component	Mounting configuration
Structural	Frame	-
OBC	OBC	$\mathcal{O}_b$
EPS	Power management	$\mathcal{O}_b$
EPS	battery	$\mathcal{O}_b$
EPS	Solar panels BIG	Z+
EPS	Solar panels SMALL	X+ and X-
Comms	Radio Antenna	Y+
Payload	GNSS-R Receiver	Z-
Payload	GNSS Receiver	Z+
AOCS	Propulsion System	X-
AOCS	RWs	$\mathcal{O}_b$
Misc	Wires and harnesses	-

Each component was modelled using parameters from existing flight-proven or commercially available CubeSat-compatible hardware. This was done to keep the simulated spacecraft representative of what could realistically be accommodated within a 6U platform, rather than relying on idealized subsystem assumptions. The selected hardware was therefore used as a basis for the spacecraft mass budget, power budget, propulsion capability, and component mounting configuration. The resulting spacecraft model provides a system-level approximation of a realistic GNSS-R formation-flying satellite, while still remaining simplified enough to support long-horizon simulation.

Table 5.1.2: Desired face orientation in the different operating modes

Operational mode	Primary face	Desired direction	Secondary face	Desired direction
`Coast`	Z+	Sun	Y+	Velocity
`Charge`	Z+	Sun	Y+	Velocity
`Capture`	Z-, Z+	Nadir, zenith	X+	Sun
`Comms`	Y+	Ground station	Z+	Sun
`Burn transit`	X+	Burn direction	Z+	Sun
`Burn`	X+	Burn direction	Z+	Sun
`Emergency`	–	–	–	–

5.1.1 Mass Budget & Inertia Estimate

Table 5.1.3: Spacecraft mass budget. The full component parameter overview is shown in the Appendix Table C.0.1

Subsystem	Component	Mass [g]	Source
Structural	Frame	665	GomSpace NanoStructure 6U
OBC	OBC	240	GomSpace NanoMind HP MK3
EPS	Power management	80	GomSpace NanoPower (1ACU & 1PCU)
EPS	battery	500	GomSpace Nanopower BPX 100 Wh
EPS	Solar panels BIG	392	Endurosat 6U Solar panel
EPS	Solar panels SMALL	2×147	Endurosat 3U Solar panel
Comms	Radio Antenna	2	Endurosat X-band patch antenna
Payload	Payload system	1200	Scout-2 HydroGNSS payload system
AOCS	Propulsion System Dry	1430	VACCO ArgoMoon propulsion system
AOCS	RWs	4×144	Cubespace CW0162
Misc	Wires and mounting	100	
	System mass:	5479	
	20% margin:	1096	
	Total dry mass:	6575	
	Propellant:	635	
	Total launch mass:	7210	

Assuming uniform mass distribution, a perfect cuboid shape, and the body frame definition illustrated in Figure 5.1.1, the spacecraft moment of inertia matrix can be estimated to:

$$J = \begin{bmatrix} 0.060081667 & 0 & 0 \\ 0 & 0.030040833 & 0 \\ 0 & 0 & 0.078106167 \end{bmatrix} \tag{5.1}$$

5.2 Shared Numerical Integration Configuration

The dynamics in both Basilisk-based simulators are propagated using the same numerical integration configuration. This choice is based on the numerical sensitivity analysis conducted in the specialization project [8], where several Basilisk integration settings were compared for the same orbital propagation scenario. The results showed that the low-step-size RK4 configurations and the variable-step RKF45 and RKF78 integrators converged toward the same Earth-relative and Formation-relative solutions. In contrast, fixed-step RK4 configurations with step sizes larger than 5 seconds produced more visible deviations. The study therefore concluded that numerical integration uncertainty was insignificant compared to the differences introduced by orbital perturbation selection over a 96-hour time span.

Although these results suggest that several of the tested numerical integration configurations would be sufficient for the translational propagation problem alone, the simulators developed in this thesis impose stricter requirements on the numerical setup. In the Comparative Propagation Simulator, the inclusion of the NRLMSISE-00 atmosphere model introduces atmospheric density variations that are not present in the simpler propagation configurations considered previously. In the GNSS-R Formation Flight Simulator, the same orbital propagation is further coupled to fast attitude dynamics, actuator models, power generation, and autonomous mode switching. These additional models introduce faster dynamics, making it necessary to choose a conservative integration configuration that remains robust across both simulators.

Based on these considerations, both Basilisk-based simulators use the RKF78 variable-step integrator with maximum step size of 1.0 seconds. A variable-step integrator is selected because it provides high accuracy during dynamically demanding parts of the simulation, such as attitude transitions and formation-keeping maneuvers, while allowing longer internal step sizes during less dynamic coasting or idle periods. This adaptivity is important for reducing computational cost over the long simulation horizons considered in this thesis. The RKF78 method is chosen because a high-precision integration method is desirable when propagating spacecraft trajectories over long time intervals, and because formation-control performance depends directly on accurate relative positioning between the leader and follower satellites. This justifies the increased computational cost per integration step. Finally, the maximum step size is limited to 0.5 seconds as a numerical safeguard against instability in the reaction wheel actuator dynamics used in the GNSS-R Formation Flight Simulator. Although this constraint is only required for the full formation-flight simulator, the same maximum step size is also used in the Comparative Propagation Simulator to ensure that results from the two simulators remain directly comparable.

CHAPTER

SIX

MISSION ANALYSIS METHODOLOGY

6.1 Comparative Propagation Analysis Method

The comparative propagation analysis quantifies the divergence between the Basilisk translational propagator and an observation-informed Skyfield-SGP4 reference. The purpose is not to validate Basilisk against a true physical ground truth, but to benchmark the thesis propagation configuration also used in the GNSS-R Formation Flying Simulator, against a continuously updating TLE-based reference to evaluate its prediction uncertainty, and whether the final Basilisk configuration improves over the configuration used in the specialization project. This provides a propagation-accuracy context for interpreting the long-horizon GNSS-R formation flight lifetime analysis.

The analysis is performed by propagating the same leader-follower scenario using Skyfield-SGP4 and several Basilisk configurations. The continuously updating Skyfield-SGP4 trajectory is used as the comparison baseline, while the Basilisk cases are selected to isolate the effect of the improved atmospheric modelling and the final thesis perturbation configuration. The resulting trajectories are compared using both same-spacecraft state- and same-formation drift differences, allowing the analysis to assess both absolute Earth-relative propagation, and formation drift accuracy.

6.1.1 Comparison Scenario

The comparative propagation analysis is performed using the Comparative Propagation Simulator described in Chapter 3. Each propagator is configured to simulate two spacecraft separated by an initial in-plane angle of 1°. The leading and trailing spacecraft are denoted as the leader and follower, respectively, although no active formation-keeping is applied in this comparison. The Basilisk spacecraft use the configuration described in Section 5.1, but with active subsystems disabled. The resulting model represents a passive 6U spacecraft with realistic mass and geometry for a remote-sensing, formation-capable satellite. The Skyfield spacecraft corresponds to the real spacecraft associated with the continuously updating TLE sequence, which is the 6U HYPSO-1 hyperspectral imaging satellite briefly described in [38]. The Basilisk spacecraft configuration is therefore selected to approximate the size and mass properties of the reference spacecraft, in an attempt to ensure that they are equally influenced by orbital perturbations. Additionally, as explained in Section 3.1, precautions are made to ensure that all spacecraft are initialized using the same state vectors across Basilisk and Skyfield-SGP4.

Both propagators are initialized at 23.10.2025 22:19:30 UTC, corresponding to the epoch of the initial TLE used by the Skyfield propagator. Using the same real epoch is important because the Basilisk environmental models depend on time-varying quantities, including the Sun and Moon positions, solar radiation pressure geometry, eclipse conditions, and space-weather-dependent atmospheric density through the NRLMSISE-00 model. This ensures that simulated Basilisk environment is as close as possible to the actual conditions experienced by the HYPSO-1 satellite during this time. The initial scenario orbit is derived from the following HYPSO-1 TLE:

```
HYPSO-1
1 51053U 22002BX  25296.93021535  .00034531  00000-0  64114-3 0  9997
2 51053  97.3166   6.7052 0004161  83.3535 276.8189 15.48851464210108
```

This corresponds to a highly inclined, near-circular LEO orbit with a mean altitude of approximately 420 km. The comparison is propagated for 14 days, which provides a long-horizon benchmark for the translational propagation environment used later in the GNSS-R formation-flight lifetime analysis.

6.1.2 Propagation Accuracy Evaluation Framework

The continuously updating Skyfield-SGP4 trajectory is used as the comparison baseline for evaluating the Basilisk propagation results. This reference is not treated as an exact truth model, but as an observation-updated SGP4-based trajectory constructed from a sequential TLE series to continuously correct for accumulating divergence from the true state. This reduces the long-term accumulation of prediction error that would occur if a single TLE were propagated over the full comparison interval. The resulting Skyfield trajectory is therefore interpreted as an approximate reference against which the Basilisk configurations can be benchmarked. All propagation differences are evaluated relative to this baseline.

The primary configuration evaluated in this analysis is the highest-fidelity Basilisk configuration used later in the GNSS-R formation flight lifetime analysis. To contextualize its performance, two additional Basilisk configurations are included. The first corresponds to the highest-fidelity configuration used in the specialization project in [8], where atmospheric drag was modelled using an exponential atmosphere that was not tuned for its orbital altitude. The second uses the same general Basilisk perturbation configuration, but replaces the old atmospheric tuning with a retuned exponential atmosphere model by adjusting its scale height, H_0. These additional cases make it possible to distinguish whether differences between the specialization project and the present thesis are primarily caused by the improved atmospheric modelling, or by the general transition to the final thesis propagation configuration. The comparison scenario is otherwise kept identical for all Basilisk cases.

Two complementary accuracy perspectives are considered. The first is the same-spacecraft state difference, which describes the difference between the state of the same spacecraft in a Basilisk simulation and in the Skyfield reference. For a spacecraft translational state vector $\mathbf{x}$, this is defined as

$$\mathbf{x}_{\text{sss}} = \mathbf{x}_{\text{bsk}} - \mathbf{x}_{\text{skf}}. \tag{6.1}$$

This quantity is used to assess the Earth-relative translational divergence of a single spacecraft, both through the magnitude of the position and velocity differences and through decomposition of the position difference in the RTN frame. The second perspective is the same-formation drift difference, which compares how the leader-follower separation evolves in each Basilisk case relative to the Skyfield baseline. Let the formation-relative state in propagator p be defined as

$$\Delta\mathbf{x}_{\text{form},p} = \mathbf{x}_{\text{F},p} - \mathbf{x}_{\text{L},p}. \tag{6.2}$$

The corresponding same-formation drift difference is then

$$\mathbf{x}_{\text{sfd}} = \Delta\mathbf{x}_{\text{form,bsk}} - \Delta\mathbf{x}_{\text{form,skf}}. \tag{6.3}$$

This provides a measure of how accurately each Basilisk configuration reproduces the passive leader-follower drift predicted by the reference model, which is especially relevant for interpreting the later formation flight lifetime analysis.

6.1.3 Propagation Benchmarking Campaign

The propagation benchmarking campaign is performed to evaluate the difference between the Basilisk propagation configurations and the continuously updating Skyfield-SGP4 reference defined in the previous subsections. The campaign consists of one Skyfield-SGP4 reference run and three Basilisk runs, as summarized in Table 6.1.1. The Skyfield-SGP4 run provides the updating-TLE comparison baseline, while the primary Basilisk run represents the highest-fidelity thesis configuration used later in the GNSS-R formation flight lifetime analysis. The remaining Basilisk runs are included to contextualize the result by comparing against the highest-fidelity specialization-project configuration and an additional re-tuned exponential-atmosphere configuration.

All runs are propagated for 14 days from the same initial scenario described in Section 6.1.1. In the first run, the Skyfield-SGP4 trajectory is generated using the continuously updating TLE sequence. The primary Basilisk run is then initialized from the Skyfield state at the initial epoch, ensuring that the two propagation frameworks start from the same translational state. The subsequent Basilisk runs reuse the previously generated Skyfield reference data, which ensures that all Basilisk configurations are compared against the same baseline and initialized consistently. The Basilisk configurations use the same deactivated spacecraft configuration described in Section 5.1, but differ in their atmospheric density model. The primary thesis configuration uses the NRLMSISE-00 atmosphere model, while the two reference configurations use exponential atmosphere models with different scale heights.

Table 6.1.1: Basilisk propagation benchmarking campaign.

Run #	**1**	**2**	**3**	**4**
Propagator framework	Skyfield-SGP4	Basilisk	Basilisk	Basilisk
Role	Updating-TLE reference	Primary thesis configuration	Specialization project reference	Re-tuned exponential atm. reference
Initial epoch	23.10.2025, 22:19:30 UTC	Same as run 1	Same as run 1	Same as run 1
Duration	14 days	Same as run 1	Same as run 1	Same as run 1
Spacecraft	HYPSO-1 from TLE	Deactivated configuration from 5.1	Same as run 2	Same as run 2
Initial state	Initial TLE evaluated at epoch	Initial Skyfield-SGP4 state	Same as run 2	Same as run 2
Gravity	SGP4 model	Spherical harmonics, 4th degree and order	Same as run 2	Same as run 2
Atmospheric density model	SGP4 model	NRLMSISE-00	Un-tuned exponential	Tuned exponential
Exponential scale height H_0	–	–	7200.0	15180.0
Sun and Moon 3rd body pull	SGP4 model	Enabled	Same as run 2	Same as run 2
SRP	SGP4 model	Enabled	Same as run 2	Same as run 2
Numerical integration	SGP4 model	RKF78 w. 1.0 sec max step	Same as run 2	Same as run 2

The output from the campaign is used to compute the same-spacecraft state- and same-formation drift differences defined in Section 6.1.2. Together, these outputs provide the basis for assessing whether the thesis Basilisk configuration improves over the specialization-project configuration and for quantifying the propagation uncertainty relevant to the later GNSS-R formation flight lifetime analysis.

6.1.4 Assumptions & Limitations

The comparative propagation analysis uses the continuously updating Skyfield-SGP4 trajectory as a comparison baseline, not as a true physical ground truth. The TLE sequence is treated as an observation-updated reference for benchmarking the Basilisk propagation configurations, but the result remains subject to the assumptions and uncertainties of the TLE/SGP4 framework. The comparison therefore quantifies the difference between selected Basilisk propagation configurations and with the selected updating-TLE reference rather than absolute orbit prediction accuracy.

The Basilisk spacecraft used in the comparison is represented by the deactivated spacecraft configuration described in Section 5.1. The spacecraft is propagated passively, without active attitude control, but instead assumes a fixed attitude relative to the ECI frame. Furthermore, incident direction, and therefore attitude dependent, drag coefficient is assumed to be constant.

Finally, the benchmarking campaign is limited to the selected HYPSO-1 initial orbit, comparison epoch, 14-day propagation horizon, and the three Basilisk configurations listed in Table 6.1.1. The analysis does not include a full perturbation-by-perturbation decomposition, stochastic uncertainty analysis, or comparison against independent orbit-determination products.

6.2 GNSS-R Formation Flight Lifetime Analysis Method

The GNSS-R formation flight lifetime analysis estimates the fuel-limited mission lifetime of the selected leader-follower formation scenario using the GNSS-R Formation Flight Simulator. The analysis focuses on the follower spacecraft, which is responsible for acquiring and maintaining the desired concentric formation relative to the leader. Since the mission lifetime definition in Section 2.6.4 is based on the remaining usable fuel mass, the methodology is built around estimating how fuel is consumed during formation acquisition and subsequent formation maintenance.

The lifetime estimation is therefore divided into two simulation campaigns. The *deployment velocity fuel sensitivity campaign* estimates the acquisition fuel cost function $m_{\mathrm{acq}}(\Delta v_{\mathrm{dep}})$ by varying the initial leader-follower deployment velocity difference. The *formation steady-state fuel consumption campaign* estimates the post-acquisition cumulative maintenance fuel consumption function $m_{\mathrm{maint}}(t)$ from a longer-duration simulation. Together, these two contributions provide the fuel-consumption model used to estimate the GNSS-R formation flight mission lifetime for the selected spacecraft configuration, formation geometry, disturbance environment, and operational constraints.

6.2.1 Formation flight Scenario

The GNSS-R formation flight lifetime analysis is performed using the GNSS-R Formation Flight Simulator described in Chapter 4, with the full spacecraft configuration introduced in Section 5.1. The simulated scenario consists of two spacecraft: one leader and one follower. Both spacecraft are initialized from a shared deployer orbit, but each with a configurable additional velocity component representing the ejection velocity. The deployer orbital elements are derived from the same initial HYPSO-1 TLE used to initialize the comparative propagation scenario, so that the lifetime analysis is conducted in the same LEO orbital regime. The spacecraft are subject to the same disturbance environment as in the primary thesis configuration introduced in the comparative propagation study, including fourth-degree-and-order spherical harmonics, atmospheric drag with density estimated by the NRLMSISE-00 model, third-body perturbations from the Sun and Moon, and SRP.

The follower spacecraft is controlled to acquire and maintain a concentric relative orbit around the leader with a radius of 400m. As presented in Section 2.6.2, a concentric formation normally represents a multi-spacecraft configuration, where several followers occupy projected circular relative orbits with increasing radii around a central spacecraft. In this thesis, the configuration is reduced to a single leader-follower pair due to computational constraints, while preserving the essential fuel-consuming task of acquiring and maintaining a bounded relative orbit. Ideally, a 100m spacing would have been chosen to coincide with the HydroSwarm concentric concept presented in [7], but a 400m separation was instead chosen to prevent the implemented formation controller from diverging due to poorly defined orbital elements in the scenario's near-circular orbit. In addition to the formation control task, the spacecraft are subject to operational constraints imposed by the autonomous mode-switching logic described in Section 4.2.2. These constraints determine whether each spacecraft charges, communicates, captures payload data, coasts, or performs formation-control burns by adjusting its attitude and activating relevant components to enable the corresponding behavior. The constant operational mode transition constraints, attitude controller gains and module update rates used by all GNSS-R formation lifetime analyses are shown in Table E.0.1.

6.2.2 Formation Acquisition Definition

Formation acquisition refers to the process of applying orbital maneuvers to achieve a desired formation geometry. Since the implemented formation controller is based on desired classical orbital element differences, acquisition is evaluated in terms of the follower's orbital element difference relative to the leader. The formation is considered *acquired* when the deviation between the actual and desired leader-follower orbital element differences remains within prescribed tolerances for a minimum duration dwell time. The acquisition condition is evaluated using tolerances on the semimajor axis, eccentricity and inclination differences. The tolerances used in all formation acquisition campaigns are

$$\Delta a_{\mathrm{tol}} = 1.0 \cdot 10^{-5}, \qquad \Delta e_{\mathrm{tol}} = 2.0 \cdot 10^{-4}, \qquad \Delta i_{\mathrm{tol}} = 2.0 \cdot 10^{-5}. \tag{6.4}$$

The formation is considered acquired only if all three conditions are satisfied continuously for a dwell time of

$$t_{\mathrm{dwell}} = 0.5 \text{ h}. \tag{6.5}$$

The acquisition time t_{acq} is defined as the first time at which the formation orbital element difference enters the prescribed tolerance set, provided that it remains within this set continuously for the entire dwell time duration. Finally, the selected tolerances and dwell time were tuned through preliminary simulations to ensure the most reliable acquisition detection and to prevent short transient crossings of the target state from being counted as successful acquisition.

6.2.3 Lifetime Estimation Framework

The fuel-limited lifetime definition introduced in Section 2.6.4 defines the GNSS-R formation flight mission lifetime as the time at which the remaining usable fuel mass reaches the reserved fuel threshold,

$$m_{\text{fuel}}(T_{\text{life}}) = m_{\text{res}}. \tag{6.6}$$

For this spacecraft and scenario configuration, m_{res} is set to zero such that mission end is reached only when the available propellant for a required spacecraft has been fully depleted. This is based on the assumption that the spacecraft will de-orbit naturally and burn up in the atmosphere because of the LEO altitude and its small size, such that reserve fuel capacity is not needed for this purpose.

For the leader-follower deployment scenario considered in this thesis, the formation flying mission is separated into two phases. The first phase is the formation acquisition phase, which lasts from deployment until the acquisition time t_{acq} defined in the previous subsection. During this phase, fuel is consumed to remove the initial deployment-induced relative motion and insert the follower into the desired concentric formation geometry. The second phase is the formation-maintenance phase, where fuel is consumed to counteract the long-term relative drift caused by differential perturbations. To evaluate the fuel-limited lifetime, the fuel expenditure in these two phases is represented through a model for the remaining follower fuel mass. The remaining spacecraft fuel mass m_{fuel} at any given time $t \geq 0$ is modelled as

$$m_{\text{fuel}}(t) = m_{\text{fuel},0} - m_{\text{acq}}(\Delta v_{\text{dep}}) - m_{\text{maint}}(t), \qquad \Delta v_{\text{dep}} = \|\mathbf{v}_{\text{L},0} - \mathbf{v}_{\text{F},0}\|_2 \tag{6.7}$$

Here, $m_{\text{fuel},0}$ is the initial spacecraft fuel mass, and $m_{\text{maint}}(t)$ is the cumulative post-acquisition formation-maintenance fuel consumption defined for any given time $t \geq t_{\text{acq}}$. The differential deployment speed Δv_{dep} is defined as the Euclidean norm of the initial deployment-velocity difference between the leader and follower, where $\mathbf{v}_{\text{L},0}$ and $\mathbf{v}_{\text{F},0}$ denote the initial leader and follower deployment velocity vectors, respectively. The acquisition fuel consumption $m_{\text{acq}}(\Delta v_{\text{dep}})$ is modelled as a function of this differential deployment speed. The lifetime estimation is therefore reduced to estimating the two fuel-consumption contributions in the remaining-fuel model. The *deployment velocity fuel sensitivity campaign* described in Section 6.2.4 estimates $m_{\text{acq}}(\Delta v_{\text{dep}})$, while the *steady-state formation-maintenance campaign* detailed in Section 6.2.5 estimates $m_{\text{maint}}(t)$. These two function estimates are then combined to predict the fuel-limited GNSS-R formation flight mission lifetime for the selected spacecraft configuration, formation geometry, and deployment scenario.

6.2.4 Deployment Velocity Fuel Sensitivity Campaign

The deployment velocity fuel sensitivity campaign is performed to estimate the fuel required to acquire the desired concentric formation as a function of the initial differential deployment speed, $m_{\text{acq}}(\Delta v_{\text{dep}})$. Each simulation begins with the leader and follower initialized from the same deployer orbit, but with different deployment velocity offsets. The acquisition fuel consumption is then computed as the follower fuel consumed from deployment until the acquisition condition defined in Section 6.2.2 is satisfied. This makes the campaign directly connected to the first fuel-consumption contribution in the lifetime estimation framework. The shared configuration used for all simulations in the campaign is summarized in Table **??**. This includes the formation type, target formation radius, simulation duration, operational constraints, disturbance environment, and numerical integration settings.

Due to computational constraints, the campaign is limited to cases where the leader and follower are ejected in opposite directions from the shared deployer. This represents a conservative deployment scenario because the relative velocity between the spacecraft is maximized for a given individual deployment speed. The deployment velocities are applied along the inertial $\hat{\mathbf{x}}^n$ direction, with the leader ejected in the positive direction and the follower ejected in the negative direction. This direction was selected after preliminary tests of deployment along the inertial principal axes, where the $\hat{\mathbf{x}}^n$ case produced the largest perturbation in the leader-follower orbital element differences. The selected direction is therefore used as a worst-case deployment direction for the acquisition analysis.

Table 6.2.1: Shared configuration for the deployment velocity fuel sensitivity campaign.

Parameter	**Value**
Primary metric	Acquisition fuel $m_{\text{acq}}(\Delta v_{\text{dep}})$
Duration	48 h
Number of spacecraft	2: one leader and one follower
Initial epoch	23.10.2025, 22:19:30 UTC
Initial state	Shared deployer initial state + ejection velocity
Formation type	Concentric/circular relative orbit
Target formation radius	400 m
Acquisition criterion	OED tolerances with dwell-time requirement
Formation-maintenance Control	Enabled
Attitude GNC	Enabled
Operational mode switching	Enabled
EPS	Enabled
Initial battery charge	80%
Disturbance environment	Full thesis Basilisk perturbation model
Numerical integration	RKF78 with 1.0 s maximum step

Table 6.2.2: Run-specific deployment velocities for the deployment sensitivity campaign.

Run #	**0**	**1**	**2**	...	**8**
Role	Baseline	Low-speed	Medium-speed	...	Max-speed
Leader speed in $\hat{\mathbf{x}}^n$ [m/s]	0.0	0.25	0.5	...	2.0
Follower speed in $\hat{\mathbf{x}}^n$ [m/s]	0.0	-0.25	-0.5	...	-2.0
Differential speed Δv_{dep} [m/s]	0.0	0.5	1.0	...	4.0

The simulated deployment speed range is chosen to represent realistic 6U CubeSat deployment conditions. Several commercial deployer specifications indicate deployment speeds below approximately 2 m/s for 6U-class satellites [39, 40, 41]. The campaign therefore consists of nine 48-hour simulations, where the individual deployment speed magnitude is varied across the range summarized in Table 6.2.2. Since the leader and follower are deployed in opposite directions, the differential deployment speed is twice the individual deployment speed magnitude,

$$\Delta v_{\text{dep}} = ||\mathbf{v}_{\text{L},0} - \mathbf{v}_{\text{F},0}||_2. \tag{6.8}$$

The campaign therefore provides a discrete mapping between differential deployment speed and acquisition fuel consumption. These results are later used to estimate $m_{\text{acq}}(\Delta v_{\text{dep}})$ for the lifetime estimation framework.

6.2.5 Formation Steady-State Fuel Consumption Campaign

The formation steady-state fuel consumption campaign is performed to estimate the cumulative fuel consumed after the follower has acquired the desired concentric formation, $m_{\text{maint}}(t)$. Unlike the deployment velocity fuel sensitivity campaign, this campaign does not aim to evaluate the influence of initial deployment conditions. Instead, it is designed to observe the long-term fuel consumption trend associated with maintaining the selected leader-follower formation under the full thesis disturbance environment and operational constraints. Due to computational constraints, the campaign consists of a single simulator run over a 90 day duration. The configuration for this campaign is summarized in Table 6.2.3.

Table 6.2.3: Shared configuration for the formation steady-state fuel consumption campaign

Parameter	Value
Primary metric	Acquisition fuel $m_{\text{maint}}(t)$
Duration	90 days
Number of spacecraft	2: one leader and one follower
Initial epoch	23.10.2025, 22:19:30 UTC
Initial state	Shared deployer initial state
Formation type	Concentric/circular relative orbit
Target formation radius	400 m
Acquisition criterion	OED tolerances with dwell-time requirement
Formation-maintenance Control	Enabled
Attitude GNC	Enabled
Operational mode switching	Enabled
EPS	Enabled
Disturbance environment	Full thesis Basilisk perturbation model
Numerical integration	RKF78 with 1.0 s maximum step

The resulting follower fuel history is used to estimate the cumulative post-acquisition formation-maintenance fuel consumption. After the acquisition phase has been excluded, the remaining fuel-consumption history is converted into cumulative fuel expenditure and fitted with a continuous curve. This fitted curve provides the simulation-based estimate of $m_{\text{maint}}(t)$ used later in the fuel-limited lifetime estimate.

6.2.6 Assumptions & Limitations

The GNSS-R formation flight lifetime analysis is limited to the two-spacecraft leader-follower scenario defined in Section 6.2.1. The simulated formation is therefore a reduced representation of a larger concentric GNSS-R formation, and only the follower spacecraft is used as the limiting spacecraft for the fuel-limited lifetime estimate. The formation geometry is also limited by the single selected concentric orbit radius and the implemented orbital-element-difference-based formation controller. Other formation geometries, alternative control strategies, and larger multi-spacecraft configurations are not evaluated.

The deployment velocity fuel sensitivity campaign is limited to the selected deployment direction and deployment-speed range. The leader and follower are initialized from the same deployer orbit and are ejected in opposite directions along the inertial $\hat{\mathbf{x}}^n$ direction. The resulting acquisition fuel model $m_{\mathrm{acq}}(\Delta v_{\mathrm{dep}})$ is therefore based only on this set of deterministic deployment cases. Other deployment directions, release dispersions, navigation errors, actuator errors, or stochastic uncertainty in the initial conditions are not included.

Although the formation steady-state fuel consumption campaign was intended for a 90-day simulation horizon in order to capture the long-term formation-maintenance fuel trend, the final completed run was limited to 24 days due to unresolved memory overflow issues caused by accumulated simulation data in the GNSS-R Formation Flight Simulator. The post-acquisition fuel consumption model $m_{\mathrm{maint}}(t)$ is therefore estimated by fitting the available 24-day fuel history and extrapolating this fitted model beyond the simulated interval. The analysis does not include multiple steady-state Monte Carlo runs or uncertainty bounds on the fitted maintenance fuel model.

The simulator do not include propagated GNSS constellation positions, meaning specular reflection point estimation and observation geometry is not modeled. Thus, each spacecraft will assume GNSS-R measurements are always possible, and will be unconstrained normal pointing requirements. Instead, the spacecraft assumes a sufficient attitude is pointing the GNSS-R receiver in the nadir direction, while pointing the GNSS receiver in zenith direction.

CHAPTER

SEVEN

RESULTS

7.1 Comparative Propagation Analysis Results

This section presents the results from the comparative propagation benchmarking campaign described in Section 6.1. The continuously updating Skyfield-SGP4 propagation is used as the comparison baseline, while three Basilisk configurations are evaluated against it: the primary thesis configuration with NRLMSISE-00 atmospheric density, the specialization-project configuration with an un-tuned exponential atmosphere model, and a re-tuned exponential atmosphere configuration. All reported differences are computed as Basilisk minus Skyfield-SGP4 and should be interpreted as differences relative to the updating-TLE reference, not as errors relative to a true physical ground truth.

The results are first presented using same-spacecraft state differences for the leader spacecraft, followed by an RTN decomposition of the position difference, altitude differences, ground-track offsets, and same-formation drift differences for the passive leader-follower pair. Together, these quantities describe both the Earth-relative divergence of each Basilisk configuration and the corresponding difference in predicted formation drift. This provides the propagation benchmark used to contextualize the GNSS-R formation flight lifetime analysis.

7.1.1 Same-Spacecraft State Magnitude Difference

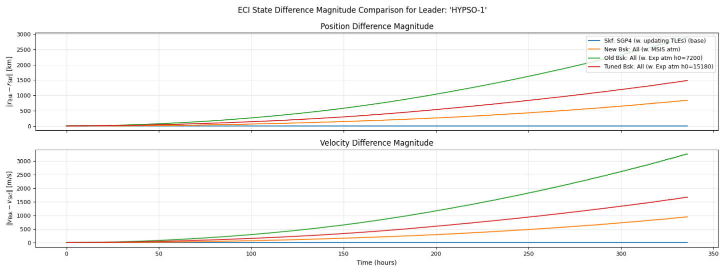

Figure 7.1.1: Leader spacecraft ECI position and velocity difference magnitudes between various Basilisk configuration and the updating TLE Skyfield-SGP4 baseline

Figure 7.1.1 shows the same-spacecraft ECI position and velocity difference magnitudes for the leader spacecraft over the 14-day propagation benchmarking campaign. The differences are computed between each Basilisk configuration and the continuously updating Skyfield-SGP4 baseline. Both the position and velocity difference magnitudes increase throughout the propagation interval for all Basilisk configurations. The ordering of the configurations is the same for both translational states: the primary thesis Basilisk configuration with the NRLMSISE-00 atmosphere model has the smallest divergence, the re-tuned exponential atmosphere configuration has an intermediate divergence, and the specialization-project exponential atmosphere configuration has the largest divergence.

At the end of the 14-day propagation interval, the primary thesis Basilisk configuration has reached a leader position difference of approximately 900 km relative to the Skyfield-SGP4 baseline. The re-tuned exponential atmosphere configuration reaches approximately 1500 km, while the specialization-project exponential atmosphere configuration reaches approximately 2900 km. The velocity difference magnitudes show the same relative ordering.

7.1.2 Decomposition of Same-Spacecraft Position Difference

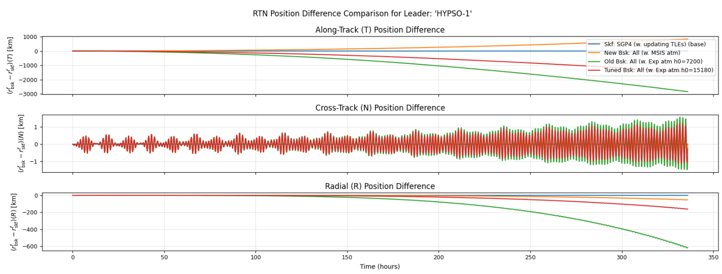

Figure 7.1.2: Decomposition of the same-spacecraft position difference in the RTN frame

Figure 7.1.2 shows the same-spacecraft position difference for the leader spacecraft decomposed in the Skyfield-SGP4 baseline leader's RTN frame. The along-track component is the dominant contributor to the position difference for all Basilisk configurations. The primary thesis configuration with the NRLMSISE-00 atmosphere model diverges in the positive along-track direction, while both exponential atmosphere configurations diverge in the negative along-track direction. By the end of the 14-day comparison interval, the old specialization-project exponential atmosphere configuration has the largest along-track difference, followed by the re-tuned exponential atmosphere configuration and the primary thesis configuration.

The cross-track separation remains small compared with the along-track and radial components throughout the comparison. The plotted cross-track component remains oscillatory and within a few kilometers for all Basilisk configurations. The radial component is also smaller than the along-track component, but shows a clear separation between the configurations. The old specialization-project exponential atmosphere configuration has the largest radial difference, while the primary thesis configuration has the smallest radial difference. The ordering of the RTN components is therefore consistent with the ECI state-difference magnitudes shown in Figure 7.1.1.

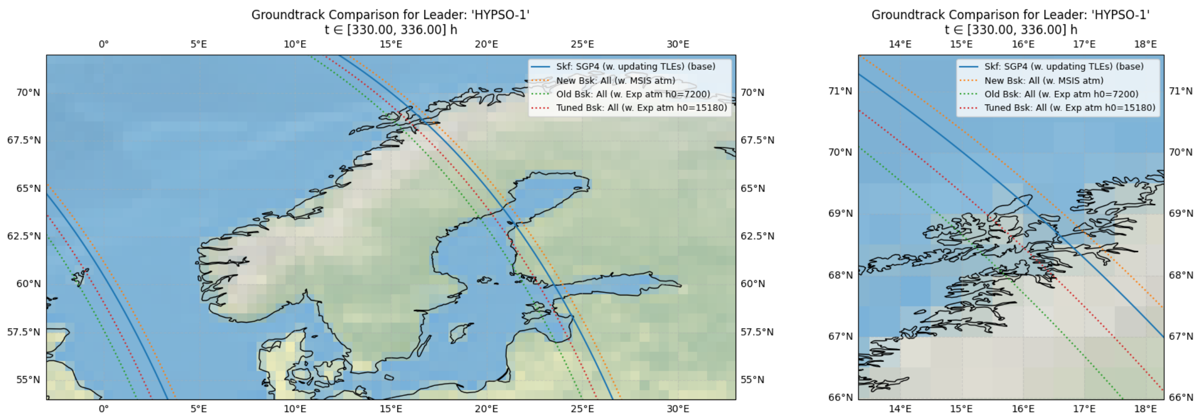

Figure 7.1.3: Surface projections for the leader satellite position over time propagated by each Basilisk configuration and the updating TLE Skyfield-SGP4 baseline. The zoomed-out view is shown on the left, while the zoomed-in view of the north-western coast of Norway is shown on the right.

Figure 7.1.3 shows the corresponding ground-track tracings for the same leader trajectories. The left subplot shows the ground-track comparison over the Nordic countries during the final six hours of the simulation, while the right subplot shows a zoomed-in view near the north-western coast of Norway. The visible separation between the projected tracks is largest for the old specialization-project exponential atmosphere configuration and smallest for the primary thesis configuration. The estimated ground-track offset relative to the Skyfield-SGP4 baseline is approximately 17 km for the primary thesis configuration, 32 km for the re-tuned exponential atmosphere configuration, and 60 km for the old specialization-project configuration.

The ground-track offset shown in Figure 7.1.3 is not equivalent to the cross-track separation shown in Figure 7.1.2. The cross-track separation is the orbit-normal component of the three-dimensional position difference in the RTN frame, while the ground-track offset is the separation between the Earth-projected spacecraft trajectories. The RTN decomposition shows that the three-dimensional position difference is dominated by along-track drift, whereas the ground-track comparison shows the surface projection of this accumulated drift onto the Earth-fixed latitude-longitude map.

7.1.3 Altitude Difference Comparison

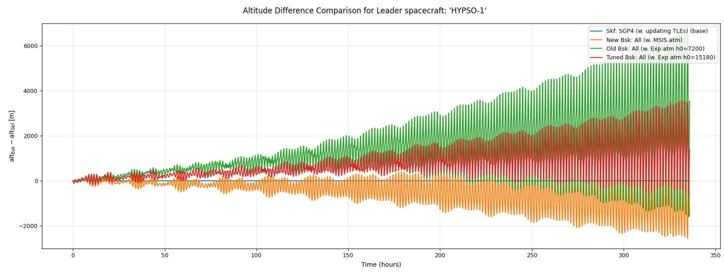

Figure 7.1.4: Leader spacecraft orbital altitudes for each Basilisk configuration compared against the updating TLE Skyfield-SGP4 baseline

Figure 7.1.4 shows the leader spacecraft altitude difference for each Basilisk configuration relative to the continuously updating Skyfield-SGP4 baseline. All Basilisk configurations exhibit short-period oscillations in altitude difference, with amplitudes that increase over the 14-day propagation interval. The primary thesis configuration with the NRLMSISE-00 atmosphere model remains closest to the Skyfield-SGP4 baseline throughout the comparison, while the re-tuned exponential atmosphere configuration shows a larger altitude difference and the old specialization-project exponential atmosphere configuration shows the largest altitude difference.

The long-term altitude trend differs between the atmosphere models. The primary thesis configuration gradually decreases in altitude relative to the Skyfield-SGP4 baseline, while both exponential atmosphere configurations diverge in the positive altitude direction. The old specialization-project exponential atmosphere configuration has the largest positive altitude difference by the end of the comparison interval. The ordering of the altitude differences is therefore consistent with the previous same-spacecraft state difference results: the primary thesis configuration gives the smallest deviation from the Skyfield-SGP4 baseline, followed by the re-tuned exponential atmosphere configuration and then the old specialization-project configuration.

7.1.4 Decomposition of Same-Formation Drift Difference

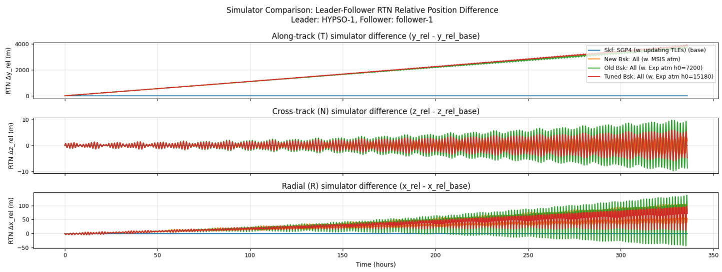

Figure 7.1.5: Leader-follower pair same-formation position drift difference for each Basilisk configuration compared against the updating TLE Skyfield-SGP4 baseline.

Figure 7.1.5 shows the same-formation drift difference for the leader-follower pair, decomposed in the leader RTN frame. The plotted quantities compare the leader-relative follower position predicted by each Basilisk configuration against the corresponding leader-relative follower position predicted by the continuously updating Skyfield-SGP4 baseline. All three RTN components exhibit short-period oscillations with amplitudes that increase throughout the 14-day comparison interval.

The along-track component is the dominant same-formation drift difference. All Basilisk configurations show a positive along-track drift difference relative to the Skyfield-SGP4 baseline, reaching approximately 4 km by the end of the propagation interval. The three Basilisk configurations remain close to each other in this component. Although more oscillatory, the old specialization-project exponential atmosphere configuration shows slightly smaller final along-track drift difference than the primary thesis and re-tuned exponential configurations, although it also exhibits stronger short-period oscillations.

The cross-track and radial components remain much smaller than the along-track component. In these components, the same ordering observed in the same-spacecraft state difference results is present: the primary thesis configuration has the smallest divergence, the re-tuned exponential atmosphere configuration is intermediate, and the old specialization-project exponential atmosphere configuration has the largest oscillation amplitude. The radial drift difference remains below approximately 150 m, while the cross-track drift difference remains within approximately ± 10 m over the full comparison interval.

7.2 Deployment Velocity Acquisition Fuel Sensitivity

This section presents the results from the deployment velocity fuel sensitivity campaign described in Section 6.2.4. The campaign consists of nine 48-hour GNSS-R formation flight simulations where one leader and one follower are initialized from the same deployer orbit, but with different opposite deployment velocity offsets. The differential deployment speed Δv_{dep} is varied from 0 to 4 m/s. For each case, the acquisition time t_{acq} is determined using the OED acquisition criterion defined in Section 6.2.2, and the corresponding acquisition fuel consumption m_{acq} is computed as the follower fuel consumed from deployment until this acquisition time.

The results are first presented using leader-follower RTN relative-position trajectories to show the geometric formation response during and after the initial deployment transient. The acquisition condition is then evaluated using the OED tolerances that define t_{acq}, before the follower fuel consumption is used to extract m_{acq} for each deployment case. The section concludes by summarizing the resulting acquisition time, acquisition fuel, burn statistics, and operational diagnostics. Together, these results provide the simulation-based deployment sensitivity estimate of $m_{\text{acq}}(\Delta v_{\text{dep}})$ used in the GNSS-R formation flight lifetime analysis.

7.2.1 RTN Formation Trajectories

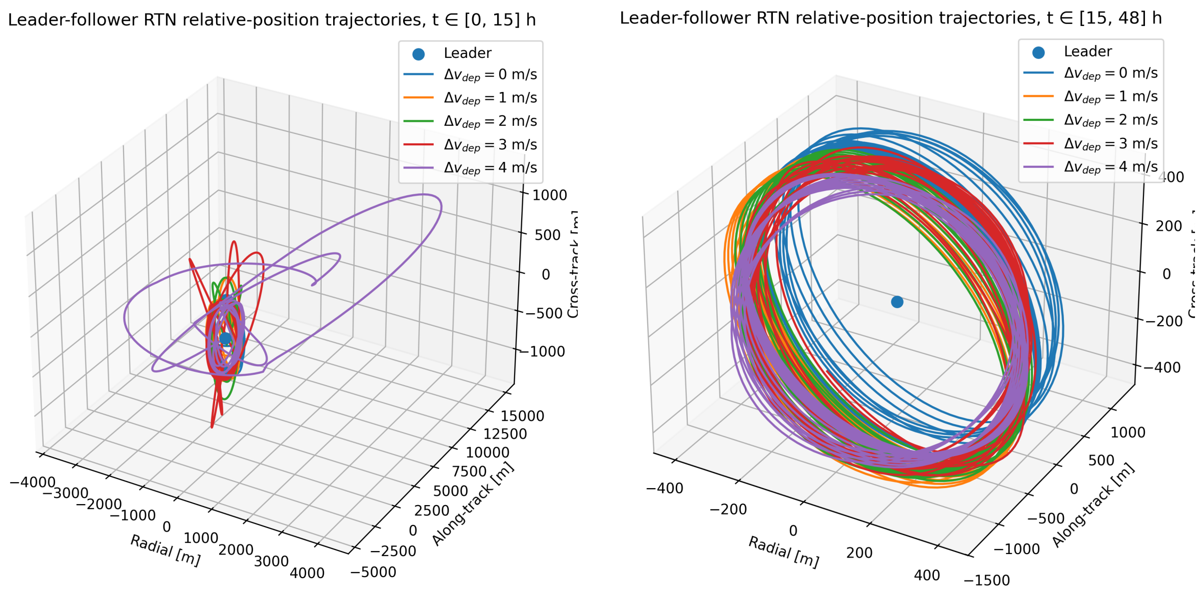

Figure 7.2.1: Leader-follower RTN relative-position trajectories for a selected set of differential deployment velocities corresponding to run 0, 2, 4, 6 and 8 detailed in 6.2.4. The left plot shows the initial transient response over $t \in [0, 15]$ h, while the right plot shows the subsequent bounded relative motion over $t \in [15, 48]$ h.

Figure 7.2.1 shows the leader-follower relative-position trajectories for selected deployment velocity cases, expressed in the leader RTN frame. The left plot shows the initial response during the first 15h after deployment, while the right plot shows the relative motion over the remaining interval from 15 to 48h. The largest initial relative-position magnitude occur for the largest differential deployment speeds, but all cases evolve into similar bounded relative-position trajectories.

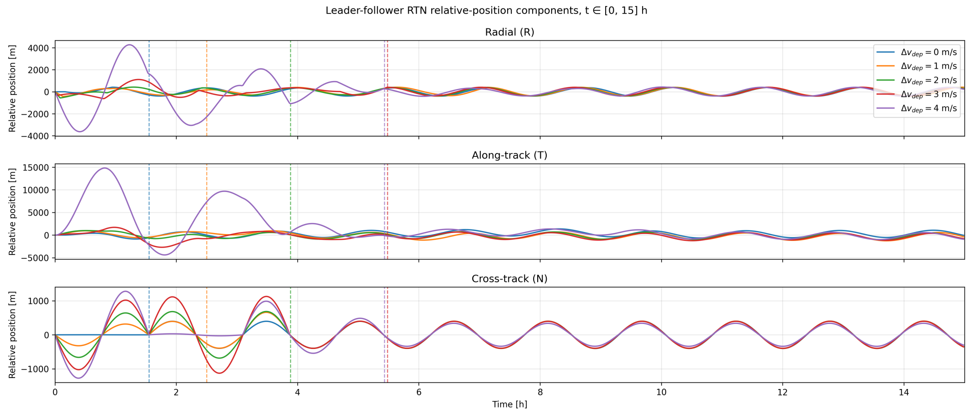

Figure 7.2.2: Decomposition of the leader-follower RTN relative-position trajectories into its individual components over $t \in [0, 15]$. Only a selected set of differential deployment velocities are included, corresponding to run 0, 2, 4, 6 and 8 detailed in 6.2.4. The vertical dashed lines show the corresponding color's acquisition time.

The component-wise RTN relative-position time series is shown in Figure 7.2.2. The radial, along-track, and cross-track components all show transient oscillations during the initial formation response. The largest transient amplitudes occur for the highest differential deployment speed cases, most clearly in the cross-track component. The $\Delta v_{\text{dep}} = 4$ m/s case reaches the largest along-track displacement, while the lower deployment speed cases remain closer to the leader during the initial interval. The vertical dashed lines in Figure 7.2.2 indicate the acquisition times determined for the corresponding deployment velocity cases. These acquisition times are evaluated formally in the next subsection using the orbital-element-difference acquisition criterion. The RTN trajectories shown here therefore present the geometric leader-follower response before and after the initial formation insertion process.

7.2.2 Formation Acquisition Determination

The formation acquisition time was determined from the time evolution of the follower-leader classical orbital element differences, shown in Figure 7.2.3 for a selected subset of all cases presented in Section 6.2.4. For this campaign, acquisition was defined by the follower remaining within the specified tolerance bounds for the selected acquisition-relevant OED components: $\Delta a/a$, Δe, and Δi, in accordance with Section 6.2.2. The tolerance ranges are shown as black dashed horizontal lines, and the vertical dashed lines show the corresponding acquisition times.

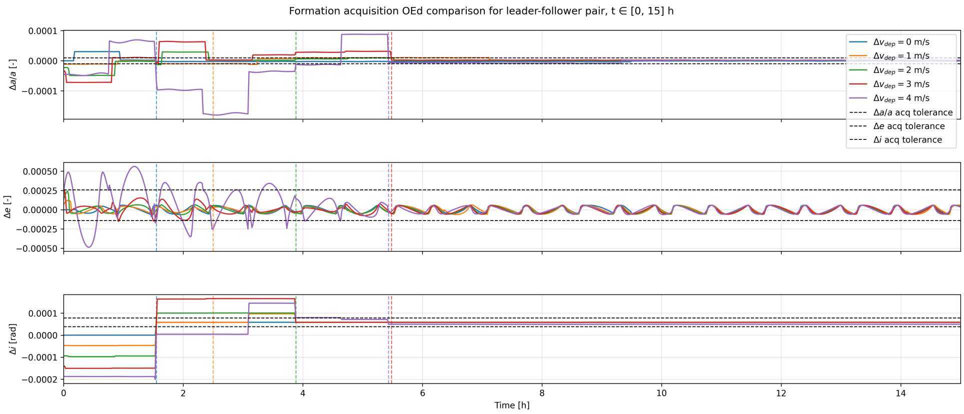

Figure 7.2.3: Classic orbital element differences between the leader-follower pair for a selected set of differential deployment velocities corresponding to run 0, 2, 4, 6 and 8 detailed in 6.2.4. The horizontal dashed lines show the OED tolerance range, while the vertical dashed lines show the acquisition times. The plotted time-horizon is reduced to $t \in [0, 15]$.

The results show that the initial differential deployment speeds produces different transient OED responses that eventually converge within the formation tolerances at different rates. The steep changes observed in the OED histories are caused by formation-control burn maneuvers, which rapidly adjust the relative orbital elements. These jumps are especially visible in $\Delta a/a$ and Δi, where the commanded burns rapidly changes the semi-major axis and inclination difference. Between these burn events, the OEDs evolve more smoothly due to the natural relative orbital motion and perturbation environment.

7.2.3 Acquisition Time & Fuel Consumption

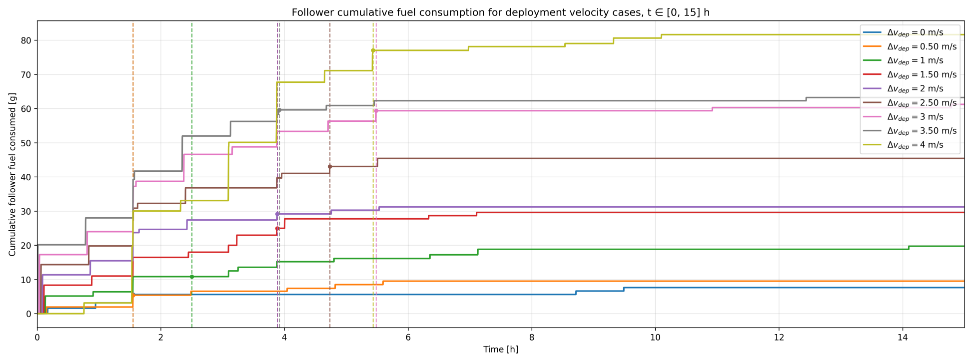

Figure 7.2.4: Cumulative formation acquisition fuel consumption for all campaign cases over $t \in [0, 15]$. The vertical dashed lines show the acquisition times.

The cumulative follower fuel consumption during the first 15 hours of the deployment sensitivity campaign is shown in Figure 7.2.4. The colored curves show the accumulated fuel consumed by the follower spacecraft, while the vertical dashed lines mark the acquisition time for the corresponding differential deployment speed case. The circular markers at the intersection between these lines show the acquisition fuel consumption m_{acq}.

The fuel time series show a clear difference between the acquisition phase and the subsequent formation-keeping phase, where the general cumulative fuel consumption rate is significantly larger during acquisition. This behavior is observed across the deployment speed cases, although the number, timing, and magnitude of the burn events vary between cases. The acquisition fuel consumption also increases with differential deployment speed. The lowest deployment speed case requires only a few grams of fuel before acquisition, while the largest differential deployment speed case requires approximately 77 g. The cumulative fuel curves therefore show that larger initial deployment velocity differences lead to a larger propellant cost before the formation is acquired. This trend is more consistent for fuel consumption than for acquisition time, as each increase in differential deployment speed generally results in a larger value of m_{acq}.

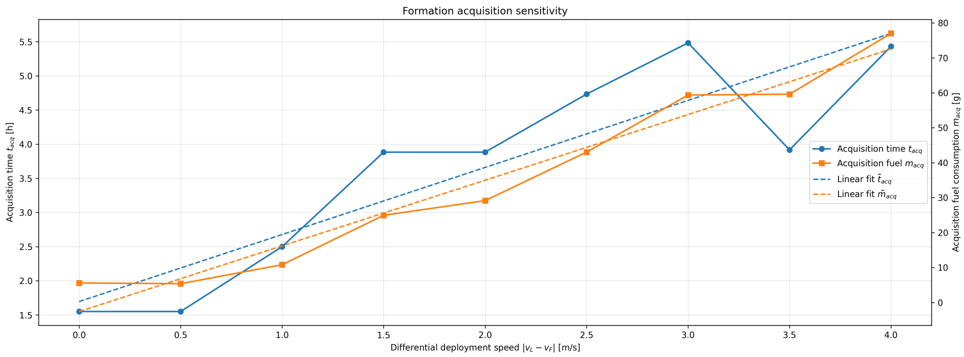

Figure 7.2.5: Formation acquisition time and cumulative fuel consumption as a function of all campaign differential deployment speed cases

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Figure 7.2.5 summarizes the acquisition time and acquisition fuel consumption as functions of the differential deployment speed. The observed acquisition time t_{acq} is plotted on the left axis, while the acquisition fuel consumption m_{acq} is plotted on the right axis. Both quantities show a strong positive correlation with differential deployment speed. The acquisition time generally increases as Δv_{dep} increases, but the trend does not strictly apply to all cases. In particular, some intermediate cases deviate from the otherwise increasing trend, showing that the exact acquisition time depends on the timing of the discrete burn sequence and on when the dwell-time acquisition criterion is first satisfied.

A first-order prediction model was estimated for both quantities using a linear least-squares fit. The resulting approximations are shown as dashed lines in Figure 7.2.5. For the acquisition time, the fitted model is

$$\hat{t}_{\text{acq}}(\Delta v_{\text{dep}}) = 0.982\Delta v_{\text{dep}} + 1.696, \tag{7.1}$$

where $\hat{t}_{\text{acq}}$ is given in hours when Δv_{dep} is given in m/s. For the acquisition fuel consumption, the fitted model is

$$\hat{m}_{\text{acq}}(\Delta v_{\text{dep}}) = 18.782\Delta v_{\text{dep}} - 2.557, \tag{7.2}$$

where $\hat{m}_{\text{acq}}$ is given in grams. The fuel-consumption fit follows the observed data closely, reflecting the approximately linear increase in acquisition fuel cost over the simulated deployment speed range. The acquisition-time fit captures the overall positive trend, but the observed acquisition times show larger deviations from a purely linear response. This is consistent with the stepwise nature of the formation-control process, where acquisition depends not only on the size of the initial deployment velocity difference, but also on the timing of the burn maneuvers and the dwell-time condition used to define successful acquisition.

7.2.4 Deployment Sensitivity Summary

The differential deployment speed formation acquisition results are summarized in Table 7.2.1. All nine deployment cases successfully acquired the desired formation within the 48-hour simulation interval. The acquisition time increased from 1.55 h for the $\Delta v_{\text{dep}} =$ 0.0 m/s and 0.5 m/s cases to 5.43 h for the $\Delta v_{\text{dep}} = 4.0$ m/s case. The acquisition fuel consumption increased from 5.60 g for the zero-deployment-speed case to 77.02 g for the largest deployment-speed case. The total follower fuel consumption after 48 hours follows the same overall trend, increasing from 10.42 g to 83.41 g across the tested deployment-speed range.

Table 7.2.1: Summary of deployment velocity formation acquisition results.

Run	Δv_{dep} **[m/s]**	**Acquired**	t_{acq} **[h]**	m_{acq} **[g]**	**Burns**	$\bar{t}_{\text{burn}}$ **[s]**	m_{48h} **[g]**
0	0.0	Yes	1.55	5.60	3	2.27	10.42
1	0.5	Yes	1.55	5.39	2	3.20	12.34
2	1.0	Yes	2.50	10.83	3	4.33	21.58
3	1.5	Yes	3.88	24.96	7	4.27	33.24
4	2.0	Yes	3.88	29.16	7	5.23	34.77
5	2.5	Yes	4.73	43.05	9	5.74	49.87
6	3.0	Yes	5.48	59.36	10	7.09	66.01
7	3.5	Yes	3.92	59.61	9	8.26	67.98
8	4.0	Yes	5.43	77.02	11	8.37	83.41

The burn statistics in Table 7.2.1 also show that the increased acquisition fuel consumption is accompanied by both an increased number of formation-control burns and longer average thruster on-times. The number of burns during acquisition increases from 3 in the zero-deployment-speed case to 11 in the $\Delta v_{\text{dep}} = 4.0$ m/s case. The average thruster on-time during acquisition increases from 2.27 s to 8.37 s over the same range.

Table 7.2.2: Additional deployment acquisition diagnostics.

Run	Δv_{dep} **[m/s]**	$\bar{\sigma}_{\mathrm{err,burn}}$ **[deg]**	$e_{\mathrm{batt,min}}$ **[%]**	**Burns denied**
0	0.0	26.55	37.90	No
1	0.5	10.17	37.18	No
2	1.0	18.53	37.43	No
3	1.5	18.61	37.82	No
4	2.0	12.08	37.18	No
5	2.5	13.31	37.01	No
6	3.0	11.06	37.57	No
7	3.5	11.39	37.32	No
8	4.0	13.93	37.50	No

The additional diagnostics in Table 7.2.2 show that no burn requests were denied in any of the deployment cases. The average attitude tracking error during burns remains case-dependent, but no case is marked as burn-limited by the internal subsystem logic. The follower EPS overview shown in Appendix F.1.4 shows the battery-energy history and instantaneous net power for all deployment cases during the acquisition interval. The operational-mode in Appendix F.1.3 histories show the transitions between burn, capture, charge, communication, and coast modes, while the burn diagnostics in Appendix F.1.2 summarize the burn execution statistics. Across the tested deployment-speed cases, the operational constraints did not prevent the execution of any commanded formation-maintenance burns. Since entering emergency mode is the condition required to block a burn request, this indicates that the simulated battery state remained sufficient for the formation acquisition sequence in all cases.

7.3 Steady-State Fuel Consumption

This section presents the results from the formation steady-state fuel consumption campaign described in Section 6.2.5. The campaign uses the GNSS-R Formation Flight Simulator to propagate the selected leader-follower concentric formation after deployment, with formation control, attitude control, operational mode switching, EPS modelling, and the full thesis disturbance environment enabled. The purpose of the campaign is to estimate the cumulative post-acquisition formation-maintenance fuel consumption $m_{\mathrm{maint}}(t)$ from the follower fuel history.

The results are first presented using the follower cumulative fuel consumption over the available long-horizon simulation interval, together with fitted continuous models for $m_{\mathrm{maint}}(t)$. The section then summarizes the operational feasibility of the run using EPS, mode, pointing, and burn-execution metrics for both spacecraft. Together, these results provide the steady-state fuel-consumption model and subsystem feasibility check used in the GNSS-R formation flight lifetime estimate.

7.3.1 Post-Acquisition Fuel Consumption

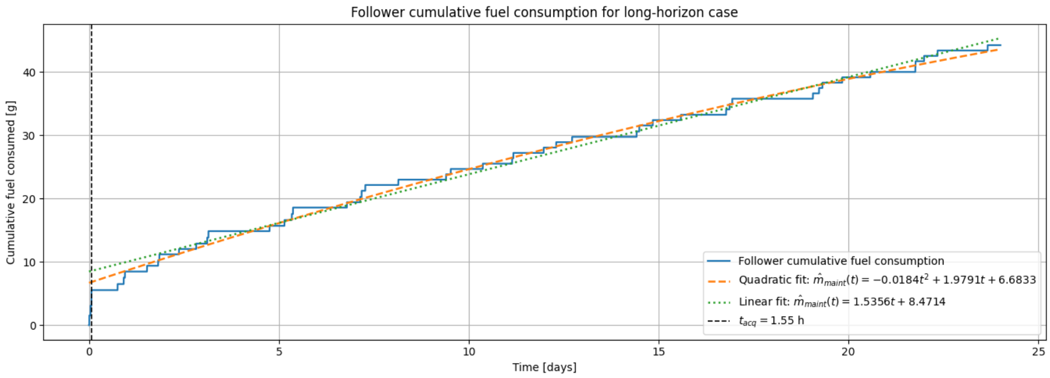

Figure 7.3.1: Cumulative fuel consumption for a zero differential deployment speed case over 24 days. A quadratic and linear model has been fitted to the cumulative fuel consumption using samples in the timespan $t \in [t_{\text{acq}}, t_{\text{end}}]$

Figure 7.3.1 shows the cumulative follower fuel consumption from the long-horizon steady-state fuel consumption campaign. The follower is initialized with zero relative deployment velocity with respect to the leader, and the vertical dashed line marks the acquisition time $t_{\text{acq}} = 1.55$ h. After 24 days of simulation time, the follower has consumed approximately 44 g of propellant during acquisition and subsequent formation maintenance.

Two continuous models are fitted to the cumulative fuel-consumption history in the period after the acquisition time. The quadratic fit is

$$\hat{m}_{\text{maint}}(t) = -0.018t^2 + 1.979t + 6.683, \tag{7.3}$$

while the linear fit is

$$\hat{m}_{\text{maint}}(t) = 1.536t + 8.471. \tag{7.4}$$

Here, t is measured in days and the fitted fuel consumption is given in grams. The fitted curves provide the simulation-based approximation of the cumulative post-acquisition formation-maintenance fuel consumption used in the lifetime estimate.

7.3.2 Steady-State Operational Feasibility

Table 7.3.1 summarizes the operational feasibility metrics from the steady-state fuel consumption campaign. The simulation was propagated for 24 days for both spacecraft. The leader and follower show nearly identical EPS statistics, with a minimum battery energy of approximately 36%, a mean battery energy of approximately 42%, and a maximum battery energy of approximately 80%. The mean net power is close to zero for both spacecraft, while the instantaneous net power varies between approximately -28 W and 20 W. Both spacecraft spend most of the simulation time in CAPTURE mode, followed by CHARGE, COAST, and COMMS. No burn requests were denied for either spacecraft during the campaign.

Table 7.3.1: Steady-state campaign operational feasibility metrics

Metric	Leader	Follower
Simulation duration [days]	24.00	24.00
Minimum battery energy [%]	36.16	36.14
Mean battery energy [%]	42.48	42.48
Maximum battery energy [%]	79.89	79.89
Mean net power [W]	-0.06	-0.06
Minimum net power [W]	-23.08	-28.12
Maximum net power [W]	18.25	20.41
Time in BURN mode [h]	0.00 (0.00%)	0.02 (0.00%)
Time in CAPTURE mode [h]	328.35 (57.01%)	328.17 (56.97%)
Time in CHARGE mode [h]	161.63 (28.06%)	161.73 (28.08%)
Time in COMMS mode [h]	35.90 (6.23%)	35.87 (6.23%)
Time in COAST mode [h]	50.12 (8.70%)	50.21 (8.72%)
Burns	0	49
Mean burn duration [s]	0	1.28
Mean pointing error [deg]	0.04	0.36
Burn denied	No	No

Figure 7.3.2 shows the EPS and operational-mode behaviour during the final 24 h of the long-horizon simulation. The leader and follower battery-energy time series, net-power profiles, and operational-mode transitions are nearly identical over this interval. Both spacecraft exhibit a repeated mode sequence dominated by CAPTURE and CHARGE, with shorter intervals in COMMS and COAST. A single follower BURN-mode event occurs during the displayed interval, while the leader remains without burn activity. The similarity between the leader and follower EPS time series and mode transitions indicates that the two spacecraft remain synchronized in their steady-state operational cycle at the end of the simulation.

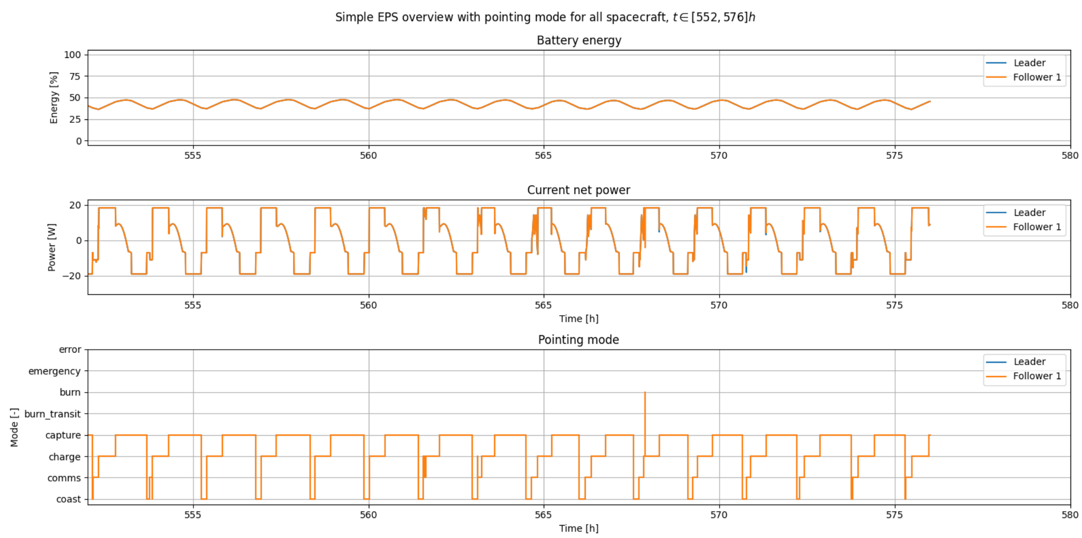

Figure 7.3.2: Operational mode transitions and EPS parameters for the leader and follower spacecraft during the final hours of the steady-state campaign

7.4 GNSS-R Formation Flight Lifetime Estimation

The fuel-limited lifetime estimate is obtained by combining the acquisition fuel model from Section 7.2.3 with the post-acquisition maintenance fuel model from Section 7.3.1. The remaining follower fuel mass is modelled as

$$\hat{m}_{\mathrm{fuel}}(t, \Delta v_{\mathrm{dep}}) = m_{\mathrm{fuel},0} - \hat{m}_{\mathrm{acq}}(\Delta v_{\mathrm{dep}}) - \hat{m}_{\mathrm{maint}}(t), \tag{7.5}$$

where $m_{\mathrm{fuel},0} = 635$ g is the initial follower fuel mass. The fitted acquisition fuel model is

$$\hat{m}_{\mathrm{acq}}(\Delta v_{\mathrm{dep}}) = 18.782\Delta v_{\mathrm{dep}} - 2.557, \tag{7.6}$$

with Δv_{dep} given in m/s and $\hat{m}_{\mathrm{acq}}$ in grams. The post-acquisition fuel consumption was fitted using both a quadratic and a linear model. The quadratic model is

$$\hat{m}_{\mathrm{maint,q}}(t) = -0.018t^2 + 1.979t + 6.683, \tag{7.7}$$

while the linear model is

$$\hat{m}_{\mathrm{maint,l}}(t) = 1.536t + 8.471. \tag{7.8}$$

Here, t is measured in days and $\hat{m}_{\mathrm{maint}}$ is given in grams. Substituting the two maintenance models into the remaining fuel expression gives

$$m_{\mathrm{fuel}} \approx \hat{m}_{fuel} = \begin{cases} \hat{m}_{\mathrm{fuel,q}}(t, \Delta v_{\mathrm{dep}}) = 0.018t^2 - 1.979t - 18.782\Delta v_{\mathrm{dep}} + 630.874 \\ \text{or} \\ \hat{m}_{\mathrm{fuel,l}}(t, \Delta v_{\mathrm{dep}}) = -1.536t - 18.782\Delta v_{\mathrm{dep}} + 629.086. \end{cases} \tag{7.9}$$

The predicted fuel-limited lifetime is evaluated using both remaining-fuel models and the reserve condition $m_{\mathrm{res}} = 0$. The quadratic lifetime estimate $\hat{T}_{\mathrm{life,q}}$ is defined as the time at which $\hat{m}_{\mathrm{fuel,q}}(t, \Delta v_{\mathrm{dep}})$ reaches zero, while the linear lifetime estimate $\hat{T}_{\mathrm{life,l}}$ is defined as the time at which $\hat{m}_{\mathrm{fuel,l}}(t, \Delta v_{\mathrm{dep}})$ reaches zero. Table 7.4.1 summarizes the resulting lifetime estimates for the deployment speeds considered in the acquisition fuel sensitivity campaign, together with the corresponding simulated and estimated acquisition fuel consumption.

Table 7.4.1: Predicted GNSS-R formation flight lifetime as a function of differential deployment speed using the quadratic and linear remaining fuel models.

Δv_{dep} [m/s]	m_{acq} [g]	$\hat{m}_{\mathrm{acq}}$ [g]	$\hat{T}_{\mathrm{life,q}}$ [days]	$\hat{T}_{\mathrm{life,l}}$ [days]
0.0	5.60	-2.56	–	413.58
0.5	5.39	6.83	–	409.14
1.0	10.83	16.23	–	403.06
1.5	24.96	25.62	–	397.01
2.0	29.16	35.01	–	390.89
2.5	43.05	44.40	–	384.81
3.0	59.36	53.79	–	378.73
3.5	59.61	63.18	–	372.55
4.0	77.02	72.57	–	366.50

The quadratic model does not produce a finite lifetime estimate for any of the deployment-speed cases. This is indicated by the undefined entries for $\hat{T}_{\mathrm{life,q}}$ in Table 7.4.1. The undefined values occur because the fitted quadratic remaining-fuel model does not reach zero fuel. Instead, the positive quadratic term causes the remaining-fuel estimate to eventually increase after reaching a minimum. The linear model therefore provides the finite lifetime estimates reported in the final column of the table.

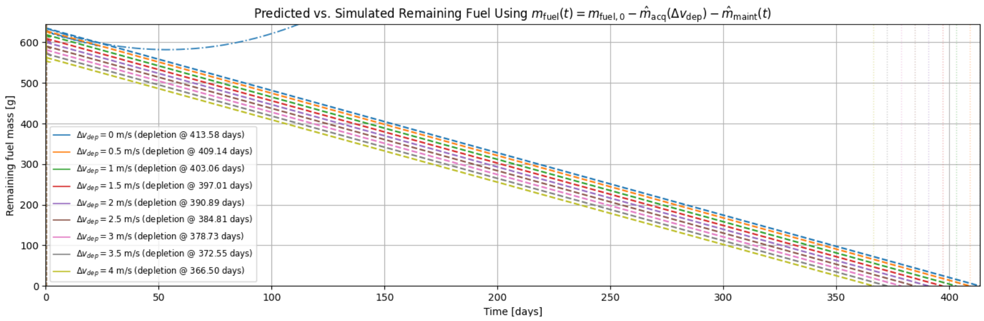

Figure 7.4.1: Linear remaining fuel model applied to each differential deployment speed case considered in the deployment sensitivity campaign, plotted for the estimated lifetime duration. The quadratic remaining fuel model is only applied to the zero-deployment-speed case

Figure 7.4.1 shows the predicted remaining follower fuel mass over the extrapolated lifetime interval for each differential deployment speed. The dashed curves correspond to the linear remaining-fuel model and reach zero fuel at the predicted linear lifetime $\hat{T}_{\mathrm{life,l}}$. The vertical markers indicate the corresponding depletion times, which decrease from 413.58 days for the zero-deployment-speed case to 366.50 days for the $\Delta v_{\mathrm{dep}} = 4.0$ m/s case. The quadratic model is shown for the $\Delta v_{\mathrm{dep}} = 0$ m/s case and illustrates why $\hat{T}_{\mathrm{life,q}}$ is undefined.

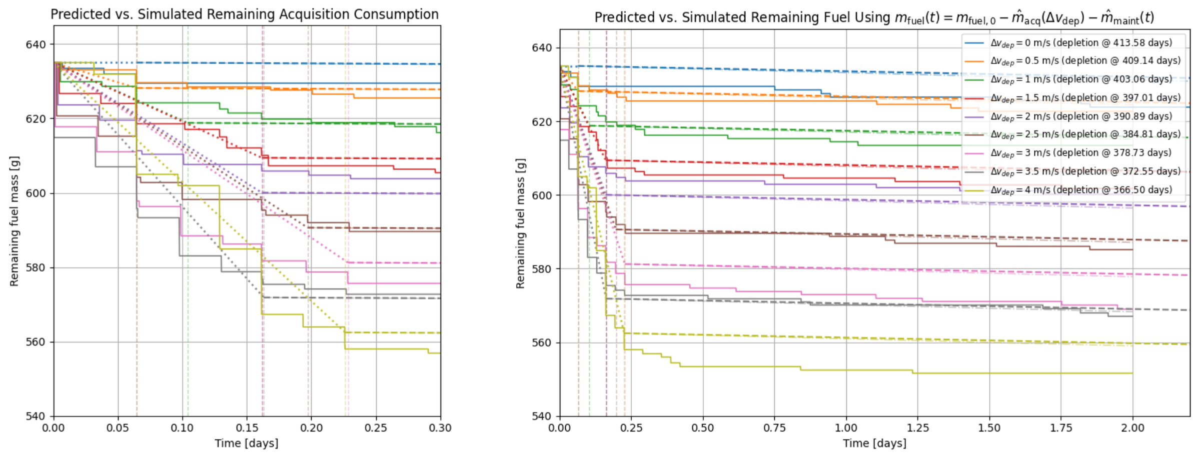

Figure 7.4.2: Comparison between the linear remaining fuel model and the simulated differential deployment cases in Section 7.2. The left plot highlights fuel acquisition estimate accuracy, while the right plot shows the remaining fuel model compared to the full 48 h simulation duration.

Figure 7.4.2 compares the predicted remaining fuel time seris against the simulated fuel histories during the acquisition interval. The solid curves show the simulated follower fuel mass from the deployment velocity sensitivity campaign, while the dashed curves show the model prediction using the fitted acquisition fuel model and the linear post-acquisition fuel model. The zoomed view highlights the deployment-speed-dependent fuel offset introduced by $\hat{m}_{\text{acq}}(\Delta v_{\text{dep}})$ during the first part of the simulation.

CHAPTER

EIGHT

DISCUSSION

8.1 Propagation Benchmark Interpretation

8.1.1 Basilisk-Skyfield-SGP4 Divergence Interpretation

Across the same-spacecraft comparison results, the primary thesis Basilisk configuration shows the smallest divergence from the continuously updating Skyfield-SGP4 baseline. Figure 7.1.1 shows that the specialization-project exponential atmosphere configuration reaches approximately three times the position difference of the primary thesis configuration after the 14-day propagation interval, while the re-tuned exponential atmosphere configuration reaches approximately 1.5 times the position difference. Since the main difference between the evaluated Basilisk configurations is the atmospheric density model, the observed ordering can be interpreted primarily through the different drag environments experienced by the spacecraft.

The altitude difference in Figure 7.1.4 supports this interpretation. The primary thesis configuration with the NRLMSISE-00 atmosphere model gradually drops below the Skyfield-SGP4 reference altitude, while both exponential atmosphere configurations drift toward higher relative mean altitude. This indicates that the primary thesis configuration experiences stronger effective drag than the exponential atmosphere configurations over the comparison interval. The re-tuned exponential atmosphere model reduces the altitude divergence relative to the specialization-project configuration, but it does not reproduce the same altitude evolution as the NRLMSISE-00 configuration.

The RTN decomposition in Figure 7.1.2 shows that the resulting same-spacecraft position difference is dominated by radial and along-track drift. Because we are effectively considering the relative drift between two deactivated spacecraft affected by different amounts of drag, the relative drift mechanics covered in Section 2.5 apply. Sustained differential drag will accumulate into substantial along-track drift. This is because a sustained altitude difference changes the relative orbital period, which accumulates into along-track separation over time. The exponential atmosphere configurations maintain a higher relative mean altitude than the Skyfield-SGP4 baseline and therefore increasingly lag behind the reference trajectory. This produces the negative along-track difference observed in the RTN decomposition. In contrast, the primary thesis configuration drops to a lower relative mean altitude, corresponding to a shorter relative orbital period, and therefore drifts ahead of the Skyfield-SGP4 baseline in the positive along-track direction.

The radial component in Figure 7.1.2 should be interpreted together with the dominant along-track separation. Since the RTN frame is defined at the Skyfield-SGP4 leader position, the radial component represents the direction along its position vector away from the Earth. For large along-track separations, this radial projection is not equivalent to the altitude difference, but rather the circular curvature of the orbit. Thus, a sufficiently large along-track difference correspond to a negative radial deviation for spacecraft in the same approximate near-circular orbit.

The same differential-drag interpretation also applies to the same-formation drift difference shown in Figure 7.1.5. All Basilisk configurations predict a larger leader-follower along-track drift than the Skyfield-SGP4 baseline. However, the ordering between the Basilisk configurations is reversed and less separated compared to the same-satellite state comparison. The primary thesis configuration gives a slightly larger same-formation along-track drift difference, while the exponential configurations remain close to it. This indicates that the atmospheric model has a much stronger effect on the absolute Earth-relative propagation of each spacecraft than on the passive leader-follower drift over the same interval. The primary thesis configuration likely diverges more than the others, because its differential drag drift is stronger due to each spacecraft being subject to higher intensity drag. Also, the very small relative divergence between the Basilisk configurations are probably due to the ECI-fixed spacecraft attitude assumption, where the difference in incident flow in each body-fixed frame are only slightly different due to the initial 1° in-plane separation angle.

8.1.2 Updating-TLE SGP4 as a Comparison Baseline

The interpretation of the comparative propagation results depends on the quality of the continuously updating Skyfield-SGP4 baseline. This baseline is not a true physical ground truth, but an observation-updated reference generated from a sequence of TLEs. Since the propagated TLE is replaced regularly throughout the comparison interval, the SGP4 prediction error is not allowed to accumulate over the full 14-day horizon in the same way as it would for a static-TLE propagation. This makes the updating-TLE baseline a more useful long-horizon comparison reference than the single-TLE baseline used in the specialization project [8].

The frequent TLE updates also explain why no distinct correction discontinuities are visible in the comparison figures. Over short propagation intervals, SGP4 is generally expected to remain sufficiently close to the estimated spacecraft state that the transition between adjacent TLEs does not dominate the plotted state histories. The resulting baseline is therefore best interpreted as a sequence of short SGP4 predictions tied together by updated orbit estimates, rather than as one uninterrupted long-horizon physical propagation.

However, the TLEs used by the Skyfield-SGP4 baseline are tied to the real HYPSO-1 spacecraft. The orbital evolution represented by the TLE sequence therefore includes the effect of real perturbations acting on HYPSO-1, including effects associated with its actual attitude and operational history. Since HYPSO-1 is an Earth-observation satellite with a body-fixed imaging payload, its attitude may vary over time due to science objectives. This means that the effective drag experienced by the real spacecraft may differ from the drag experienced by the simplified Basilisk spacecraft, which is propagated with a fixed attitude in the comparative propagation analysis. As a result, differences between Basilisk and the updating-TLE Skyfield-SGP4 would still be present with a perfect atmosphere model simply because of different experienced attitude-coupled drag.

The updating-TLE Skyfield-SGP4 trajectory is therefore a valuable comparison baseline, but it should not be interpreted as exact truth. It provides a practical reference for evaluating whether the primary thesis Basilisk configuration improves over the specialization-project configuration and for quantifying the scale of the remaining propagation divergence. The comparison result should consequently be interpreted as agreement with an observation-updated SGP4 reference, rather than as an absolute validation of the Basilisk propagator.

8.1.3 Implications for the Lifetime Analysis

The comparative propagation analysis indicates that the primary thesis Basilisk configuration is the most accurate of the evaluated Basilisk configurations relative to the continuously updating Skyfield-SGP4 baseline. Across the same-spacecraft state comparisons, the NRLMSISE-00 configuration produces substantially smaller divergence than both exponential-atmosphere configurations. This supports the use of the primary thesis configuration as the propagation environment for the GNSS-R formation flight lifetime analysis. However, the primary thesis configuration still accumulates a significant same-satellite difference over the 14-day comparison interval, dominated by radial and along-track drift.

Perhaps more importantly for the lifetime analysis, is the same-formation position drift difference. Although the major drift difference between the primary thesis configuration and the Skyfield-SGP4 baseline remain unexplained, the increased along-track formation drift will require additional formation-maintenance maneuvers. The resulting lifetime estimations will therefore likely be pessimistic in its predictions. However, the updating-TLE Skyfield-SGP4 baseline also limits how strongly the propagation comparison can be interpreted. Since the baseline is an observation-updated SGP4 reference rather than a precise orbit-determination truth model, the observed Basilisk-Skyfield differences are not direct physical prediction errors. The comparison still provides useful evidence that the thesis configuration improves over the specialization-project configuration, but it does not fully validate the long-horizon Basilisk trajectory.

8.2 Formation Fuel Consumption

8.2.1 Acquisition Consumption

The deployment sensitivity results show summarized in Figure 7.2.5, show that acquisition time is generally positively correlated with differential deployment speed, but that the relationship is not strictly increasing. The most visible exception is the $\Delta v_{\text{dep}} = 3.5$ m/s case, which acquired the formation after 3.92 h, compared with 5.48 h for the $\Delta v_{\text{dep}} = 3.0$ m/s case and 5.43 h for the $\Delta v_{\text{dep}} = 4.0$ m/s case. Since t_{acq} is determined directly from the OED tolerance and dwell-time criterion, this non-monotonic behavior indicates that the selected acquisition criterion does not provide a perfectly smooth measure of the acquisition transient across the tested deployment cases.

The OED histories shown in Figure 7.2.3 suggest that the acquisition criterion may be somewhat relaxed for the purpose of distinguishing the end of acquisition from the beginning of maintenance. For example, the $\Delta v_{\text{dep}} = 4.0$ m/s case performs several additional burns after its acquisition time, which further reduces the semi-major-axis difference toward the center of the tolerance band. Similarly, the eccentricity difference is not fully synchronized across all deployment cases until approximately 7.6 h, which is later than the largest recorded acquisition time. The cumulative fuel-consumption time series in Figure 7.2.4 also show several post-acquisition burn events of non-negligible size. These observations suggest that the exact numerical value of t_{acq} is sensitive to the selected OED tolerances and dwell time, although the plots alone do not provide a strict separation between late acquisition burns and early maintenance burns.

The acquisition fuel consumption shows a more uniform dependence on differential deployment speed than the acquisition time. The only from this strictly increasing behavior is in the change from $\Delta v_{\text{dep}} = 0.0$ m/s to $\Delta v_{\text{dep}} = 0.5$ m/s, where the measured acquisition fuel consumption decreases with differential deployment speed. The slightly lower acquisition fuel consumption for the 0.5 m/s case may indicate that a small non-zero differential deployment speed can reduce the fuel needed to initiate the concentric formation. This makes sense considering this removes the need for an initial burn to separate the leader and follower spacecraft. For the remaining cases, the increasing trend is consistent with the larger velocity correction required to remove the initial deployment-induced relative motion and insert the follower into the target formation.

Because the acquisition time is on the order of a couple of hours for the cases studied in this thesis, the time-horizon is too short for any small differential orbital perturbations to take any noticeable effect, and cannot be considered a major driver for acquisition fuel consumption. Thus, the main driver of acquisition fuel consumption is the initial differential deployment condition and the corrective burns required to reach the desired formation. Modeling this cumulative fuel consumption purely as a function of Δv_{dep} is therefore a reasonable simplification. However, as discussed initially, the largest uncertainty in the extracted acquisition fuel values is likely associated with the acquisition-time definition itself, since changing the OED tolerances or dwell-time requirement would shift the point at which cumulative fuel consumption is sampled.

8.2.2 Steady-State Maintenance Consumption

The steady-state fuel consumption result in Figure 7.3.1 show that the steady-state cumulative fuel consumption rate is not perfectly constant, but appears to decrease slightly over the 24-day simulation. This trend is physically reasonable, since a lower remaining propellant mass reduces the spacecraft mass, such that a given orbital correction requires slightly less thrust impulse, assuming that the required formation-maintenance corrections remain approximately the same. This behavior differs from the acquisition phase, where fuel consumption is dominated by a short sequence of corrective burns shortly after deployment. Regardless of how the acquisition time is defined, the deployment sensitivity fuel time series in Figure 7.2.4 clearly show that the burn frequency is strongly reduced after the initial formation insertion period. The remaining fuel consumption is therefore associated with slower formation-maintenance corrections, rather than the removal of the initial deployment-induced relative motion.

This behavior differs from the acquisition phase, where fuel consumption is dominated by a short sequence of corrective burns shortly after deployment. Regardless of how the acquisition time is defined, the deployment sensitivity fuel time series in Figure 7.2.4 clearly show that the burn frequency is strongly reduced after the initial formation insertion period. The remaining fuel consumption is therefore associated with slower formation-maintenance corrections, rather than the removal of the initial deployment-induced relative motion.

The steady-state drift mechanism is not expected to be dominated by large ballistic-coefficient differences between the spacecraft. As described in Section 2.5.3, the differential atmospheric drag depends on differences in ballistic coefficient, which is affected by the drag coefficient and flow-normal cross-sectional area. In the simulated formation, both spacecraft use the same drag coefficient, and Figure 7.3.2 shows that the leader and follower are synchronized in their operational modes during steady-state operation. Since the operational modes determine the attitude profile, the two spacecraft therefore spend most of the time in similar attitudes. This reduces differences in flow-normal area and gives the leader and follower very similar ballistic coefficients.

Some differential drag is still present because the follower follows an oscillatory relative trajectory around the leader. The radial component in Figure 7.2.2 has an amplitude of approximately 400 m, which for small separation distances, correspond to a small periodic altitude difference between the leader and follower. The follower therefore periodically occupies slightly higher and lower atmospheric density regions than the leader. However, this separation is small compared with the orbital altitude, and the formation controller also limits long-term along-track drift by keeping the semimajor axis difference close to zero.

The full OED time series in Appendix F.1.1 show that the RAAN difference increases over time. This indicates a gradual relative precession of the follower orbit plane with respect to the leader. Since the spacecraft are in a highly inclined low Earth orbit, this behavior is consistent with differential J_2 perturbation effects from Earth's non-spherical gravity field. Within the simulated steady-state scenario, this makes differential gravitational perturbations a likely contributor to the long-term formation drift that produces the observed maintenance burns.

8.3 Lifetime Estimate Reliability

8.3.1 Acquisition Fuel Model Reliability

The acquisition fuel model used in the lifetime estimate is based on the nine deployment velocity cases shown in Figure 7.2.5. The fitted linear model captures the overall increase in acquisition fuel consumption with differential deployment speed, but it does not reproduce all features of the simulated data. The deviations between the predicted and simulated acquisition fuel consumption are shown in Figure 7.4.2 and summarized numerically in Table 7.4.1. Some of non-systematic deviation from the linear model may be explained by the uncertainty related to the determination of the acquisition time, as discussed in Subsection 8.2.1. However, the largest deviation occurs for the zero-deployment-speed case, where the fitted model predicts an acquisition fuel consumption of -2.56 g, compared with the simulated value of 5.60 g. This gives a prediction error of 8.16 g, which is both a relatively large deviation, and is physically unrealistic since acquisition fuel consumption cannot be negative.

The poor zero-speed prediction is consistent with the non-monotonic behavior between the $\Delta v_{\mathrm{dep}} = 0.0$ m/s and $\Delta v_{\mathrm{dep}} = 0.5$ m/s cases discussed in Section 8.2.1. Since the remaining data points follow a more regular increasing trend, the fitted linear model is more representative for the higher deployment-speed cases than for the lowest part of the tested range. This indicates that the acquisition fuel dependence on Δv_{dep} is not fully captured by a single unconstrained linear fit across the complete interval.

A higher-order model could potentially represent the low-speed nonlinearity more accurately, but the available data do not clearly support a single quadratic model over the full deployment-speed range. Such a fit would force a continuously curved relationship, even though the simulated acquisition fuel consumption is approximately linear for most of the tested interval. A piecewise model, with a separate low-speed representation below approximately 1 m/s and a linear model at higher differential deployment speeds, might better represent the observed behavior. However, additional differential deployment speed cases would be required to determine this speculated dynamics more closely.

8.3.2 Steady-State Fuel Model Reliability

The steady-state maintenance fuel model is based on the post-acquisition cumulative fuel consumption shown in Figure 7.3.1. A quadratic model was fitted in order to capture the slight decrease in apparent fuel consumption rate over the completed simulation interval due to mass reduction. Although this model follows the simulated fuel trajectory over the available data interval, its extrapolated behavior in Figure 7.4.1 shows that it is not suitable for predicting fuel depletion over approximately 40 days. The corresponding remaining-fuel model does not reach zero fuel, which makes the quadratic lifetime estimate undefined.

A linear steady-state fuel model was therefore also fitted. The comparison between the predicted and simulated remaining fuel time series in Figure 7.4.2 shows that, after accounting for the initial offset caused by acquisition-fuel model deviations, the linear model remains approximately parallel to the simulated post-acquisition fuel time series. This indicates that the linear model captures the average steady-state consumption rate over the short post-acquisition interval covered by the deployment sensitivity campaign cases.

The reliability of the linear steady-state model is limited by the available simulation horizon. The deployment sensitivity simulations only provide a few days of post-acquisition comparison, while the long-horizon steady-state campaign was limited to 24 days. As stated in Section 6.2.6, the campaign was originally intended for a 90-day simulation horizon, but all longer attempts ended prematurely due to persistent memory overflow issues in the GNSS-R Formation Flight Simulator. Longer completed steady-state simulations would be required to model the fuel rate reduction over time due to decreasing stored propellant mass.

8.3.3 Interpretation of the Predicted Lifetime

The linear remaining-fuel model gives a predicted fuel-limited lifetime range of 366.5-413.6 days across the evaluated deployment-speed cases, as shown in Figure 7.4.1. The spread in predicted lifetime is caused by the deployment-speed-dependent acquisition fuel consumption, where larger differential deployment speeds reduce the remaining fuel available for long-term formation maintenance. However, the dominant source of uncertainty in the lifetime estimate is arguably the steady-state maintenance fuel consumption rate. This is because the estimated lifetime is extrapolated over hundreds of days, while the acquisition phase lasts only a few hours. For example, reducing the fitted linear maintenance rate from 1.536 g/day to 1.436 g/day increases the predicted lifetime by approximately 28.4 days when zero acquisition fuel consumption is assumed. This is comparable to the lifetime difference between the $\Delta v_{\text{dep}} = 0$ m/s and $\Delta v_{\text{dep}} = 2.5$ m/s cases.

The acquisition fuel model also introduces uncertainty, but its effect on the predicted lifetime is smaller. The largest deviation between fitted and simulated acquisition fuel consumption is 8.16 g, which corresponds to a lifetime change of approximately 5.3 days when using the fitted linear steady-state maintenance rate. The lifetime estimate is therefore more sensitive to the steady-state maintenance model than to the acquisition model. This also means that the limited duration of the completed steady-state campaign is a central limitation of the final lifetime prediction.

The linear steady-state model does not capture the observed reduction in fuel consumption rate over the completed long-horizon simulation. Since the spacecraft mass decreases as propellant is consumed, the required propellant mass for equivalent orbital corrections may also decrease over time. This makes the linear steady-state model conservative with respect to this effect. In addition, the comparative propagation analysis showed that the NRLMSISE-00 Basilisk configuration used in the GNSS-R Formation Flight Simulator experienced stronger drag than the updating-TLE Skyfield-SGP4 baseline. Although this effect is reduced in the considered leader-follower formation scenario, as discussed Subsection 8.2.2, the same-formation drift comparison in Section 8.1.3 showed larger simulated leader-follower drift in Basilisk than in the Skyfield-SGP4 baseline. This indicates that the steady-state fuel consumption may be overestimated, although the increased formation drift is only partially explained by the available comparison results.

Based on these considerations, the overall predicted lifetime range should be interpreted as a conservative fuel-limited estimate for the selected spacecraft configuration, formation geometry, controller, and disturbance environment. The results indicate that the formation can likely remain fuel-capable for over one year under the simulated assumptions, but the exact depletion time is uncertain. If we assume that the linear steady-state fuel consumption rate were overestimated by a maximum of 10%, the predicted lifetime would increase by roughly 45 days. Combined with the acquisition fuel model uncertainty due to the sensitivity of t_{acq} to the selected OED tolerances, the estimated lifetime is likely accurate to within 2 months of the true lifetime, heavily biased in the positive direction.

8.4 GNSS-R Formation Flight Mission Realism

8.4.1 Representativeness of the Spacecraft Model

The spacecraft configuration used in the GNSS-R formation flight lifetime analysis was designed to satisfy the main operational constraints identified for a cooperative GNSS-R mission in Section 2.6.3. These constraints include the need to observe both direct and reflected GNSS signals, to generate and store electrical power, to communicate with ground stations, and to perform attitude- and formation-control maneuvers. As shown in Table 5.1.1, the direct GNSS receiver and the nadir-pointing GNSS-R payload receiver are mounted on opposite spacecraft faces. This enables the spacecraft to orient the payload side toward the Earth during data capture while maintaining visibility toward the GNSS constellation through the opposite face.

The EPS model also includes the main power-producing and power-consuming elements required for the simulated mission operations. The spacecraft includes solar panels, battery storage, battery heaters, an OBC, a communication system, reaction wheels, a propulsion system, and payload-related power consumption. The payload power model includes the steady-state power consumption associated with the capture and processing pipeline, rather than only the passive GNSS and GNSS-R receiver components. In addition, solar-panel power generation depends on spacecraft attitude, and reaction-wheel power consumption depends on the commanded actuation. The remaining power sinks are represented using mode-dependent constant power draws, which simplifies the component dynamics while still including the main subsystem-level power demands.

The same component set is also used to construct the mass budget in Table 5.1.3. The total spacecraft mass is therefore based on representative 6U-compatible hardware rather than an idealized point-mass spacecraft. However, the inertia estimate is limited by the assumption of uniform density, which produces a diagonal inertia matrix and does not capture the detailed effect of individual component placement on the inertia tensor. Despite this simplification, the spacecraft model includes the major subsystems required for a propulsion-based cooperative GNSS-R formation flight mission. The resulting configuration is therefore representative at the system level, while not being a detailed hardware-resolved spacecraft design.

8.4.2 Operational Mode Realism

The GNSS-R Formation Flight Simulator implements the operational modes needed to represent the main spacecraft-level constraints identified in Section 2.6.3. The spacecraft can switch between `coast`, `comms`, `charge`, `capture`, and burn-related modes depending on the internal battery state and the surrounding communication, illumination, and formation-maintenance opportunities. The steady-state operational behavior shown in Figure 7.3.2 demonstrates that both spacecraft periodically switch between `coast`, `comms`, `charge`, and `capture`, while maintaining a bounded oscillation in battery energy. This indicates that the implemented mode logic is able to reproduce a recurring operational cycle rather than allowing uninterrupted payload operation or unconstrained maneuver execution.

The observed battery oscillation is mainly a consequence of the battery thresholds used by the autonomous mode switching logic. During nominal operation, `capture` is prioritized, but the spacecraft is not allowed to enter this mode if the battery energy is below the lower capture threshold. The spacecraft must then charge until the upper capture threshold is reached before returning to `capture`. These thresholds are listed in Appendix E.0.1. The chosen values were selected to ensure that the spacecraft entered all mission-relevant modes while maximizing the time spent in `capture`. The exact distribution of time between modes is therefore not a unique mission result, but depends on the selected operational thresholds.

Three important limitations remain in the implemented operational mode model. First, `capture` does not evaluate the actual GNSS-R observation geometry or the corresponding payload pointing opportunity. Consequentially, this mode should not be interpreted as a period where the spacecraft is actively performing payload observations, but rather a timeframe where observations are possible. The full payload power consumption is nevertheless active during `capture`, so the mode still contributes to the simulated power budget as a payload-operation state.

Second, cooperative capture scheduling is not implemented. This limitation is less restrictive for the concentric SAR imaging scenario than it would be for a beamforming mode focused on a single shared specular point. However, it still means that the simulator does not verify whether the leader and follower are simultaneously observing a compatible target region for cooperative processing. However, it removes the ability to see when potential cooperative measurements would have been possible

Third, data accumulation from payload observation and data transfer rates during downlink are not modeled. An accurate model of data accumulation would first require an expanded `capture` mode that detects and performs payload observations whenever the constellation and spacecraft geometry makes it feasible. However, a minimum communication event parameter has been implemented to ensure that each feasible `comms` operational mode transfer lasts for a minimum of 10 min, or until the link is broken due to loss of sufficient elevation or line of sight.

8.4.3 Formation Realism

Arguably, the largest mission-realism limitation in the formation flight analysis is related to the achieved relative formation accuracy. Hill et al. explicitly state in [7] that the close-formation concept requires maintaining formations within a 10 m accuracy to enable the cooperative imaging mode. However, Figures 7.2.1 and 7.2.2 show that the acquired leader-follower relative trajectories in this thesis vary by more than 10 m after acquisition, especially in the along-track direction. This means that the simulated formation captures the fuel-consuming task of acquiring and maintaining a bounded relative orbit, but does likely not fully demonstrate the formation accuracy required for a complete concentric cooperative imaging mission.

This comparison should still be interpreted carefully, since the simulated scenario is not a full six-spacecraft concentric formation, as proposed by Hill et al. In their concept, spacecraft are distributed on several concentric relative orbits with increasing radius, while this thesis considers only one leader-follower pair. The 10 m requirement is therefore not applied to the exact same formation architecture. Nevertheless, the observed post-acquisition relative-position variations indicate that achieving a required formation accuracy would require a more precise formation-control implementation than the one used in this thesis.

Several factors may contribute to the observed formation variation. First, Table 7.2.2 shows that the average attitude tracking error during burns varies between approximately 10.2° and 26.6° across the deployment cases. Since the burn-transit mode is intended to align the spacecraft before burn execution, these errors can only be caused by rapidly changing desired attitudes during short burn intervals. Any thrust applied with a pointing error will reduce the accuracy of the resulting formation correction. Second, the implemented formation-maintenance controller regulates desired orbital-element differences between the leader and follower. Since some classical orbital elements are weakly defined for near-circular orbits, this may reduce the robustness of the control setpoint target in the near-circular orbit considered in this formation scenario. These effects limit how directly the maintained formation can be interpreted as a flight-ready GNSS-R imaging formation.

8.5 Main Limitations and Future Work

One of the biggest thesis limitations is that the formation flight lifetime analysis is limited to one formation scenario, a two-spacecraft leader-follower concentric formation. The resulting lifetime estimate is therefore specific to the selected formation radius, spacecraft configuration, controller, and disturbance environment. The lifetime estimates for this formation scenario therefore cannot be generalized directly to other cooperative GNSS-R formations, larger multi-spacecraft constellations, or different relative CPO radii without additional formation studies. A natural extension would be to repeat the lifetime analysis for additional formation geometries and compare their formation-acquisition and steady-state maintenance fuel costs. This would make it possible to evaluate whether the selected concentric case is favorable compared with alternative cooperative formation types.

The achieved formation accuracy is also a major limitation for its applicability to real cooperative GNSS-R missions. The implemented Basilisk *spacecraftReconfig* formation-maintenance module was able to maintain a bounded relative trajectory, but it did not maintain the approximately 10 m relative-position accuracy associated with the close-formation SAR imaging mode discussed in Section ??. Future work should therefore investigate alternative formation-control strategies that are better suited for precise formation maintenance in the near-circular orbital regime considered in this thesis. This may require replacing the classical OED control target with a controller based on relative states.

The operational modelling should also be expanded to better represent a real GNSS-R mission. In the present simulator, the `capture` mode does not evaluate actual GNSS-R observation opportunities. Future work should include GNSS constellation modelling and specular-point computation, such that payload observations are only performed when the transmitter-surface-receiver geometry supports a relevant measurement. In addition, payload data accumulation and on-board and downlink-limited storage should be modelled, such that the spacecraft can be constrained by available onboard storage and required to downlink data before continuing observations.

The attitude dynamics model includes attitude control and reaction-wheel actuation, but it does not include environmental disturbance torques. As a result, the reaction wheels do not experience realistic long-term momentum build-up and do not require desaturation. This removes an important operational constraint for a LEO spacecraft. Future work should include aerodynamic torque, gravity-gradient torque, solar-radiation-pressure torque, magnetic disturbance torque, and a corresponding momentum-management or desaturation strategy.

Finally, the fuel-limited lifetime estimate is primarily limited by the duration of the completed steady-state simulation. The steady-state campaign was intended to cover a longer horizon, but the final completed run was limited to 24 days because of unresolved memory overflow issues in the GNSS-R Formation Flight Simulator. This forced the mission lifetime to be extrapolated far beyond the simulated interval. Future work should therefore reduce the simulator memory accumulation, improve long-horizon data logging, or run the simulation on hardware capable of supporting longer campaigns. Longer completed runs would make it possible to assess whether the steady-state fuel consumption rate continues to decrease as propellant mass is depleted, and whether the apparent low-speed deployment-fuel behavior observed in the acquisition campaign is systematic.

CHAPTER

NINE

CONCLUSIONS

This thesis has investigated the use of a Basilisk simulation framework for estimating the fuel-limited lifetime of a simplified GNSS-R formation flying mission. The work was divided into two main parts. First, the long-horizon translational behavior of the selected Basilisk propagation configuration was benchmarked against a continuously updating Skyfield-SGP4 reference. Second, the simulator was expanded into a GNSS-R formation flight mission simulator, including spacecraft subsystems, operational mode switching, attitude control, formation control, propulsion, power consumption, and fuel depletion.

The first research question asked how the final high-fidelity Basilisk propagation configuration compares against a continuously updating Skyfield-SGP4 reference over a long-horizon propagation interval. The comparative propagation analysis showed that the primary thesis Basilisk configuration, using the NRLMSISE-00 atmosphere model, diverged less from the updating-TLE Skyfield-SGP4 baseline than both exponential-atmosphere Basilisk configurations. The specialization-project exponential atmosphere configuration produced the largest same-spacecraft state difference, while the re-tuned exponential atmosphere case reduced the divergence but did not outperform the NRLMSISE-00 configuration.

These results indicate that the thesis Basilisk configuration is the most suitable of the evaluated configurations for the subsequent formation flight lifetime analysis. However, the comparison does not constitute a strict validation of absolute orbit prediction accuracy, since the Skyfield-SGP4 baseline should be treated as a self-correcting reference rather than a true orbit-determination ground truth. The propagation benchmark therefore provides confidence and uncertainty context for the lifetime analysis, rather than exact long-horizon prediction error bounds.

The second research question asked what fuel-limited mission lifetime is predicted for the selected GNSS-R leader-follower formation when formation acquisition, steady-state formation maintenance, spacecraft subsystem constraints, and operational mode switching are included in the simulation. To answer this, two separate simulation campaigns were performed. The deployment velocity sensitivity campaign was used to estimate the acquisition fuel consumption as a function of differential deployment speed, while the steady-state fuel consumption campaign was used to estimate the post-acquisition formation-maintenance fuel consumption over time.

In the final lifetime estimate, the acquisition and steady-state fuel consumption models were combined into a remaining-fuel model. This model was then used to predict the fuel-limited lifetime of the GNSS-R formation flight mission for the differential deployment speed cases considered in the deployment sensitivity campaign. The resulting linear lifetime estimates ranged from 413.6 days for the most optimistic zero-differential-deployment case to 366.5 days for the $\Delta v_{\mathrm{dep}} = 4$ m/s case.

The acquisition fuel model contains uncertainty related to the definition of acquisition time. The extracted acquisition fuel depends on the selected orbital element difference tolerances and dwell-time criterion, and some post-acquisition burns still contributed noticeably to the cumulative fuel consumption. However, the steady-state fuel consumption model is the dominant source of lifetime estimation uncertainty. This is because the predicted lifetime is extrapolated over several hundred days, while the completed steady-state campaign lasted only 24 days. The linear model used for the final lifetime estimate also does not capture the observed reduction in fuel consumption rate over the simulated interval. Since a decreasing stored propellant mass reduces the propellant required for equivalent orbital corrections, this makes the linear lifetime estimate likely pessimistic with respect to this effect.

The comparative propagation analysis gives a second indication that the lifetime estimate may be pessimistic. The same-formation comparison showed larger leader-follower drift in the Basilisk configurations than in the updating-TLE Skyfield-SGP4 baseline. Since formation-maintenance fuel is driven by the correction of relative drift, an overestimated drift rate would also lead to overestimated steady-state fuel consumption. However, this interpretation is limited by the fact that the Skyfield-SGP4 baseline is not a true physical ground truth, and by the fact that the same-formation comparison was performed in a deactivated open-loop propagation scenario rather than in the closed-loop formation-maintenance simulator.

The main limitations of the thesis are tied to the considered formation scenario, the available simulation duration, and the implemented formation controller. The lifetime analysis considered only a two-spacecraft leader-follower concentric formation. The resulting lifetime estimate is therefore specific to the selected formation geometry and cannot be generalized directly to other cooperative GNSS-R formations, larger satellite groups, or additional followers in the concentric formation. More simulation campaigns would be required to determine how the acquisition and maintenance fuel consumption change for other formation types, other radii, and more spacecraft.

The simulation duration is another major limitation. The deployment sensitivity campaign was limited to 48 h per case, while the steady-state campaign was intended for a longer horizon but produced a final completed run of only 24 days due to unresolved memory overflow issues. The final lifetime estimate therefore depends on extrapolating a fitted maintenance fuel model far beyond the simulated interval. Longer completed steady-state simulations would be required to determine whether the observed reduction in fuel consumption rate continues, stabilizes, or changes over longer mission durations.

Finally, the achieved formation accuracy limits how directly the simulated mission can be interpreted as a complete cooperative GNSS-R imaging mission. The implemented formation controller is based on classical orbital-element differences and was able to maintain a bounded relative trajectory, but it did not achieve the approximately 10 m relative-position accuracy associated with the close-formation cooperative processing benefits discussed for HydroSwarm-like concentric formations. Future work should therefore investigate alternative control formulations, such as relative-state tracking, relative orbital elements better suited for near-circular orbits, or direct RTN tracking. Further work should also include realistic navigation uncertainty, longer-duration simulation capability, larger formation scenarios, and explicit GNSS-R observation opportunity modelling.

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APPENDIX

A -

GITHUB REPOSITORY

All simulator code used in the thesis is provided by the GitHub repository linked below. Note that both the Comparative Propagation Simulator and GNSS-R Formation Flight Simulator are contained within the *Simulator Repository* link, but reside in different directories.

A.1 Simulator Repository

- https://github.com/chystad/formation-flight-gnssr-simulator

APPENDIX

B -

SIMULATOR FLOW DIAGRAMS

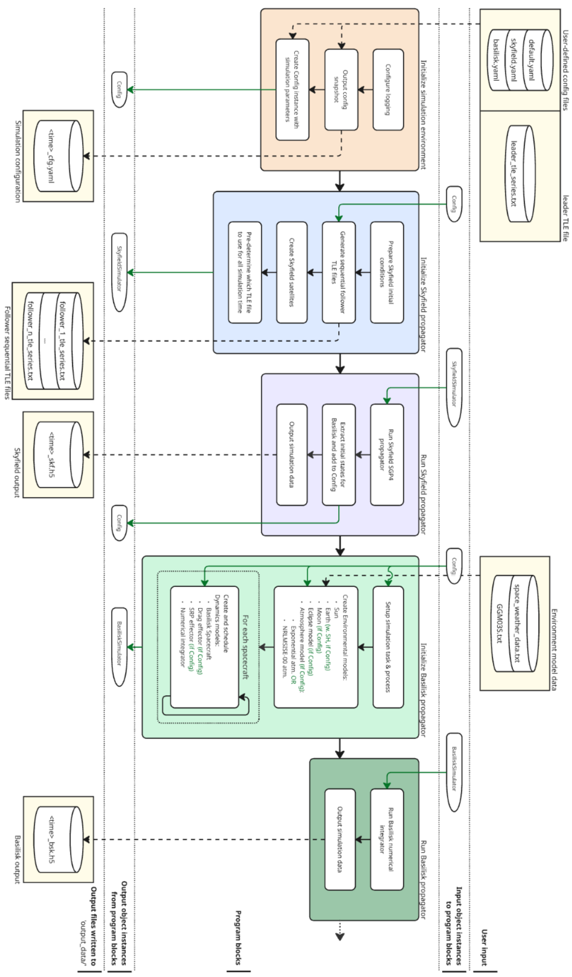

Figure B.0.1: Full size detailed Comparative Propagation Simulator flow diagram

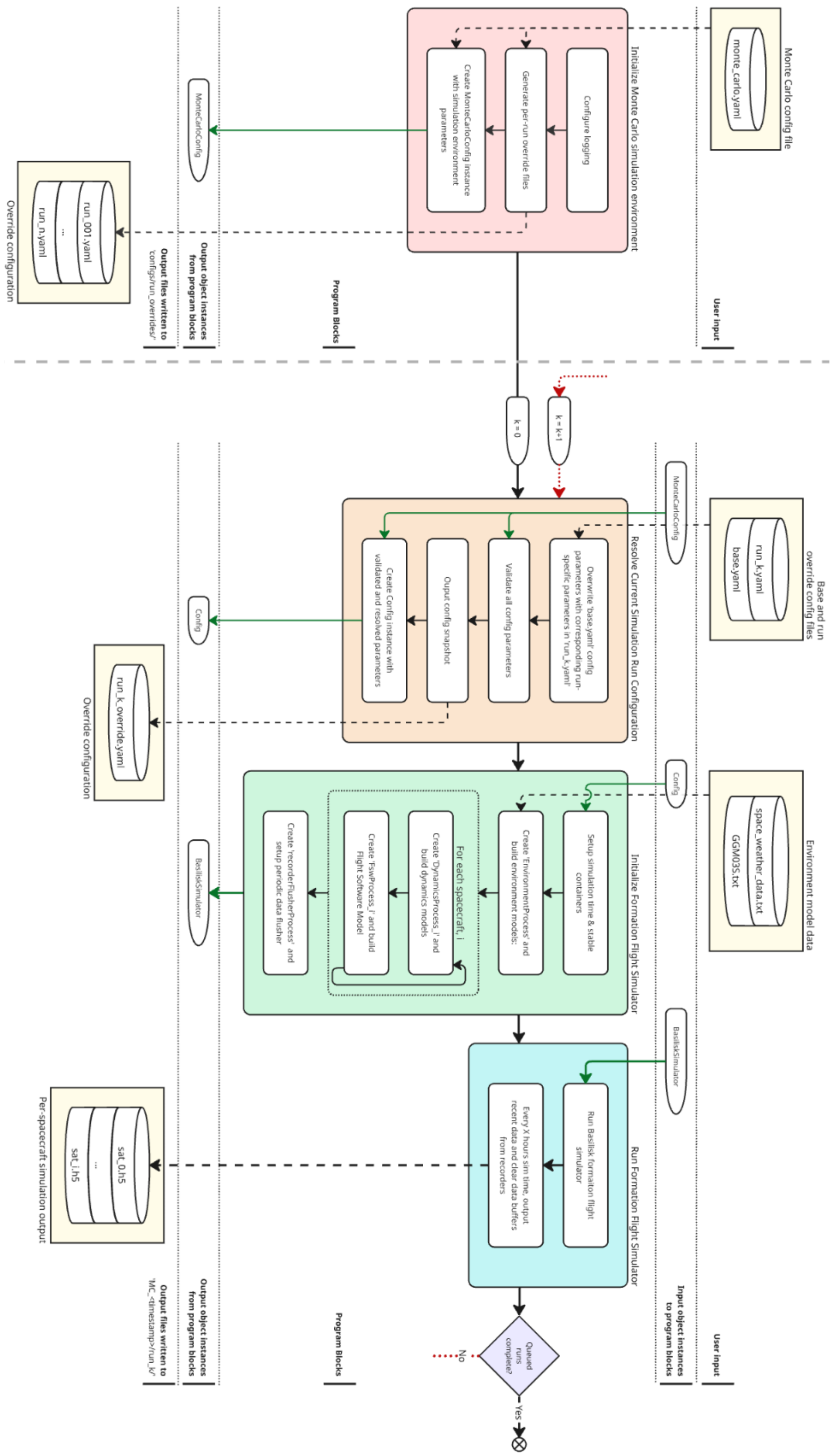

Figure B.0.2: Full size detailed GNSS-R Formation Flight Simulator flow diagram

APPENDIX

C -

SPACECRAFT COMPONENT PARAMETERS

Table C.0.1: Overview of spacecraft components and relevant parameters.

Component	**Mass** [g]	**Power consumptio**	**Important parameters**	**Based on**
Frame (Structural)	665	–	• Dim. [mm]: $226.3 \times 340.5 \times 100.0$	GomSpace NanoStructure 6U
Power distribution (EPS)	80	–	• Dim. [mm]: $90 \times 96 \times 19.8$ • Only support one single connected battery	GomSpace NanoPower dock + 1ACU + 1PCU
Battery (EPS)	500	6 W (Heater)	• Dim. [mm]: $93 \times 86 \times 41$ • 100 Wh, with 86 Wh in desired operating range • 3500 mAh, with 3000 mAh in desired operating range	NanoPower BPX 100 Wh
Solar panels BIG (EPS)	392	19.2 W (Max)	• Dim. [mm]: 200×300 • 30% efficiency	EnduroSat 6U solar panel
Solar panels SMALL (EPS)	2×147	8.4 W (Max)	• Dim. [mm]: 100×300 • 30% efficiency	EnduroSat 3U solar panel
RWs (AOCS)	4×144	0.336 W (avg. at 2000 rpm) 7.2 W (Max)	• Dim. [mm]: $46 \times 46 \times 24$ • 16.2 mNms at 6 krpm • 10,000 rpm max • 7 mNm max torque • 4 in pyramid configuration	CubeSpace CW0162

Continued on next page

Component	Mass [g]	Power consumptic	Important parameters	Based on
Propulsion system (AOCS)	1430 dry 2065 wet	< 1 W (idle) 20 W (warmup) 4.3 W (operating)	• Dim. [mm]: $99.8 \times 92.8 \times 143.5$ • Hybrid Micro Propulsion System (MiPS) • 1 100 mN green monopropellant thruster • 425 mN cold gas thrusters • 783 Ns total impulse • ~133 Isp	VACCO ArgoMoon propulsion system
OBC	240	2 W (idle)	• Dim. [mm]: $95 \times 95 \times 24.2$	GomSpace NanoMind HP MK3
Radio antenna (Comms)	2.2	4 W (max)	• Dim. [mm]: $24 \times 24 \times 5.6$ • Single X-band patch • 74° half power beamwidth • 6 dBi gain • Up to 4 W	EnduroSat X-band patch antenna
Zenith GNSS antenna (Payload)	114	–	• Dim. [mm]: $70 \times 70 \times 10$ • L1+L2 • 100–105 deg HPBW • 3.3–4.0 dBi gain • RHCP • Passive	GNSS patch antenna L1+L2
Nadir GNSS-R antenna (Payload)	–	–	• 12 dBi gain • 3×2 microstrip patch array • LHCP	–

APPENDIX

D -

SIMULATOR BASILISK PROCESS-TASK-MODEL STRUCTURE

D.1 GNSS-R Formation Flight Simulator

```
GNSS-R Formation Flight Simulator
    EnvironmentProcess
        EnvironmentTask
            [20] spiceObj
            [20] eclipseObj
            [20] groundStation(s)
            [20] atmObj
        MsisInputUpdaterTask
            [20] msisInputUpdater
    DynamicsProcess_<sat_idx>
        DynamicsTask_<sat_idx>
            scObj
            [20] dragEffector
            [20] srpEffector
            [20] solarPanel(s)
            [20] rwEffector
            [20] thrusterEffector
            [20] fuelTankEffector
            [20] battery
            [20] obcPowerSink
            [20] rwPower(s)
            [10] recorders
    FswProcess_<sat_idx>
        FswTask_<sat_idx>
            [20] scheduler
            [10] recorders
    RecorderFlusherProcess
        [20] RecorderFlusherTask
            [20] recorderFlusher
```

Figure D.1.1: Process-task-Model hierarchy structure for the GNSS-R Formation Flight Simulator.

APPENDIX

E -

GNSS-R FORMATION FLIGHT CAMPAIGN SUBSYSTEM CONSTANTS

Table E.0.1: GNSS-R Formation Flight Simulator update rate, controller gain and operational mode transition constants shared by all lifetime analysis campaigns

Category	Variable	Value	Description
Module update rate	ENV_RATE	1.0 s^{-1}	Environment model update rate
	DYN_RATE	0.1 s^{-1}	Dynamics model update rate
	FSW_RATE	0.1 s^{-1}	Flight software update rate
	MSIS_RATE	30 s^{-1}	MSIS input update rate
	FLUSH_RATE	24 h	Recorder output and flush interval
Attitude controller gain	MRP_K	0.05	Gain on attitude error
	MRP_P	0.035	Gain on attitude-rate error
	MRP_KI	-1	Integral gain, where -1 disables integral action
Operational mode transition constraints	LOW_BATTERY_THRESHOLD	0.3	Upper battery threshold for low-charge state
	CRITICAL_BATTERY_THRESHOLD	0.2	Upper battery threshold for critically low-charge state
	EMERGENCY_BATTERY_EXIT_THRESHOLD	0.7	Minimum battery level required to exit emergency mode
	CAPTURE_BATTERY_ENTER_THRESHOLD	0.45	Battery threshold for entering capture mode
	CAPTURE_BATTERY_EXIT_THRESHOLD	0.38	Battery threshold for exiting capture mode
	COMMS_BATTERY_ENTER_THRESHOLD	0.35	Battery threshold for entering communication mode
	COMMS_BATTERY_EXIT_THRESHOLD	0.28	Battery threshold for exiting communication mode
	SHADOWFAC_ENTER_THRESHOLD	0.6	Minimum illumination factor required to enter charge mode
	SHADOWFAC_EXIT_THRESHOLD	0.4	Maximum illumination factor required to exit charge mode
	MIN_MINUTES_COM_TIME	10 min	Minimum duration of a communication event, if feasible
	MAX_HOURS_SINCE_LAST_COM_THRESHOLD	12 h	Maximum time since last communication event
	BURN_ATT_ADJUSTMENT_TIME_SEC	15 s	Fixed time between burn request and burn execution

APPENDIX

F -

SUPPLEMENTARY RESULT FIGURES

F.1 Deployment Velocity Acquisition Fuel Sensitivity

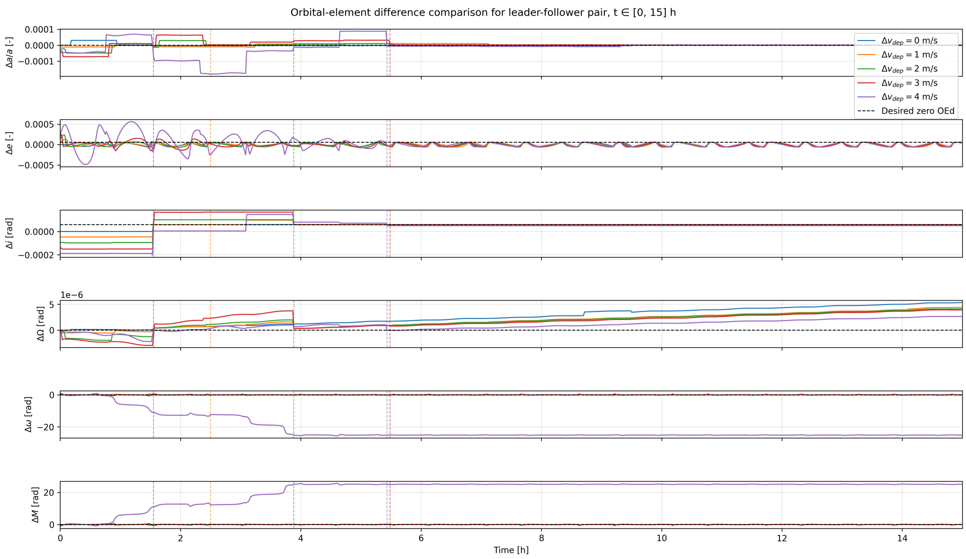

Figure F.1.1: All six OEDs for a selected set of cases in the differential deployment speed fuel sensitivity campaign.

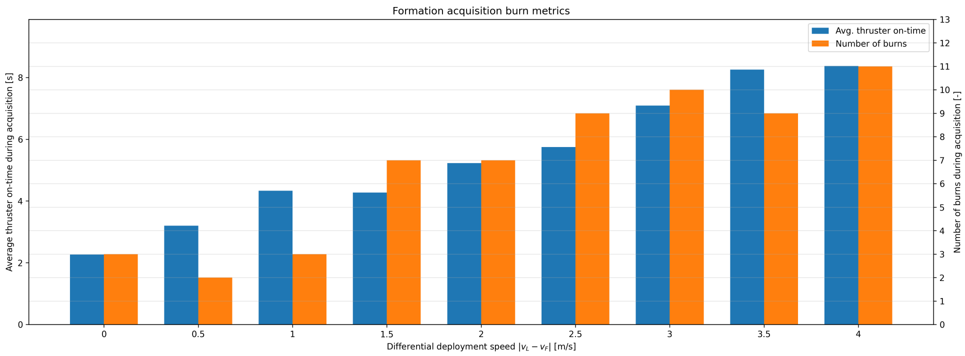

Figure F.1.2: Burn statistics for every case in the differential deployment speed fuel sensitivity campaign. The blue column shows the average duration each burn lasted, while the orange column shows the number of insertion/maintenance burns

.

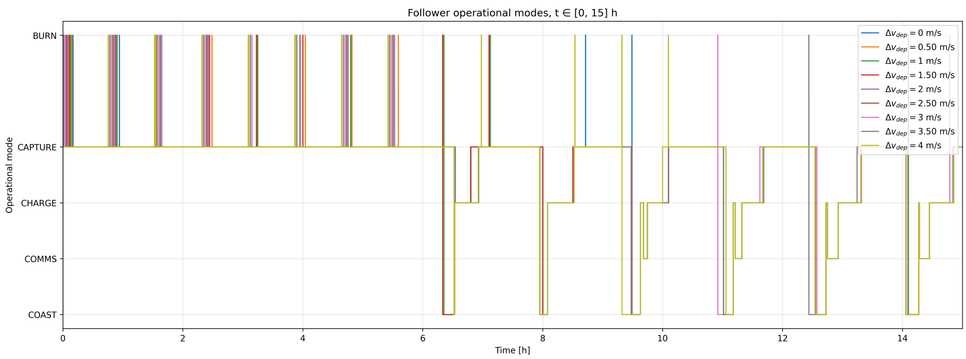

Figure F.1.3: Operational modes shown for all deployment cases considered in the differential deployment sensitivity analysis

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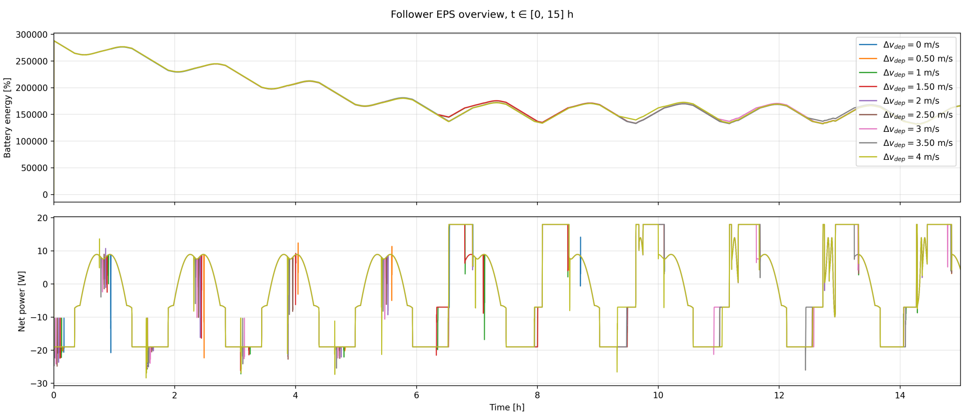

Figure F.1.4: Battery percentage and operational modes for the first 15 hours for all deployment cases considered in the differential deployment sensitivity analysis

.